

# Solutions to Dummit & Foote's Abstract Algebra 3<sup>rd</sup> Edition

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October 27<sup>th</sup>, 2025

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## Preface

This is a collection of solutions to the exercises in Dummit and Foote's *Abstract Algebra*, 3<sup>rd</sup> edition. These solutions were written by me as I worked through the book, and are intended to serve as a reference for myself and others who are studying abstract algebra. I have made every effort to ensure the correctness of these solutions, but I cannot guarantee that they are free of errors. If you find any mistakes or have suggestions for improvement, please feel free to contact me.

I've attempted to format the solutions in a clear and consistent manner, using LaTeX for typesetting. Each chapter's exercises are included in separate files for better organization if viewing on GitHub. Moreover, solutions to exercises only utilize techniques that are available to an advanced undergraduate student or beginning graduate student, in line with the intended audience of the textbook. As an aside, I've graduated from a bachelor's program in mathematics but have not pursued graduate studies, so my background is primarily at the undergraduate level.

Some suggestions I've been given to improve the guide are citing particularly important exercises that will prove useful in reading subsequent chapters, and providing writeups/discussions for why I choose a particular method of solution (such as functions/homomorphisms that may appear out of nowhere). Most problems are enclosed in the following environment:

**Exercise 0.0.1**

Example exercise.

However, there are some exercises that I found to be either challenging, interesting, or useful for the development of later material. These exercises are enclosed in a special environment:

**(\*) Exercise 0.0.2**

Example special exercise.

This is to highlight these exercises for future readers who may want to focus on them.

Many thanks to the authors, David S. Dummit and Richard M. Foote, for writing such an excellent textbook that has been a valuable resource for my recreational studies in abstract algebra. I've received requests on Reddit for individuals to assist me in completing this project, but at this time I prefer to work on it independently. However, I welcome feedback and suggestions from anyone who is interested in contributing to the project in the future.

## 2 Subgroups

### 2.1 Definitions and Examples

#### Exercise 2.1.1

In each of (a)-(e) prove that the specified subset is a subgroup of the given group:

- (a) the set of complex numbers of the form  $a + ai$ ,  $a \in \mathbb{R}$  (under addition)
- (b) the set of complex numbers of absolute value 1, i.e., the unit circle in the complex plane (under multiplication)
- (c) for fixed  $n \in \mathbb{Z}^+$  the set of rational numbers whose denominators divide  $n$  (under addition)
- (d) for fixed  $n \in \mathbb{Z}^+$  the set of rational numbers whose denominators are relatively prime to  $n$  (under addition)
- (e) the set of nonzero real numbers whose square is a rational number (under multiplication)

**Solution.** Let  $G$  be the group and  $H$  be the subset in question.

- (a) Since  $0 + 0i \in H$ , then  $H$  is nonempty. For  $a + ai, b + bi \in H$ ,

$$(a + ai) - (b + bi) = (a - b) + (a - b)i \in H$$

so that  $H \leq G$ .

- (b) Since  $|1| = 1$ , then  $1 \in H$  so that it is nonempty. For  $z, w \in H$ , we first note that  $|w^{-1}| = |\bar{w}|/|w|^2 = 1/1 = 1$ , where  $\bar{w}$  is the complex conjugate of  $w$ . Then

$$|zw^{-1}| = |z||w^{-1}| = 1 \cdot 1 = 1$$

so that  $zw^{-1} \in H$ , hence  $H \leq G$ .

- (c) Fix  $n \in \mathbb{Z}^+$ . Since  $0 = 0/k$  for some  $k$  such that  $k \mid n$ , then  $0 \in H$  so that  $H$  is nonempty. Let  $a/b, c/d \in H$  such that  $(a, c) = (b, d) = 1$ , and  $b \mid n$  and  $d \mid n$ . Then there exists  $x, y \in \mathbb{Z}^+$  such that  $bx = n$  and  $dy = n$ . Then

$$\frac{a}{b} - \frac{c}{d} = \frac{ax - cy}{n}$$

where reducing this fraction still yields a denominator that divides  $n$ . Then  $H \leq G$ .

- (d) Fix  $n \in \mathbb{Z}^+$ . Since  $0 = 0/k$  for some  $k$  such that  $(k, n) = 1$ , then  $0 \in H$  so that  $H$  is nonempty. Let  $a/b, c/d \in H$  such that  $(a, c) = 1$ ,  $(b, n) = 1$ , and  $(d, n) = 1$ . Then

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

Assume, by way of contradiction, that  $(bd, n) \neq 1$ . Then there exists some prime  $p$  such that  $p \mid bd$  and  $p \mid n$ . Then  $p \mid b$  or  $p \mid d$ , contradicting our assumption. Then  $(bd, n) = 1$  so that  $H \leq G$ .

- (e) Since  $1 = 1^1 = 1/1$ , then  $1 \in H$  so that  $H$  is nonempty. For  $x, y \in H$ , then  $x^2, y^2 \in \mathbb{Q}$ . Then

$$\left(\frac{x}{y}\right)^2 = \frac{x^2}{y^2} \in \mathbb{Q}$$

so that  $x/y \in H$ , hence  $H \leq G$ . ■

**Exercise 2.1.2**

In each of (a)–(e) prove that the specified subset is *not* a subgroup of the given group:

- (a) the set of 2-cycles in  $S_n$  for  $n \geq 3$
- (b) the set of reflections in  $D_{2n}$  for  $n \geq 3$
- (c) for  $n$  a composite integer  $> 1$  and  $G$  a group containing an element of order  $n$ , the set  $\{x \in G \mid |x| = n\} \cup \{1\}$
- (d) the set of (positive and negative) odd integers in  $\mathbb{Z}$  together with 0
- (e) the set of real numbers whose square is a rational number (under addition)

**Solution.** Let  $G$  be the group and  $H$  be the subset in question.

- (a) For  $n = 3$ , then  $(1\ 2), (1\ 3) \in H$ , but  $(1\ 2)(1\ 3) = (1\ 3\ 2) \notin H$ .
- (b) For  $n = 3$ , then  $s, sr \in H$ , but  $s(sr) = r \notin H$ .
- (c) Let  $n = ab$  for some  $a, b \in \mathbb{Z}^+$  with  $1 < a, b < n$ . Let  $x \in G$  with  $|x| = n$ . Then  $x \in H$  so that  $x^a \in H$ , but  $(x^a)^b = x^n = 1$  implies that  $|x^a| \leq b$  so that  $x^a \notin H$ .
- (d) Observe that  $1 \in H$ , but  $1 + 1 = 2 \notin H$ .
- (e) Observe that  $\sqrt{2}, \sqrt{3} \in H$ , but

$$(\sqrt{2} + \sqrt{3})^2 = 2 + 2\sqrt{6} + 3 \notin \mathbb{Q}$$

so that  $\sqrt{2} + \sqrt{3} \notin H$ . ■

**Exercise 2.1.3**

Show that the following subsets of the dihedral group  $D_8$  are actually subgroups:

- (a)  $\{1, r^2, s, sr^2\}$
- (b)  $\{1, r^2, sr, sr^3\}$

**Solution.**

(a)

| $\circ$ | 1      | $r^2$  | $s$    | $sr^2$ |
|---------|--------|--------|--------|--------|
| 1       | 1      | $r^2$  | $s$    | $sr^2$ |
| $r^2$   | $r^2$  | 1      | $sr^2$ | $s$    |
| $s$     | $s$    | $sr^2$ | 1      | $r^2$  |
| $sr^2$  | $sr^2$ | $s$    | $r^2$  | 1      |

(b)

| $\circ$ | 1      | $r^2$  | $sr$   | $sr^3$ |
|---------|--------|--------|--------|--------|
| 1       | 1      | $r^2$  | $sr$   | $sr^3$ |
| $r^2$   | $r^2$  | 1      | $sr^3$ | $sr$   |
| $sr$    | $sr$   | $sr^3$ | 1      | $r^2$  |
| $sr^3$  | $sr^3$ | $sr$   | $r^2$  | 1      |

Both tables show closure, since no product is an element outside of the subset. ■

**Exercise 2.1.4**

Give an explicit example of a group  $G$  and an infinite subset  $H$  of  $G$  that is closed under the group operation but is not a subgroup of  $G$ .

**Solution.** Let  $G = \mathbb{Z}$  and  $H = \mathbb{Z}^+$ . Then for any  $m, n \in \mathbb{Z}^+$ , we have  $m + n \in \mathbb{Z}^+$  but  $m - n \notin \mathbb{Z}^+$  when  $m < n$ . Hence,  $H$  is not a subgroup of  $G$ . ■

**Exercise 2.1.5**

Prove that  $G$  cannot have a subgroup  $H$  with  $|H| = n - 1$ , where  $n = |G| > 2$ .

**Solution.** Assume, by way of contradiction, that  $G$  has a subgroup  $H$  where  $|H| = n - 1$ . By Lagrange's Theorem,  $|H|$  divides  $|G|$ , so there exists some  $k \in \mathbb{Z}^+$  such that  $k(n - 1) = kn - k = n$ . Equivalently,  $n = k/(k - 1)$ . Since  $n > 2$ , then  $k > 2$ . Then  $k/(k - 1)$  is not an integer, a contradiction. Hence, no such subgroup  $H$  exists. ■

**(\*) Exercise 2.1.6**

Let  $G$  be an abelian group. Prove that  $\{g \in G \mid |g| < \infty\}$  is a subgroup of  $G$  (called the torsion subgroup of  $G$ ). Give an explicit example where this set is not a subgroup when  $G$  is non-abelian.

**Solution.** Denote the set as  $\text{Tor}(G)$ . Since  $|1| = 1$ , then  $1 \in \text{Tor}(G)$  so that  $\text{Tor}(G)$  is nonempty. Suppose  $g, h \in \text{Tor}(G)$  where  $|g| = m$  and  $|h| = n$  for  $m, n \in \mathbb{Z}^+$ . Then

$$(gh^{-1})^{mn} = g^{mn}(h^{-1})^{mn} = (g^m)^n((h^n)^{-1})^m = 1^n(1^{-1})^m = 1$$

where we use the fact that  $G$  is abelian to rearrange the terms. Then  $|gh^{-1}| \leq mn < \infty$  so that  $gh^{-1} \in \text{Tor}(G)$ , hence  $\text{Tor}(G) \leq G$ .

For a non-abelian example, let  $G = \text{Aut}(\mathbb{R})$ . Consider the elements  $f, g \in G$  defined as  $f(x) = -x$  and  $g(x) = 1 - x$ . Note that  $|f| = 2$  and  $|g| = 2$ , but

$$(fg)(x) = f(g(x)) = f(1 - x) = x - 1$$

It is easy to show by induction that  $(fg)^n(x) = x - n$  for all  $n \in \mathbb{Z}^+$ , so that  $|fg| = \infty$ . Then  $fg \notin \text{Tor}(G)$ , hence  $\text{Tor}(G)$  is not a subgroup of  $G$ . ■

**Exercise 2.1.7**

Fix some  $n \in \mathbb{Z}$  with  $n > 1$ . Find the torsion subgroup (cf. the previous exercise) of  $\mathbb{Z} \times (\mathbb{Z}/n\mathbb{Z})$ . Show that the set of elements of infinite order together with the identity is *not* a subgroup of this direct product.

**Solution.** Observe that the only element of finite order in  $\mathbb{Z}$  is 0. Moreover, every element of  $\mathbb{Z}/n\mathbb{Z}$  is of finite order. Then

$$\text{Tor}(\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}) = \{(0, \bar{x}) \mid \bar{x} \in \mathbb{Z}/n\mathbb{Z}\}$$

Now consider the set of elements of infinite order together with the identity. Denoting this set as  $H$ , observe that  $(1, 1), (-1, 0) \in H$ , but

$$(1, 1)(-1, 0) = (0, 1) \notin H$$

since  $|(0, 1)| = n < \infty$ . Hence,  $H$  is not a subgroup of  $\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ . ■

**Exercise 2.1.8**

Let  $H$  and  $K$  be subgroups of  $G$ . Prove that  $H \cup K$  is a subgroup if and only if either  $H \subseteq K$  or  $K \subseteq H$ .

**Solution.**

( $\Rightarrow$ ) Suppose  $H \cup K$  is a subgroup. If  $H \subseteq K$ , then we are done. Suppose that  $H \not\subseteq K$ . Then there exists some  $h \in H$  such that  $h \notin K$ . Since  $H \cup K$  is a subgroup, then for any  $k \in K$ , we have  $hk \in H \cup K$ .

Observe that  $k^{-1} \in K$  since  $K \leq G$ . If  $hk \in K$ , then  $h = (hk)k^{-1} \in K$ , contradicting our assumption. It must be that  $hk \in H$ . Since  $h \in H$ , then  $h^{-1}H = H \leq G$ . Then  $h^{-1}(hk) = k \in H$ , hence  $K \subseteq H$ .

( $\Leftarrow$ ) If  $H \subseteq K$ , then  $H \cup K = K \leq G$ . Similarly, if  $K \subseteq H$ , then  $H \cup K = H \leq G$ . ■

(\*) **Exercise 2.1.9**

Let  $G = \text{GL}_n(F)$ , where  $F$  is any field. Define

$$\text{SL}_n(F) = \{A \in \text{GL}_n(F) \mid \det(A) = 1\}$$

(called the special linear group). Prove that  $\text{SL}_n(F) \leq \text{GL}_n(F)$ .

**Solution.** Since  $\det(I_n) = 1$ , then  $I_n \in \text{SL}_n(F)$  so that  $\text{SL}_n(F)$  is nonempty. Suppose  $A, B \in \text{SL}_n(F)$ . Then

$$\det(AB^{-1}) = \det(A) \det(B^{-1}) = \frac{\det(A)}{\det(B)} = \frac{1}{1} = 1$$

so that  $AB^{-1} \in \text{SL}_n(F)$ . It follows that  $\text{SL}_n(F) \leq \text{GL}_n(F)$ . ■

**Exercise 2.1.10**

- (a) Prove that if  $H$  and  $K$  are subgroups of  $G$  then so is their intersection  $H \cap K$ .
- (b) Prove that the intersection of an arbitrary nonempty collection of subgroups of  $G$  is again a subgroup of  $G$  (do not assume the collection is countable).

**Solution.**

- (a) Let  $L = H \cap K$ . Since  $1 \in H$  and  $1 \in K$  because  $H \leq G$  and  $K \leq G$ , then  $1 \in L$  so that  $L$  is nonempty. Suppose  $a, b \in L$ . Then  $a, b \in H$  and  $a, b \in K$ . Since  $H$  and  $K$  are subgroups of  $G$ , then  $ab^{-1} \in H$  and  $ab^{-1} \in K$ . Hence,  $ab^{-1} \in L$ , and  $L \leq G$ .
- (b) Let  $I$  be an indexing set, and consider the collection of subgroups  $\{H_i\}_{i \in I}$  of  $G$ . Let

$$H = \bigcap_{i \in I} H_i$$

Since  $1 \in H_i$  for all  $i \in I$ , then  $1 \in H$ . Suppose  $a, b \in H$ . Then  $a, b \in H_i$  for all  $i \in I$ . Since each  $H_i$  is a subgroup of  $G$ , then  $ab^{-1} \in H_i$  for all  $i \in I$ . Hence,  $ab^{-1} \in H$ , and  $H \leq G$ . ■

**Exercise 2.1.11**

Let  $A$  and  $B$  be groups. Prove that the following sets are subgroups of the direct product  $A \times B$ :

- (a)  $\{(a, 1) \mid a \in A\}$
- (b)  $\{(1, b) \mid b \in B\}$
- (c)  $\{(a, a) \mid a \in A\}$ , where we assume  $B = A$  (called the diagonal subgroup)

**Solution.** Let  $C$  be the set in each part.

- (a) Since  $1 \in A$ , then  $(1, 1) \in C$  so that  $C$  is nonempty. Suppose  $(a_1, 1), (a_2, 1) \in C$ . Then

$$(a_1, 1)(a_2, 1)^{-1} = (a_1, 1)(a_2^{-1}, 1) = (a_1 a_2^{-1}, 1) \in C$$

since  $a_1 a_2^{-1} \in A$ . Hence,  $C \leq A \times B$ .

- (b) Since  $1 \in B$ , then  $(1, 1) \in C$  so that  $C$  is nonempty. Suppose  $(1, b_1), (1, b_2) \in C$ . Then

$$(1, b_1)(1, b_2)^{-1} = (1, b_1)(1, b_2^{-1}) = (1, b_1 b_2^{-1}) \in C$$

since  $b_1 b_2^{-1} \in B$ . Hence,  $C \leq A \times B$ .

- (c) Since  $1 \in A$ , then  $(1, 1) \in C$  so that  $C$  is nonempty. Suppose  $(a_1, a_1), (a_2, a_2) \in C$ . Then

$$(a_1, a_1)(a_2, a_2)^{-1} = (a_1, a_1)(a_2^{-1}, a_2^{-1}) = (a_1 a_2^{-1}, a_1 a_2^{-1}) \in C$$

since  $a_1 a_2^{-1} \in A$ . Hence,  $C \leq A \times B$ . ■

**Exercise 2.1.12**

Let  $A$  be an abelian group and fix some  $n \in \mathbb{Z}$ . Prove that the following sets are subgroups of  $A$ :

- (a)  $\{a^n \mid a \in A\}$
- (b)  $\{a \in A \mid a^n = 1\}$

**Solution.** Let  $B$  be the sets in question.

(a) Since  $1 = 1^n$ , then  $1 \in B$  so that  $B$  is nonempty. Suppose  $a^n, b^n \in B$ . Then

(b)

$$(a^n)(b^n)^{-1} = a^n b^{-n} = (ab^{-1})^n \in B$$

where the last equality follows from the fact that  $A$  is abelian. Hence,  $B \leq A$ .

(c) Since  $1^n = 1$ , then  $1 \in B$  so that  $B$  is nonempty. Suppose  $a, b \in B$ . Then

$$(ab^{-1})^n = a^n (b^{-1})^n = a^n (b^n)^{-1} = 1 \cdot 1^{-1} = 1$$

where the second equality follows from the fact that  $A$  is abelian. Hence,  $B \leq A$ . ■

**Exercise 2.1.13**

Let  $H$  be a subgroup of the additive group of rational numbers with the property that  $1/x \in H$  for every nonzero element  $x$  of  $H$ . Prove that  $H = 0$  or  $\mathbb{Q}$ .

**Solution.** Let  $H \leq \mathbb{Q}$ . Then  $0 \in H$ . If no other elements are in  $H$ , then  $H = 0$ , and we are done. Suppose there exists some nonzero  $a/b \in H$  where  $a, b \in \mathbb{Z}$  and  $b \neq 0$ . Moreover, we may assume without loss of generality that  $a/b > 0$ , for otherwise we may take  $-a/b$  since  $H$  contains additive inverses. We now use  $a/b$  to generate  $\mathbb{Q}$ .

Since  $a/b \in H$ , then  $b(a/b) = a \in H$  so that  $1/a \in H$ . Then  $a(1/a) = 1 \in H$ . From this, we can see that  $\mathbb{Z} \subseteq H$ . Now let  $c/d \in \mathbb{Q}$  for some  $c, d \in \mathbb{Z}$  with  $d \neq 0$ . Since  $1 \in H$ , then  $d(1) = d \in H$  so that  $1/d \in H$ . Then  $c(1/d) = c/d \in H$ . Since  $c/d$  was arbitrary, then  $\mathbb{Q} \subseteq H$ . Hence,  $H = \mathbb{Q}$ . ■

**Exercise 2.1.14**

Show that  $\{x \in D_{2n} \mid x^2 = 1\}$  is *not* a subgroup of  $D_{2n}$  (here  $n \geq 3$ ).

**Solution.** Let  $H$  be the set in question. Clearly,  $s \in H$ . Moreover,

$$(sr)^2 = sr sr = sr r^{-1} r = 1$$

so that  $|sr| = 2$  and  $sr \in H$ . However,  $s(sr) = r \notin H$  since  $|r| = n > 2$ . Hence,  $H$  is not a subgroup of  $D_{2n}$ . ■

**Exercise 2.1.15**

Let  $H_1 \subseteq H_2 \subseteq \cdots$  be an ascending chain of subgroups of  $G$ . Prove that  $\bigcup_{i=1}^{\infty} H_i$  is a subgroup of  $G$ .

**Solution.** Let

$$H = \bigcup_{i=1}^{\infty} H_i$$

Since  $H_i \leq G$  for each  $i \in \mathbb{Z}^+$ , then  $1 \in H_i$  for all  $i \in \mathbb{Z}^+$ , hence  $1 \in H$  so that  $H$  is nonempty. Suppose  $a, b \in H$ . Then there exists some  $m, n \in \mathbb{Z}^+$  such that  $a \in H_m$  and  $b \in H_n$ . Let  $k = \max(m, n)$ . Since  $H_m \subseteq H_k$  and  $H_n \subseteq H_k$ , then  $a, b \in H_k$ . Since  $H_k \leq G$ , then  $ab^{-1} \in H_k$ , hence  $ab^{-1} \in H$ . It follows that  $H \leq G$ . ■

(\*) **Exercise 2.1.16**

Let  $n \in \mathbb{Z}^+$  and let  $F$  be a field. Prove that the set  $\{(a_{ij}) \in \text{GL}_n(F) \mid a_{ij} = 0 \text{ for all } i > j\}$  is a subgroup of  $\text{GL}_n(F)$  (called the group of upper triangular matrices).

**Solution.** Let  $\text{UT}_n(F)$  denote the set of  $n \times n$  upper triangular matrices with entries from  $F$ . Observe that  $I_n$  contains 0's everywhere except the diagonal, hence  $I_n \in \text{UT}_n(F)$  so that  $\text{UT}_n(F)$  is nonempty. Suppose  $A, B \in \text{UT}_n(F)$ , and consider  $C = AB$ . Then the  $(i, j)$ -th entry of  $C$  is given by

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

Consider the case when  $i > j$ , which is below the main diagonal.

- If  $i > k$ , then  $a_{ik} = 0$  since  $A$  is upper triangular.
- If  $k > j$ , then  $b_{kj} = 0$  since  $B$  is upper triangular.

Hence, for all  $k$ , either  $a_{ik} = 0$  or  $b_{kj} = 0$ , so that  $c_{ij} = 0$ . It follows that  $C \in \text{UT}_n(F)$ .

Let  $A \in \text{UT}_n(F)$ , and let  $D \in \text{GL}_n(F)$  such that  $AD = DA = I_n$ . We show by induction that  $D \in \text{UT}_n(F)$ . For the base case, consider  $n = 2$ . Since we know  $A \in \text{UT}_n(F)$  by assumption but do not know about  $D$ , we write

$$A = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix}, \quad D = \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix}, \quad AD = \begin{pmatrix} a_{11}d_{11} + a_{12}d_{21} & a_{11}d_{12} + a_{12}d_{22} \\ a_{22}d_{21} & a_{22}d_{22} \end{pmatrix} = I_2$$

Observe that  $a_{22}d_{21} = 0$ . Since  $a_{22} \neq 0$  because  $A \in \text{UT}_2(F) \subseteq \text{GL}_2(F)$ , then  $d_{21} = 0$ , hence  $D \in \text{UT}_2(F)$ . Now suppose that the inverse of any upper triangular matrix in  $\text{GL}_n(F)$  is also upper triangular, and consider  $A \in \text{UT}_{n+1}(F)$  and  $D \in \text{GL}_{n+1}(F)$  such that  $AD = DA = I_{n+1}$ . Using block matrices, we may write this as

$$A = \begin{pmatrix} A_0 & \mathbf{a}_{12} \\ 0 & a_{22} \end{pmatrix}, \quad D = \begin{pmatrix} D_0 & \mathbf{d}_{12} \\ \mathbf{d}_{21}^T & d_{22} \end{pmatrix}, \quad AD = \begin{pmatrix} A_0D_0 & A_0\mathbf{d}_{12} + \mathbf{a}_{12}d_{22} \\ a_{22}\mathbf{d}_{21}^T & a_{22}d_{22} \end{pmatrix} = I_{n+1}$$

where  $\mathbf{a}_{12}, \mathbf{d}_{12}, \mathbf{d}_{21}^T$  are  $n \times 1$  column vectors, and  $a_{22}, d_{22} \in F$ . We can see that  $A_0D_0 = I_n$  so that  $D_0 \in \text{UT}_n(F)$  by assumption. Moreover,  $a_{22}\mathbf{d}_{21}^T = \mathbf{0}_{n \times 1}^T$ . Since  $a_{22} \neq 0$  because  $A \in \text{UT}_{n+1}(F)$ , then  $\mathbf{d}_{21}^T = \mathbf{0}_{n \times 1}^T$ . By induction, then  $D \in \text{UT}_{n+1}(F)$ , hence the inverse of any upper triangular matrix is also upper triangular. ■

**Exercise 2.1.17**

Let  $n \in \mathbb{Z}^+$  and let  $F$  be a field. Prove that the set  $\{(a_{ij}) \in \text{GL}_n(F) \mid a_{ij} = 0 \text{ for all } i > j, \text{ and } a_{ii} = 1 \text{ for all } i\}$  is a subgroup of  $\text{GL}_n(F)$ .

**Solution.** Let  $H$  be the set in question. Since the diagonal of  $I_n$  is only 1's, then  $I_n \in H$ . Suppose  $A, B \in H$ . By the previous exercise,  $A, B \in \text{UT}_n(F)$ , so it remains to show that  $a_{ii} = b_{ii} = 1$  for all  $1 \leq i \leq n$ . Then

$$(AB)_{ii} = \sum_{k=1}^n a_{ik} b_{ki}$$

Since  $a_{ik} = 0$  for  $k < i$  and  $b_{ki} = 0$  for  $k > i$ , then the sum degrades to  $a_{ii}b_{ii} = 1$ . Then  $H$  is closed under multiplication. Moreover, suppose  $D \in \text{UT}_n(F)$  such that  $DA = I_n$ . Then

$$1 = (DA)_{ii} = d_{ii}a_{ii}$$

where we use the above to collapse the  $ii$ -th term of  $DA$ . Then  $d_{ii} = 1$ , hence  $D \in H$ , and  $H$  is closed under inverses. Hence,  $H \leq \text{GL}_n(F)$ . ■



## 2.2 Centralizers and Normalizers, Stabilizers and Kernels

### Exercise 2.2.1

Prove that  $C_G(A) = \{g \in G \mid g^{-1}ag = a \text{ for all } a \in A\}$ .

**Solution.** For any  $a \in A \subseteq G$ , then  $g \in C_G(A)$  if and only if  $gag^{-1} = a$  if and only if  $ga = ag$  if and only if  $g^{-1}ag = a$ . Hence,  $C_G(A) = \{g \in G \mid g^{-1}ag = a \text{ for all } a \in A\}$ . ■

### Exercise 2.2.2

Prove that  $C_G(Z(G)) = G$  and deduce that  $N_G(Z(G)) = G$ .

**Solution.** By definition,  $C_G(Z(G)) \subseteq G$ . Now let  $g \in G$  and  $z \in Z(G)$ . Then  $gz = zg$ , or  $gzg^{-1} = z$ . Then  $g \in C_G(Z(G))$ , hence  $G \subseteq C_G(Z(G))$ . It follows that  $C_G(Z(G)) = G$ . Since  $C_G(Z(G)) \leq N_G(Z(G)) \leq G$ , then  $N_G(Z(G)) = G$ . ■

### Exercise 2.2.3

Prove that if  $A$  and  $B$  are subsets of  $G$  with  $A \subseteq B$  then  $C_G(B)$  is a subgroup of  $C_G(A)$ .

**Solution.** Pick  $g \in C_G(B)$ . Then  $gbg^{-1} = b$  for all  $b \in B$ . In particular,  $gag^{-1} = a$  for all  $a \in A$  because  $A \subseteq B$ . Then  $C_G(B) \subseteq C_G(A)$ . Since  $C_G(B)$  and  $C_G(A)$  are subgroups of  $G$ , then  $C_G(B) \leq C_G(A)$ . ■

### (\*) Exercise 2.2.4

For each of  $S_3$ ,  $D_8$ , and  $Q_8$  compute the centralizers of each element and find the center of each group. Does Lagrange's Theorem (Exercise 1.7.19) simplify your work?

**Solution.** To compute the center of a group  $G$ , we need to compute the centralizer of each element in  $G$  and then take the intersection of all these centralizers. The proof of this is trivial.

We begin with  $S_3$ . It is trivial to see that  $C_G(1) = G$  for any group  $G$ , since 1 commutes with every element of  $g$ . Now consider  $(1\ 2) \in S_3$ , and consider the subgroup  $A = \{1, (1\ 2)\}$ . Since  $A$  is a subgroup, then  $A \leq C_{S_3}((1\ 2))$ . By Lagrange's Theorem, then 2 divides  $|C_{S_3}((1\ 2))|$  and  $|C_{S_3}((1\ 2))|$  divides 6. Since  $(1\ 3)(1\ 2) = (1\ 2\ 3) \neq (1\ 3\ 2) = (1\ 2)(1\ 3)$ , then  $(1\ 3) \notin C_{S_3}((1\ 2))$ . By Exercise 1.3.20, we know that  $(1\ 2)$  and  $(1\ 3)$  generate  $S_3$  so that  $|C_{S_3}((1\ 2))| \neq 6$ . Then  $|C_{S_3}((1\ 2))| = 2$  so that  $C_{S_3}((1\ 2)) = A$ . By symmetry, the same argument holds for  $(1\ 3)$  and  $(2\ 3)$ , so that

$$C_{S_3}((1\ 3)) = \{1, (1\ 3)\}, \quad C_{S_3}((2\ 3)) = \{1, (2\ 3)\}$$

Now consider  $(1\ 2\ 3) \in S_3$ , and the subgroup  $B = \{1, (1\ 2\ 3), (1\ 3\ 2)\}$ . Since  $B$  is a subgroup, then  $B \leq C_{S_3}((1\ 2\ 3))$ . By Lagrange's Theorem, then 3 divides  $|C_{S_3}((1\ 2\ 3))|$  and  $|C_{S_3}((1\ 2\ 3))|$  divides 6. Observe that  $(1\ 2) \notin C_{S_3}((1\ 2\ 3))$  since  $(1\ 2)(1\ 2\ 3) \neq (1\ 2\ 3)(1\ 2)$ . Then  $|C_{S_3}((1\ 2\ 3))| \neq 6$ , so that  $|C_{S_3}((1\ 2\ 3))| = 3$  and  $C_{S_3}((1\ 2\ 3)) = B$ . By symmetry, the same argument holds for  $(1\ 3\ 2)$ . Moreover,  $Z(G) = \{1\}$ .

The following are the centralizers for each element in  $D_8$ :

$$\begin{aligned} C_{D_8}(1) &= D_8 & C_{D_8}(r) &= \{1, r, r^2, r^3\} \\ C_{D_8}(r^3) &= \{1, r, r^2, r^3\} & C_{D_8}(r^2) &= D_8 \\ C_{D_8}(sr) &= \{1, r^2, sr, sr^3\} & C_{D_8}(s) &= \{1, r^2, s, sr^2\} \\ C_{D_8}(sr^3) &= \{1, r^2, sr, sr^3\} & C_{D_8}(sr^2) &= \{1, r^2, s, sr^2\} \end{aligned}$$

and  $Z(D_8) = \{1, r^2\}$ . Lastly, the centralizers of  $Q_8$  are

$$\begin{aligned} C_{Q_8}(1) &= Q_8 & C_{Q_8}(-1) &= Q_8 \\ C_{Q_8}(i) &= \{1, -1, i, -i\} & C_{Q_8}(-i) &= \{1, -1, i, -i\} \\ C_{Q_8}(j) &= \{1, -1, j, -j\} & C_{Q_8}(-j) &= \{1, -1, j, -j\} \\ C_{Q_8}(k) &= \{1, -1, k, -k\} & C_{Q_8}(-k) &= \{1, -1, k, -k\} \end{aligned}$$

and  $Z(Q_8) = \{1, -1\}$ . ■

**Exercise 2.2.5**

In each of parts (a) to (c) show that for the specified group  $G$  and subgroup  $A$  of  $G$ ,  $C_G(A) = A$  and  $N_G(A) = G$ .

- (a)  $G = S_3$  and  $A = \{1, (1\ 2\ 3), (1\ 3\ 2)\}$ .
- (b)  $G = D_8$  and  $A = \{1, s, r^2, sr^2\}$ .
- (c)  $G = D_{10}$  and  $A = \{1, r, r^2, r^3, r^4\}$ .

**Solution.**

- (a) Observe that  $A$  is generated by  $(1\ 2\ 3)$ . From the previous exercise, we know that  $C_{S_3}((1\ 2\ 3)) = A$ , hence  $C_{S_3}(A) = A$ . Since  $C_G(A) \leq N_G(A)$ , we know that 3 divides  $|N_G(A)|$  and  $|N_G(A)|$  divides 6 by Lagrange's Theorem. Consider  $(1\ 2) \in S_3$ . Then

$$(1\ 2)A(1\ 2) = \{(1\ 2)(1)(1\ 2), (1\ 2)(1\ 2\ 3)(1\ 2), (1\ 2)(1\ 3\ 2)(1\ 2)\} = \{1, (1\ 3\ 2), (1\ 2\ 3)\} = A$$

so that  $(1\ 2) \in N_G(A)$ . Then  $|N_G(A)| \neq 3$ , hence  $|N_G(A)| = 6$  and  $N_G(A) = G$ .

- (b) Since  $A \subseteq C_G(A)$  is abelian and  $C_G(A) \leq G$ , then  $A \leq G$ . Since  $|A| = 4$ , then 4 divides  $|C_G(A)|$  and  $|C_G(A)|$  divides 8 by Lagrange's Theorem. Observe that  $r \notin C_G(A)$  since  $rs = sr^3 \neq sr$ . Then  $|C_G(A)| \neq 8$ , hence  $|C_G(A)| = 4$  and  $C_G(A) = A$ . Since  $C_G(A) \leq N_G(A)$ , then 4 divides  $|N_G(A)|$  and  $|N_G(A)|$  divides 8 by Lagrange's Theorem. Consider  $r \in D_8$ . Then

$$rAr^{-1} = \{r(1)r^{-1}, r(s)r^{-1}, r(r^2)r^{-1}, r(sr^2)r^{-1}\} = \{1, sr^3, r^2, sr\} = A$$

so that  $r \in N_G(A)$ . Then  $|N_G(A)| \neq 4$ , hence  $|N_G(A)| = 8$  and  $N_G(A) = G$ .

- (c) Since  $A \leq C_G(A)$  and  $|A| = 5$ , then 5 divides  $|C_G(A)|$  and  $|C_G(A)|$  divides 10 by Lagrange's Theorem. Observe that  $s \notin C_G(A)$  since  $sr = r^4s \neq rs$ . Then  $|C_G(A)| \neq 10$ , hence  $|C_G(A)| = 5$  and  $C_G(A) = A$ . Since  $C_G(A) \leq N_G(A)$ , then 5 divides  $|N_G(A)|$  and  $|N_G(A)|$  divides 10 by Lagrange's Theorem. Consider  $s \in D_{10}$ . Then

$$sAs^{-1} = \{s(1)s^{-1}, s(r)s^{-1}, s(r^2)s^{-1}, s(r^3)s^{-1}, s(r^4)s^{-1}\} = \{1, r^4, r^3, r^2, r\} = A$$

so that  $s \in N_G(A)$ . Then  $|N_G(A)| \neq 5$ , hence  $|N_G(A)| = 10$  and  $N_G(A) = G$ . ■

**Exercise 2.2.6**

Let  $H$  be a subgroup of the group  $G$ .

- (a) Show that  $H \leq N_G(H)$ . Give an example to show that this is not necessarily true if  $H$  is not a subgroup.
- (b) Show that  $H \leq C_G(H)$  if and only if  $H$  is abelian.

**Solution.**

- (a) Our goal is to any element  $h \in H$  normalizes  $H$ . To that end, fix  $h \in H$ , and consider  $x \in hHh^{-1}$ . Then there exists  $k \in H$  such that  $x = hkh^{-1}$ . Since  $H \leq G$ , then  $hkh^{-1} \in H$ , hence  $x \in H$ . It follows that  $hHh^{-1} \subseteq H$ .

Now consider  $y \in H$ . Since  $H$  is a subgroup, then  $h^{-1}yh \in H$ . Let  $k = h^{-1}yh$ . We then obtain  $y = hkh^{-1} \in hHh^{-1}$ , hence  $H \subseteq hHh^{-1}$ . It follows that  $hHh^{-1} = H$ , so that  $h \in N_G(H)$ . Since  $h$  was arbitrary, then  $H \leq N_G(H)$ .

As a counter example, consider the set  $G = D_6$  and  $H = \{1, r, s\}$ . Then

$$rHr^{-1} = \{1, r, sr^2\} \neq H$$

so that  $r$  does not normalize  $H$ , and  $H$  is not a subgroup of  $N_G(H)$ .

- (b) •( $\Rightarrow$ ) Suppose  $H \leq C_G(H)$ . Then for any  $h_1, h_2 \in H$ , we have  $h_1h_2 = h_2h_1$  since  $h_1 \in C_G(H)$ . Hence,  $H$  is abelian.  
 •( $\Leftarrow$ ) Suppose  $H$  is abelian, and let  $h_1 \in H$ . Then for any  $h_2 \in H$ , we have  $h_1h_2 = h_2h_1$ , so that  $h_1 \in C_G(H)$ . Since  $h_1$  was arbitrary, then  $H \leq C_G(H)$ . ■

**Exercise 2.2.7**

Let  $n \in \mathbb{Z}$  with  $n \geq 3$ . Prove the following:

- (a)  $Z(D_{2n}) = \{1\}$  if  $n$  is odd.
- (b)  $Z(D_{2n}) = \{1, r^k\}$  if  $n = 2k$ .

**Solution.** We claim that  $Z(D_{2n})$  contains no reflections and contains  $r^i$  if and only if  $n \mid 2i$ . To see the first part, suppose  $sr^j \in Z(D_{2n})$  for some  $0 \leq j < n$ . Since it must commute with  $r$ , then

$$sr^j r = r sr^j \implies sr^{j+1} = r^{n-1} sr^j \implies sr^{j+1} = sr^{j-1} \implies r^{j+1} = r^{j-1} \implies r^2 = 1$$

which is a contradiction since  $n \geq 3$ . Hence, no reflections are in  $Z(D_{2n})$ .

Now consider  $r^i \in Z(D_{2n})$  for some  $0 \leq i < n$ . Since it must commute with  $s$ , then

$$r^i s = sr^i \implies r^i s = sr^{n-i} \implies r^i = r^{n-i} \implies r^{2i} = 1$$

so that  $n \mid 2i$ .

- (a) If  $n$  is odd, then  $n \mid 2i$  implies that  $n \mid i$ . Since  $0 \leq i < n$ , then  $i = 0$  and  $Z(D_{2n}) = \{1\}$ .
- (b) If  $n = 2k$ , then  $n \mid 2i$  implies that  $2k \mid 2i$ . Simplifying, we have  $k \mid i$ . Since  $0 \leq i < 2k$ , then  $i = 0$  or  $i = k$ , so that  $Z(D_{2n}) = \{1, r^k\}$ . ■

**Exercise 2.2.8**

Let  $G = S_n$ , fix an  $i \in \{1, 2, \dots, n\}$  and let  $G_i = \{\sigma \in G \mid \sigma(i) = i\}$  (the stabilizer of  $i$  in  $G$ ). Use group actions to prove that  $G_i$  is a subgroup of  $G$ . Find  $|G_i|$ .

**Solution.** Since  $\text{id}(1) = 1$ , then  $\text{id} \in G_i$  so that  $G_i$  is nonempty. Suppose  $\sigma \in G_i$ . Then  $\sigma(i) = i$ . Then

$$\sigma^{-1}(i) = \sigma^{-1}(\sigma(i)) = i$$

so that  $\sigma^{-1} \in G_i$ . Now suppose  $\tau \in G_i$ . Then  $\tau(i) = i$ . It follows that

$$(\sigma\tau)(i) = \sigma(\tau(i)) = \sigma(i) = i$$

so that  $\sigma\tau \in G_i$ . Hence,  $G_i \leq G$ .

To find  $|G_i|$ , observe that any  $\sigma \in G_i$  is completely determined by its action on the set  $\{1, 2, \dots, n\} \setminus \{i\}$ , which has  $n - 1$  elements. Since  $\sigma$  is a permutation, then there are  $(n - 1)!$  ways to permute these  $n - 1$  elements. It follows that  $|G_i| = (n - 1)!$ . ■

**Exercise 2.2.9**

For any subgroup  $H$  of  $G$  and any nonempty subset  $A$  of  $G$  define  $N_H(A)$  to be the set  $\{h \in H \mid hAh^{-1} = A\}$ . Show that  $N_H(A) = N_G(A) \cap H$  and deduce that  $N_H(A)$  is a subgroup of  $H$  (note that  $A$  need not be a subset of  $H$ ).

**Solution.** Suppose  $h \in N_H(A)$ . By definition,  $hAh^{-1} = A$  and  $h \in H$ . Then  $h \in N_G(A)$  and  $h \in H$ , so that  $h \in N_G(A) \cap H$ . It follows that  $N_H(A) \subseteq N_G(A) \cap H$ .

Now suppose  $h \in N_G(A) \cap H$ . Then  $hAh^{-1} = A$  and  $h \in H$ . By definition, then  $h \in N_H(A)$ . It follows that  $N_G(A) \cap H \subseteq N_H(A)$ . Hence,  $N_H(A) = N_G(A) \cap H$ . Since  $N_G(A) \leq G$  and  $H \leq G$ , and  $N_H(A)$  is the intersection of these two subgroups, then  $N_H(A) \leq H$  by [Exercise 2.1.10](#). ■

**Exercise 2.2.10**

Let  $H$  be a subgroup of order 2 in  $G$ . Show that  $N_G(H) = C_G(H)$ . Deduce that if  $N_G(H) = G$  then  $H \leq Z(G)$ .

**Solution.** By assumption,  $H = \{1, h\}$  for some  $h \in G$  with  $h \neq 1$ . Since  $C_G(H) \leq N_G(H)$ , it suffices to show that  $N_G(H) \subseteq C_G(H)$ . To that end, pick  $g \in N_G(H)$ . Then  $gHg^{-1} = H$ , so that

$$gHg^{-1} = \{g(1)g^{-1}, ghg^{-1}\} = \{1, ghg^{-1}\} = H = \{1, h\}$$

It follows that  $ghg^{-1} = h$ , so that  $g \in C_G(H)$ . Since  $g$  was arbitrary, then  $N_G(H) \subseteq C_G(H)$ , hence  $N_G(H) = C_G(H)$ .

Now suppose  $N_G(H) = G$ . Then  $C_G(H) = G$ , so that for any  $g \in G$ , we have  $ghg^{-1} = h$ . It follows that  $hg = gh$  for all  $g \in G$ , hence  $H \leq Z(G)$ . ■

**Exercise 2.2.11**

Prove that  $Z(G) \leq N_G(A)$  for any subset  $A$  of  $G$ .

**Solution.** Let  $z \in Z(G)$ . We wish to show that  $z$  normalizes  $A$ . To that end, consider  $x \in zAz^{-1}$ . Then there exists  $a \in A$  such that  $x = z az^{-1}$ . Since  $z \in Z(G)$ , then  $z az^{-1} = a$ , hence  $x \in A$ . It follows that  $zAz^{-1} \subseteq A$ .

Now consider  $y \in A$ . Since  $z \in Z(G)$ , then  $z^{-1}yz = y$ . Let  $a = z^{-1}yz$ . We then obtain  $y = z az^{-1} \in zAz^{-1}$ , hence  $A \subseteq zAz^{-1}$ . It follows that  $zAz^{-1} = A$ , so that  $z \in N_G(A)$ . Since  $z$  was arbitrary, then  $Z(G) \leq N_G(A)$ . ■

**Exercise 2.2.12**

Let  $R$  be the set of all polynomials with integer coefficients in the independent variables  $x_1, x_2, x_3, x_4$  i.e., the members of  $R$  are finite sums of elements of the form  $ax_1^{r_1}x_2^{r_2}x_3^{r_3}x_4^{r_4}$  where  $a$  is any integer and  $r_1, \dots, r_4$  are nonnegative integers. For example,

$$12x_1^5x_2^7x_4 - 18x_2^3x_3 + 11x_1^6x_2x_3^3x_4^{23} \quad (*)$$

is a typical element of  $R$ . Each  $\sigma \in S_4$  gives a permutation of  $\{x_1, \dots, x_4\}$  by defining  $\sigma \cdot x_i = x_{\sigma(i)}$ . This may be extended to a map from  $R$  to  $R$  by defining

$$\sigma \cdot p(x_1, x_2, x_3, x_4) = p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)})$$

for all  $p(x_1, x_2, x_3, x_4) \in R$  (i.e.,  $\sigma$  simply permutes the indices of the variables). For example, if  $\sigma = (1\ 2)(3\ 4)$  and  $p(x_1, \dots, x_4)$  is the polynomial in  $(*)$  above, then

$$\begin{aligned} \sigma \cdot p(x_1, x_2, x_3, x_4) &= 12x_2^5x_1^7x_4 - 18x_1^3x_4 + 11x_2^6x_1x_4^3x_3^{23} \\ &= 12x_1^7x_2^5x_4 - 18x_1^3x_4 + 11x_1x_2^6x_3^{23}x_4^3. \end{aligned}$$

- Let  $p = p(x_1, \dots, x_4)$  be the polynomial in  $(*)$  above, let  $\sigma = (1\ 2\ 3\ 4)$  and let  $\tau = (1\ 2\ 3)$ . Compute  $\sigma \cdot p$ ,  $\tau \cdot (\sigma \cdot p)$ ,  $(\tau \circ \sigma) \cdot p$ , and  $(\sigma \circ \tau) \cdot p$ .
- Prove that these definitions give a (left) group action of  $S_4$  on  $R$ .
- Exhibit all permutations in  $S_4$  that stabilize  $x_4$  and prove that they form a subgroup isomorphic to  $S_3$ .
- Exhibit all permutations in  $S_4$  that stabilize the element  $x_1 + x_2$  and prove that they form an abelian subgroup of order 4.
- Exhibit all permutations in  $S_4$  that stabilize the element  $x_1x_2 + x_3x_4$  and prove that they form a subgroup isomorphic to the dihedral group of order 8.
- Show that the permutations in  $S_4$  that stabilize the element  $(x_1 + x_2)(x_3 + x_4)$  are exactly the same as those found in part (e). (The two polynomials appearing in parts (e) and (f) and the subgroup that stabilizes them will play an important role in the study of roots of quartic equations in Section 14.6.)

**Solution.**

- (a) Note that  $\tau \circ \sigma = (1\ 3\ 4\ 2)$  and  $\sigma \circ \tau = (1\ 3\ 2\ 4)$ . Then

$$\begin{aligned}\sigma \cdot p &= 12x_1x_2^5x_3^7 - 18x_3^3x_4 + 11x_1^{23}x_2^6x_3x_4^3 \\ \tau \cdot (\sigma \cdot p) &= 12x_1^7x_2x_3^5 - 18x_1^3x_4 + 11x_1x_2^{23}x_3^6x_4^3 \\ (\tau \circ \sigma) \cdot p &= 12x_1^7x_2x_3^5 - 18x_1^3x_4 + 11x_1x_2^{23}x_3^6x_4^3 \\ (\sigma \circ \tau) \cdot p &= 12x_1x_3^5x_4^7 - 18x_2x_4^3 + 11x_1^{23}x_2^3x_3^6x_4\end{aligned}$$

- (b) Note that  $1 \cdot p = p$  for any  $p \in R$ . Let  $\sigma, \tau \in S_4$ , then

$$\begin{aligned}\sigma \cdot (\tau \cdot p(x_1, x_2, x_3, x_4)) &= \sigma \cdot p(x_{\tau(1)}, x_{\tau(2)}, x_{\tau(3)}, x_{\tau(4)}) \\ &= p(x_{\sigma(\tau(1))}, x_{\sigma(\tau(2))}, x_{\sigma(\tau(3))}, x_{\sigma(\tau(4))}) \\ &= (\sigma \circ \tau) \cdot p(x_1, x_2, x_3, x_4)\end{aligned}$$

- (c) The permutations in  $S_4$  that stabilize  $x_4$  are those that fix 4 and permute  $\{1, 2, 3\}$ . These are exactly the elements of the subgroup

$$G = \{1, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

which is clearly isomorphic to  $S_3$ .

- (d) To stabilize  $x_1 + x_2$ , a permutation must either fix both 1 and 2 or swap them. The permutations that do this are

$$H = \{1, (1\ 2), (3\ 4), (1\ 2)(3\ 4)\}$$

which is abelian since every nonidentity element has order 2. Moreover,  $|H| = 4$ .

- (e) To stabilize  $x_1x_2 + x_3x_4$ , a permutation must either swap 1 and 2 or fix them, and either swap 3 and 4 or fix them. Additionally, it may swap the pairs (1 2) and (3 4). The permutations that do this are

$$K = \{1, (1\ 2), (3\ 4), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3), (1\ 3\ 2\ 4), (1\ 4\ 2\ 3)\}$$

It is straightforward to see that  $K$  is generated by (1 2) and (1 3 2 4). Consider the mapping  $\varphi : D_8 \rightarrow K$  defined by

$$\varphi(r) = (1\ 3\ 2\ 4), \quad \varphi(s) = (1\ 2)$$

Since  $\varphi(r)^4 = \varphi(s)^2 = 1$  and  $\varphi(s)\varphi(r) = (1\ 2)(1\ 3\ 2\ 4) = (1\ 4)(2\ 3) = \varphi(r)^{-1}\varphi(s)$ , then  $\varphi$  is a homomorphism. Since  $\varphi$  is clearly surjective and  $|D_8| = |K| = 8$ , then  $\varphi$  is an isomorphism, so that  $K \cong D_8$ .

- (f) The permutations that stabilize  $(x_1 + x_2)(x_3 + x_4)$  must either swap 1 and 2 or fix them, and either swap 3 and 4 or fix them. Additionally, it may swap the pairs (1 2) and (3 4). These are exactly the same permutations found in part (e). ■

### Exercise 2.2.13

Let  $n$  be a positive integer and let  $R$  be the set of all polynomials with integer coefficients in the independent variables  $x_1, x_2, \dots, x_n$ , i.e., the members of  $R$  are finite sums of elements of the form  $ax_1^{r_1}x_2^{r_2}\cdots x_n^{r_n}$  where  $a$  is any integer and  $r_1, \dots, r_n$  are nonnegative integers. For each  $\sigma \in S_n$  define a map

$$\sigma : R \rightarrow R \quad \text{by} \quad \sigma \cdot p(x_1, x_2, \dots, x_n) = p(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(n)})$$

Prove that this defines a (left) group action of  $S_n$  on  $R$ .

**Solution.** Since  $\text{id}(i) = i$  for all  $1 \leq i \leq n$ , then  $\text{id} \cdot p = p$  for any  $p \in R$ . Let  $\sigma, \tau \in S_n$ , then

$$\begin{aligned}\sigma \cdot (\tau \cdot p(x_1, x_2, \dots, x_n)) &= \sigma \cdot p(x_{\tau(1)}, x_{\tau(2)}, \dots, x_{\tau(n)}) \\ &= p(x_{\sigma(\tau(1))}, x_{\sigma(\tau(2))}, \dots, x_{\sigma(\tau(n))}) \\ &= (\sigma \circ \tau) \cdot p(x_1, x_2, \dots, x_n)\end{aligned}$$

Hence, this defines a (left) group action of  $S_n$  on  $R$ . ■

**Exercise 2.2.14**

Let  $H(F)$  be the Heisenberg group over the field  $F$  introduced in [Exercise 1.4.11](#). Determine which matrices lie in the center of  $H(F)$  and prove that  $Z(H(F))$  is isomorphic to the additive group  $F$ .

**Solution.** Let  $X, Y \in H(F)$  be written as

$$X = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad Y = \begin{pmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{pmatrix}$$

where  $X \in Z(H(F))$ . Then  $XY = YX$ , implying that

$$XY = \begin{pmatrix} 1 & a+d & af+b+e \\ 0 & 1 & c+f \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & d+a & dc+e+b \\ 0 & 1 & f+c \\ 0 & 0 & 1 \end{pmatrix} = YX$$

so that  $af + b + e = dc + e + b$ . Simplifying, we have  $af = dc$ . Since this must hold for all  $d, f \in F$ , then  $a = 0$  and  $c = 0$ . It follows that

$$Z(H(F)) = \left\{ \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mid x \in F \right\}$$

Moreover, the mapping  $\varphi : F \rightarrow Z(H(F))$  defined by

$$\varphi(x) = \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is clearly surjective. If  $\varphi(x) = I_3$ , then  $x = 0$ , so that  $\varphi$  is injective. Finally, for any  $x, y \in F$ , we have

$$\varphi(x+y) = \begin{pmatrix} 1 & 0 & x+y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \varphi(x)\varphi(y)$$

so that  $\varphi$  is a homomorphism. Hence,  $\varphi$  is an isomorphism, and  $F \cong Z(H(F))$ . ■

## 2.3 Cyclic Groups and Cyclic Subgroups

### Exercise 2.3.1

Find all subgroups of  $Z_{45} = \langle x \rangle$ , giving a generator for each. Describe the containments between these subgroups.

**Solution.** Recall that the containment relation is the following:

$$\langle x^a \rangle \leq \langle x^b \rangle \iff (b, 45) \mid (a, 45)$$

The subgroups of  $Z_{45}$  are:

$$\begin{aligned} Z_{45} &= \langle x \rangle > \langle x^3 \rangle, \langle x^5 \rangle, \langle x^9 \rangle, \langle x^{15} \rangle, 1 \\ \langle x^3 \rangle &> \langle x^9 \rangle, \langle x^{15} \rangle \\ \langle x^5 \rangle &> \langle x^{15} \rangle \\ \langle x^9 \rangle &> 1 \\ \langle x^{15} \rangle &> 1 \\ 1 &= \langle x^0 \rangle \end{aligned}$$

### Exercise 2.3.2

If  $x$  is an element of the finite group  $G$  and  $|x| = |G|$ , prove that  $G = \langle x \rangle$ . Give an explicit example to show that this result need not be true if  $G$  is an infinite group.

**Solution.** For some  $x \in G$  where  $|x| = |G| = n$ , then Proposition 2.2 says that  $1, x, x^2, \dots, x^{n-1}$  are all distinct elements of  $G$ . Since  $G$  has  $n$  elements only, it follows that these are the elements of  $G$ , hence  $G = \langle x \rangle$ . Moreover, this is not true if  $G$  is infinite, since  $|\mathbb{Z}| = |\mathbb{Z}| = \infty$ , but  $\langle 2 \rangle$  generates only the even integers. ■

### Exercise 2.3.3

Find all generators for  $\mathbb{Z}/48\mathbb{Z}$ .

**Solution.** Using Proposition 2.5 and  $|\mathbb{Z}/48\mathbb{Z}| = 48$ , then  $\langle \bar{a} \rangle$  generates  $\mathbb{Z}/48\mathbb{Z}$  when  $(a, 48) = 1$ . We have the integers

$$a \in \{1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 43, 47\}$$

### Exercise 2.3.4

Find all generators for  $\mathbb{Z}/202\mathbb{Z}$ .

**Solution.** Note that  $202 = 2 \cdot 101$ , which are both prime numbers. Then its generators is every number between 1 and 202, except 101 and even numbers. ■

### Exercise 2.3.5

Find the number of generators for  $\mathbb{Z}/49000\mathbb{Z}$ .

**Solution.** Let  $\varphi$  denote the Euler- $\varphi$  function. Then

$$\begin{aligned} \varphi(49000) &= \varphi(2^3)\varphi(5^3)\varphi(7^2) \\ &= 2^2(2-1)5^2(5-1)7(7-1) \\ &= 16800 \end{aligned}$$

**Exercise 2.3.6**

In  $\mathbb{Z}/48\mathbb{Z}$  write out all elements of  $\langle \bar{a} \rangle$  for every  $\bar{a}$ . Find all inclusions between subgroups in  $\mathbb{Z}/48\mathbb{Z}$ .

**Solution.** The elements of each subgroup are

$$\begin{aligned}\mathbb{Z}/48\mathbb{Z} = \langle \bar{1} \rangle &= \{\bar{0}, \bar{1}, \bar{2}, \dots, \bar{46}, \bar{47}\} & \langle \bar{2} \rangle &= \{\bar{0}, \bar{2}, \bar{4}, \dots, \bar{44}, \bar{46}\} \\ \langle \bar{3} \rangle &= \{\bar{0}, \bar{3}, \bar{6}, \dots, \bar{42}, \bar{45}\} & \langle \bar{4} \rangle &= \{\bar{0}, \bar{4}, \bar{8}, \dots, \bar{40}, \bar{44}\} \\ \langle \bar{6} \rangle &= \{\bar{0}, \bar{6}, \bar{12}, \dots, \bar{36}, \bar{42}\} & \langle \bar{8} \rangle &= \{\bar{0}, \bar{8}, \bar{16}, \bar{24}, \bar{32}, \bar{40}\} \\ \langle \bar{12} \rangle &= \{\bar{0}, \bar{12}, \bar{24}, \bar{36}\} & \langle \bar{16} \rangle &= \{\bar{0}, \bar{16}, \bar{32}\} \\ \langle \bar{24} \rangle &= \{\bar{0}, \bar{24}\} & \langle \bar{0} \rangle &= \{\bar{0}\}\end{aligned}$$

Moreover, the subgroup inclusions are

$$\begin{aligned}\langle \bar{1} \rangle &\geq \langle \bar{0} \rangle, \langle \bar{1} \rangle, \langle \bar{2} \rangle, \langle \bar{3} \rangle, \langle \bar{4} \rangle, \langle \bar{6} \rangle, \langle \bar{8} \rangle, \langle \bar{12} \rangle, \langle \bar{16} \rangle, \langle \bar{24} \rangle \\ \langle \bar{2} \rangle &\geq \langle \bar{0} \rangle, \langle \bar{2} \rangle, \langle \bar{4} \rangle, \langle \bar{6} \rangle, \langle \bar{8} \rangle, \langle \bar{12} \rangle, \langle \bar{24} \rangle \\ \langle \bar{3} \rangle &\geq \langle \bar{0} \rangle, \langle \bar{3} \rangle, \langle \bar{6} \rangle, \langle \bar{12} \rangle, \langle \bar{24} \rangle \\ \langle \bar{4} \rangle &\geq \langle \bar{0} \rangle, \langle \bar{4} \rangle, \langle \bar{8} \rangle, \langle \bar{12} \rangle, \langle \bar{16} \rangle, \langle \bar{24} \rangle \\ \langle \bar{6} \rangle &\geq \langle \bar{0} \rangle, \langle \bar{6} \rangle, \langle \bar{12} \rangle, \langle \bar{24} \rangle \\ \langle \bar{8} \rangle &\geq \langle \bar{0} \rangle, \langle \bar{8} \rangle, \langle \bar{16} \rangle, \langle \bar{24} \rangle \\ \langle \bar{12} \rangle &\geq \langle \bar{0} \rangle, \langle \bar{12} \rangle, \langle \bar{24} \rangle \\ \langle \bar{16} \rangle &\geq \langle \bar{0} \rangle, \langle \bar{16} \rangle \\ \langle \bar{24} \rangle &\geq \langle \bar{0} \rangle, \langle \bar{24} \rangle \\ \langle \bar{0} \rangle &\geq \langle \bar{0} \rangle\end{aligned}$$

■

**Exercise 2.3.7**

Let  $Z_{48} = \langle x \rangle$  and use the isomorphism  $\mathbb{Z}/48\mathbb{Z} \cong Z_{48}$  given by  $\bar{1} \mapsto x$  to list all subgroups of  $Z_{48}$  as computed in the preceding exercise.

**Solution.** The subgroups of  $Z_{48}$  are  $\langle x \rangle, \langle x^2 \rangle, \langle x^3 \rangle, \langle x^4 \rangle, \langle x^6 \rangle, \langle x^8 \rangle, \langle x^{12} \rangle, \langle x^{16} \rangle, \langle x^{24} \rangle$ , and 1. ■

**Exercise 2.3.8**

Let  $Z_{48} = \langle x \rangle$ . For which integers  $a$  does the map  $\varphi_a$  defined by  $\varphi_a : \bar{1} \mapsto x^a$  extend to an isomorphism from  $\mathbb{Z}/48\mathbb{Z}$  to  $Z_{48}$ ?

**Solution.** We know that  $\bar{1}$  generates  $\mathbb{Z}/48\mathbb{Z}$ . By Theorem 2.4,  $\varphi_a$  extends to an isomorphism if and only if  $x^a$  generates  $Z_{48}$ . By Proposition 2.5, this occurs if and only if  $(a, 48) = 1$ . Hence, the integers  $a$  such that  $\varphi_a$  extends to an isomorphism are

$$a \in \{1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 43, 47\}$$

■



**Exercise 2.3.9**

Let  $Z_{36} = \langle x \rangle$ . For which integers  $a$  does the map  $\psi_a$  defined by  $\psi_a : \bar{1} \mapsto x^a$  extend to a *well defined homomorphism* from  $\mathbb{Z}/48\mathbb{Z}$  into  $Z_{36}$ . Can  $\psi_a$  ever be a surjective homomorphism?

**Solution.** Suppose  $\bar{r}, \bar{s} \in \mathbb{Z}/48\mathbb{Z}$  such that  $\bar{r} = \bar{s}$ . Then  $r \equiv s \pmod{48}$ , or  $48 \mid (r - s)$ . If  $\psi_a$  is well defined, then

$$\psi_a(\bar{r}) = \psi_a(r \cdot \bar{1}) = x^{ar} = x^{as} = \psi_a(s \cdot \bar{1}) = \psi_a(\bar{s})$$

so that  $x^{a(r-s)} = 1$ . Since  $|x| = 36$ , then  $36 \mid a(r - s)$ . Since  $48 \mid (r - s)$ , there exists some integer  $k$  such that  $r - s = 48k$ . It follows that  $36 \mid 48ak$ , or  $3 \mid 4ak$ . Since  $3 \nmid 4$ , then  $3 \mid ak$ . If  $3 \nmid a$ , then  $3 \mid k$ . But this need not be true for all integers  $k$ . Hence, we must have  $3 \mid a$ .

Now suppose  $\psi_a$  is surjective. Then  $\langle x^a \rangle = Z_{36}$ , so that  $(a, 36) = 1$  by Proposition 2.5. But this contradicts the previous result that  $3 \mid a$ . Hence,  $\psi_a$  can never be surjective. ■

**Exercise 2.3.10**

What is the order of  $\overline{30}$  in  $\mathbb{Z}/54\mathbb{Z}$ ? Write out all of the elements and their orders in  $\langle \overline{30} \rangle$ .

**Solution.** By Proposition 2.5, we have  $|\bar{1}| = 54$ . Then

$$|\overline{30}| = |30 \cdot \bar{1}| = \frac{54}{(54, 30)} = \frac{54}{6} = 9$$

The first element of order 9 in  $\mathbb{Z}/54\mathbb{Z}$  is  $\bar{6}$ , so

$$\langle \overline{30} \rangle = \{\bar{0}, \bar{6}, \bar{12}, \bar{18}, \bar{24}, \bar{30}, \bar{36}, \bar{42}, \bar{48}\}$$

Moreover, the orders are

$$\begin{array}{lll} |\bar{0}| = 1 & |\bar{6}| = 9 & |\bar{12}| = 9 \\ |\bar{18}| = 3 & |\bar{24}| = 9 & |\bar{30}| = 9 \\ |\bar{36}| = 3 & |\bar{42}| = 9 & |\bar{48}| = 9 \end{array}$$

**Exercise 2.3.11**

Find all cyclic subgroups of  $D_8$ . Find a proper subgroup of  $D_8$  which is not cyclic.

**Solution.** The cyclic subgroups of  $D_8$  are

$$\begin{aligned} \langle 1 \rangle &= \{1\} \\ \langle r \rangle &= \langle r^3 \rangle = \{1, r, r^2, r^3\} \\ \langle r^2 \rangle &= \{1, r^2\} \\ \langle s \rangle &= \{1, s\} \\ \langle sr \rangle &= \{1, sr\} \\ \langle sr^2 \rangle &= \{1, sr^2\} \\ \langle sr^3 \rangle &= \{1, sr^3\} \end{aligned}$$

Moreover, a proper subgroup that is not cyclic is  $\langle r^2, s \rangle = \{1, r^2, s, sr^2\}$ . ■

**Exercise 2.3.12**

Prove that the following groups are *not* cyclic:

- (a)  $Z_2 \times Z_2$  (b)  $Z_2 \times \mathbb{Z}$  (c)  $\mathbb{Z} \times \mathbb{Z}$

**Solution.**

- (a) Put  $Z_2 = \langle x \rangle$ . We may inspect all four elements:

$$\begin{aligned}\langle (1, 1) \rangle &= \{(1, 1)\} \\ \langle (x, 1) \rangle &= \{(1, 1), (x, 1)\} \\ \langle (1, x) \rangle &= \{(1, 1), (1, x)\} \\ \langle (x, x) \rangle &= \{(1, 1), (x, x)\}\end{aligned}$$

No subgroup has order 4, hence no subgroup generates  $Z_2 \times Z_2$ .

- (b) If  $(a, b)$  generates  $Z_2 \times \mathbb{Z}$ , then it must be one of the forms  $(1, \pm 1)$  or  $(x, \pm 1)$ , since  $\langle \pm 1 \rangle = \mathbb{Z}$ . But  $(1, \pm 1)$  generates elements whose first component is only 1. If we consider  $(x, \pm 1)$ , then this also doesn't generate  $Z_2 \times \mathbb{Z}$  since  $(1, 1) \notin \langle (x, \pm 1) \rangle$ .
- (c) The only candidates for generators of  $\mathbb{Z} \times \mathbb{Z}$  is  $(\pm 1, \pm 1)$ . But any subgroup generated by  $(\pm 1, \pm 1)$  contain elements that differ only in sign as  $(x, y) \notin \langle (\pm 1, \pm 1) \rangle$  when  $|x| \neq |y|$ . ■

**Exercise 2.3.13**

Prove that the following pairs of groups are *not* isomorphic:

- (a)  $\mathbb{Z} \times Z_2$  and  $\mathbb{Z}$  (b)  $\mathbb{Q} \times Z_2$  and  $\mathbb{Q}$ .

**Solution.**

- (a)  $(0, x) \in \mathbb{Z} \times Z_2$  has order 2, but no element in  $\mathbb{Z}$  has order 2.
- (b) Same reason as above. ■

**Exercise 2.3.14**

Let  $\sigma = (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12)$ . For each of the following integers  $a$  compute  $\sigma^a$ :  
 $a = 13, 65, 626, 1195, -6, -81, -570$ , and  $-1211$ .

**Solution.** Since  $|\sigma| = 12$ , then we know that  $\langle \sigma \rangle$  has 12 distinct elements. We may then use the Division Algorithm to reduce the integers to their least residue:

$$\begin{aligned}\sigma^{13} &= \sigma^{12+1} = \sigma \\ \sigma^{65} &= \sigma^{5(12)+5} = \sigma^5 = (1\ 6\ 11\ 4\ 9\ 2\ 7\ 12\ 5\ 10\ 3\ 8) \\ \sigma^{626} &= \sigma^{52(12)+2} = \sigma^2 = (1\ 3\ 5\ 7\ 9\ 11)(2\ 4\ 6\ 8\ 10\ 12) \\ \sigma^{1195} &= \sigma^{99(12)+7} = \sigma^7 = (1\ 8\ 3\ 10\ 5\ 12\ 7\ 2\ 9\ 4\ 11\ 6) \\ \sigma^{-6} &= \sigma^{-1(12)+6} = \sigma^6 = (1\ 7)(2\ 8)(3\ 9)(4\ 10)(5\ 11)(6\ 12) \\ \sigma^{-81} &= \sigma^{-7(12)+3} = \sigma^3 = (1\ 4\ 7\ 10)(2\ 5\ 8\ 11)(3\ 6\ 9\ 12) \\ \sigma^{-570} &= \sigma^{-48(12)+6} = \sigma^6 \\ \sigma^{-1211} &= \sigma^{-101(12)+1} = \sigma\end{aligned}$$

**Exercise 2.3.15**

Prove that  $\mathbb{Q} \times \mathbb{Q}$  is not cyclic.

**Solution.** If it were cyclic, then all subgroups are also cyclic, by Theorem 2.7. But  $\mathbb{Z} \times \mathbb{Z}$  is not cyclic. ■

**Exercise 2.3.16**

Assume  $|x| = n$  and  $|y| = m$ . Suppose that  $x$  and  $y$  commute:  $xy = yx$ . Prove that  $|xy|$  divides the least common multiple of  $m$  and  $n$ . Need this be true of  $x$  and  $y$  do not commute? Give an example of commuting elements  $x$  and  $y$  such that the order of  $xy$  is not equal to the least common multiple of  $|x|$  and  $|y|$ .

**Solution.** Let  $\ell$  be the least common multiple of  $m$  and  $n$ . Then there exist  $a, b \in \mathbb{Z}$  such that  $\ell = am = bn$ . Then

$$(xy)^\ell = x^\ell y^\ell = x^{bn} y^{am} = 1 \cdot 1 = 1$$

so that by Proposition 2.3, then  $|xy|$  divides  $\ell$ . Moreover, this is not true if  $x$  and  $y$  do not commute. If we consider  $r, s \in D_8$ , then  $|r| = 4$  and  $|s| = 2$ , but  $|rs| = 2$  which 4 does not divide. ■

**Exercise 2.3.17**

Find a presentation for  $Z_n$  with one generator.

**Solution.**  $Z_n = \langle x \mid x^n = 1 \rangle$ . ■

**Exercise 2.3.18**

Show that if  $H$  is any group and  $h$  is an element of  $H$  with  $h^n = 1$ , then there is a unique homomorphism from  $Z_n = \langle x \rangle$  to  $H$  such that  $x \mapsto h$ .

**Solution.** Define the map  $\varphi : Z_n \rightarrow H$  as

$$\varphi(x^k) = h^k$$

Then  $\varphi(x) = h$ . To show that  $\varphi$  is well defined, suppose  $x^a = x^b$  for some  $a, b \in \mathbb{Z}$ . Then  $n \mid (a - b)$ , or  $a - b = nk$  for some  $k \in \mathbb{Z}$ . It follows that

$$\varphi(x^a) = h^a = h^{b+nk} = h^b (h^n)^k = h^b \cdot 1 = h^b = \varphi(x^b)$$

so that  $\varphi$  is well defined. Moreover,

$$\varphi(x^{a+b}) = h^{a+b} = h^a h^b = \varphi(x^a) \varphi(x^b)$$

so that  $\varphi$  is a homomorphism. Lastly, if  $\psi : Z_n \rightarrow H$  is another homomorphism such that  $\psi(x) = h$ , it must satisfy  $\psi(x^k) = \psi(x)^k = h^k$ . This shows that  $\psi = \varphi$ . ■

**Exercise 2.3.19**

Show that if  $H$  is any group and  $h$  is an element of  $H$ , then there is a unique homomorphism from  $\mathbb{Z}$  to  $H$  such that  $1 \mapsto h$ .

**Solution.** Define the map  $\varphi : \mathbb{Z} \rightarrow H$  as

$$\varphi(n) = h^n$$

Then  $\varphi(1) = h$ . Moreover, for any  $s, t \in \mathbb{Z}$ , then

$$\varphi(s+t) = h^{s+t} = h^s h^t = \varphi(s) \varphi(t)$$

so that  $\varphi$  is a homomorphism. Lastly, if  $\psi : \mathbb{Z} \rightarrow H$  is another homomorphism such that  $\psi(1) = h$ , it must satisfy  $\psi(n) = \psi(n \cdot 1) = \psi(1)^n = h^n$ . ■

**Exercise 2.3.20**

Let  $p$  be a prime and let  $n$  be a positive integer. Show that if  $x$  is an element of the group  $G$  such that  $x^{p^n} = 1$ , then  $|x| = p^m$  for some  $m \leq n$ .

**Solution.** If  $x^{p^n} = 1$ , then Proposition 2.3 says that  $|x|$  divides  $p^n$ . Since  $p$  is prime, then  $|x|$  must also be a power of  $p$  that is at most  $n$ . ■

**(\*) Exercise 2.3.21**

Let  $p$  be an odd prime and let  $n$  be a positive integer. Use the Binomial Theorem to show that  $(1+p)^{p^{n-1}} \equiv 1 \pmod{p^n}$  but  $(1+p)^{p^{n-2}} \not\equiv 1 \pmod{p^n}$ . Deduce that  $1+p$  is an element of order  $p^{n-1}$  in the multiplicative group  $(\mathbb{Z}/p^n\mathbb{Z})^\times$ .

**Solution.** By the Binomial Theorem, then

$$(1+p)^{p^{n-1}} = \sum_{k=0}^{p^{n-1}} \binom{p^{n-1}}{k} p^k = 1 + p^n + \sum_{k=2}^{p^{n-1}-2} \binom{p^{n-1}}{k} p^k$$

Observe that for  $k \geq 2$ , the binomial coefficient is

$$\binom{p^{n-1}}{k} = \frac{p^{n-1}(p^{n-1}-1) \cdots (p^{n-1}-k+1)}{k!}$$

where we note that the denominator  $k!$  contains at most  $p^{k-1}$  as a factor since  $p$  is prime. Then the numerator contains at least  $p^{n-1}$  as a factor, so that the entire term contains at least  $p^{n-1-(k-1)} = p^{n-k}$  as a factor. Since this is multiplied by  $p^k$  in the summation, then each term for  $k \geq 2$  contains at least  $p^n$  as a factor, so that  $(1+p)^{p^{n-1}} \equiv 1 \pmod{p^n}$ .

Now consider  $(1+p)^{p^{n-2}}$ . Using the Binomial Theorem again, we have

$$(1+p)^{p^{n-2}} = \sum_{k=0}^{p^{n-2}} \binom{p^{n-2}}{k} p^k = 1 + p^{n-1} + \sum_{k=2}^{p^{n-2}-2} \binom{p^{n-2}}{k} p^k$$

While the rest of the summation terms are divisible by  $p^n$  like before, the  $k=1$  term is  $p^{n-1}$  which is not divisible by  $p^n$ . Hence,  $(1+p)^{p^{n-2}} \not\equiv 1 \pmod{p^n}$ . Since the only integers that divide  $p^n$  are  $1, p, p^2, \dots, p^n$ , and the first power that results in  $1 \pmod{p^n}$  is  $p^{n-1}$ , it follows that the order of  $1+p$  in  $(\mathbb{Z}/p^n\mathbb{Z})^\times$  is  $p^{n-1}$ . ■

**(\*) Exercise 2.3.22**

Let  $n$  be an integer  $\geq 3$ . Use the Binomial Theorem to show that  $(1+2^2)^{2^{n-2}} \equiv 1 \pmod{2^n}$  but  $(1+2^2)^{2^{n-3}} \not\equiv 1 \pmod{2^n}$ . Deduce that 5 is an element of order  $2^{n-2}$  in the multiplicative group  $(\mathbb{Z}/2^n\mathbb{Z})^\times$ .

**Solution.** The process is similar to the previous exercise, where we use the Binomial Theorem to expand  $(1+2^2)^{2^{n-2}}$  and see that  $k=0$  and  $k=1$  terms are 1 and  $2^n$  respectively. For  $k \geq 2$ , we have the summand as

$$\binom{2^{n-2}}{k} 2^{2k} = \frac{2^{n-2}(2^{n-2}-1) \cdots (2^{n-2}-k+1)}{k!} 2^{2k}$$

Using the formula obtained in [Exercise 0.2.8](#), note that the largest power of 2 that  $k!$  divides is  $k-1$ . Then  $k!$  contains at most  $2^{k-1}$  as a factor, while the numerator contains at least  $2^{n-1}$  as a factor. Along with  $2^{2k}$ , the entire term contains at least  $2^{n-1-(k-1)+2k} = 2^{n+k}$  as a factor. Since  $k \geq 2$ , then  $n+k \geq n+2 > n$ , so that each term for  $k \geq 2$  contains at least  $2^n$  as a factor. Hence,  $(1+2^2)^{2^{n-2}} \equiv 1 \pmod{2^n}$  as desired. Moreover, considering  $(1+2^2)^{2^{n-3}}$ , the  $k=1$  term is  $2^{n-1}$  which is not divisible by  $2^n$ . Hence,  $(1+2^2)^{2^{n-3}} \not\equiv 1 \pmod{2^n}$ . Since the only integers that divide  $2^n$  are  $1, 2, 2^2, \dots, 2^n$ , and the first power that results in  $1 \pmod{2^n}$  is  $2^{n-2}$ , it follows that the order of 5 in  $(\mathbb{Z}/2^n\mathbb{Z})^\times$  is  $2^{n-2}$ . ■

**Exercise 2.3.23**

Show that  $(\mathbb{Z}/2^n\mathbb{Z})^\times$  is not cyclic for any  $n \geq 3$ . [Find two distinct subgroups of order 2.]

**Solution.** By Theorem 2.7, we must have one subgroup of order 2 in  $(\mathbb{Z}/2^n\mathbb{Z})^\times$  if it were cyclic. However,

$$(2^n - 1)^2 = (-1)^2 \equiv 1 \pmod{2^n}$$

and

$$(2^{n-1} + 1)^2 = 2^{2n-2} + 2^n + 1 \equiv 1 \pmod{2^n}$$

Since  $n \geq 3$ , then  $2^n - 1 \neq 2^{n-1} + 1$  but both have order 2 in  $(\mathbb{Z}/2^n\mathbb{Z})^\times$ . Hence, it cannot be cyclic. ■

**Exercise 2.3.24**

Let  $G$  be a finite group and let  $x \in G$ .

- (a) Prove that if  $g \in N_G(\langle x \rangle)$  then  $gxg^{-1} = x^a$  for some  $a \in \mathbb{Z}$ .
- (b) Prove conversely that if  $gxg^{-1} = x^a$  for some  $a \in \mathbb{Z}$  then  $g \in N_G(\langle x \rangle)$ . [Show first that  $gx^k g^{-1} = (gxg^{-1})^k = x^{ak}$  for any integer  $k$  so that  $g\langle x \rangle g^{-1} \leq \langle x \rangle$ . If  $x$  has order  $n$ , show that the elements  $gx^i g^{-1}, i = 0, 1, \dots, n-1$  are distinct, so that  $|g\langle x \rangle g^{-1}| = |\langle x \rangle| = n$  and conclude that  $g\langle x \rangle g^{-1} = \langle x \rangle$ .]

Note that this cuts down some of the work in commuting normalizers of cyclic subgroups since one does not have to check  $ghg^{-1} \in \langle x \rangle$  for every  $h \in \langle x \rangle$ .

**Solution.**

- (a) If  $g \in N_G(\langle x \rangle)$ , then  $g\langle x \rangle g^{-1} = \langle x \rangle$ . In particular,  $gxg^{-1} \in \langle x \rangle$ , so there exists some  $a \in \mathbb{Z}$  such that  $gxg^{-1} = x^a$ .
- (b) Suppose  $gxg^{-1} = x^a$  for some  $a \in \mathbb{Z}$ . We first show that  $gx^k g^{-1} = (gxg^{-1})^k$  by induction. For  $k = 1$ , the result is trivial. Suppose it holds for some  $k \geq 1$ . Then

$$gx^{k+1} g^{-1} = gx^k g^{-1} gxg^{-1} = (gxg^{-1})^k (gxg^{-1}) = (gxg^{-1})^{k+1}$$

so that the result holds for all positive integers  $k$ . For negative integers, observe that  $(gxg^{-1})^{-1} = gx^{-1} g^{-1}$ . Using the above result for all positive integers, we may extend it to negative integers as well.

Now, for any integer  $k$ , we have

$$gx^k g^{-1} = (gxg^{-1})^k = (x^a)^k = x^{ak}$$

so that  $g\langle x \rangle g^{-1} \leq \langle x \rangle$ . Since conjugation is an isomorphism, it is trivial to see that  $|x| = |gxg^{-1}|$ . If  $|x| = n$ , then the elements  $gx^i g^{-1}$  for  $i = 0, 1, \dots, n-1$  are distinct since if  $gx^i g^{-1} = gx^j g^{-1}$  for some  $0 \leq i < j \leq n-1$ , then  $x^i = x^j$  which contradicts the order of  $x$ . Hence,  $|g\langle x \rangle g^{-1}| = |\langle x \rangle| = n$ , so that  $g\langle x \rangle g^{-1} = \langle x \rangle$ . Therefore,  $g \in N_G(\langle x \rangle)$ . ■

**Exercise 2.3.25**

Let  $G$  be a cyclic group of order  $n$  and let  $k$  be an integer relatively prime to  $n$ . Prove that the map  $x \mapsto x^k$  is surjective. Use Lagrange's Theorem to prove that the same is true for any finite group of order  $n$ . (For such  $k$  each element has a  $k$ th root in  $G$ . It follows from Cauchy's Theorem in Section 3.2 that if  $k$  is not relatively prime to the order of  $G$  then the map  $x \mapsto x^k$  is not surjective.)

**Solution.** Fix  $k \in \mathbb{Z}^+$  such that  $(k, n) = 1$ . Since  $G$  is cyclic, then  $G = \langle x \rangle$  for some  $x \in G$ . For any  $y \in G$ , there exists some integer  $m$  such that  $y = x^m$ . Since  $(k, n) = 1$ , there exist integers  $s, t$  such that  $ks + nt = 1$ . Then

$$\varphi(x^{ms}) = x^{kms} = x^{m(1-nt)} = x^m (x^n)^{-mt} = x^m \cdot 1 = y$$

so that  $\varphi$  is surjective.

Now let  $G$  be any finite group of order  $n$ . For any  $y \in G$ , consider the subgroup  $\langle y \rangle$ . If  $|\langle y \rangle| = m$ , then by Lagrange's Theorem,  $m \mid n$ . Since  $(k, n) = 1$ , then  $(k, m) = 1$  as well. By the previous result, there exists some  $z \in \langle y \rangle$  such that  $z^k = y$ . Hence, the map  $x \mapsto x^k$  is surjective. ■

(\*) **Exercise 2.3.26**

Let  $Z_n$  be a cyclic group of order  $n$  and for each integer  $a$  let

$$\sigma_a : Z_n \rightarrow Z_n \quad \text{by} \quad \sigma_a(x) = x^a \text{ for all } x \in Z_n$$

- (a) Prove that  $\sigma_a$  is an automorphism of  $Z_n$  if and only if  $a$  and  $n$  are relatively prime.
- (b) Prove that  $\sigma_a = \sigma_b$  if and only if  $a \equiv b \pmod{n}$ .
- (c) Prove that every automorphism of  $Z_n$  is equal to  $\sigma_a$  for some integer  $a$ .
- (d) Prove that  $\sigma_a \circ \sigma_b = \sigma_{ab}$ . Deduce that the map  $\bar{a} \mapsto \sigma_a$  is an isomorphism of  $(\mathbb{Z}/n\mathbb{Z})^\times$  onto the automorphism group of  $Z_n$  (so  $\text{Aut}(Z_n)$  is an abelian group of order  $\varphi(n)$ ).

**Solution.**

- (a) • ( $\Rightarrow$ ) Suppose  $\sigma_a \in \text{Aut}(Z_n)$ , and let  $d = (n, a)$ . Assume, by way of contradiction, that  $d \neq 1$ . Then there are  $s, t \in \mathbb{Z}$  such that  $n = ds$  and  $a = dt$ . Let  $Z_n = \langle x \rangle$ . Then

$$\varphi(x^s) = x^{as} = x^{dts} = x^{ns} = 1$$

If  $x^s \neq 1$ , then  $\ker \sigma_a$  would be non-trivial, contradicting that  $\sigma_a$  is an automorphism. Hence,  $x^s = 1$ , and since  $|x| = n$  and  $n \mid s$ , then  $s = n$  so that  $d = 1$ .

- ( $\Leftarrow$ ) Suppose that  $(a, n) = 1$ . By the previous exercise, then  $\sigma_a$  is a surjective map. Since  $Z_n$  is finite, then  $\sigma_a$  is also bijective. Moreover,

$$\sigma_a(xy) = (xy)^a = x^a y^a = \sigma_a(x) \sigma_a(y)$$

for some  $x, y \in Z_n$ . Then  $\sigma_a$  is also a homomorphism, hence  $\sigma_a \in \text{Aut}(Z_n)$ .

- (b) • ( $\Rightarrow$ ) Suppose  $\sigma_a = \sigma_b$ . Let  $Z_n = \langle x \rangle$ . Then

$$x^a = \sigma_a(x) = \sigma_b(x) = x^b$$

so that  $n \mid (a - b)$ , or  $a \equiv b \pmod{n}$ .

- ( $\Leftarrow$ ) Suppose  $a \equiv b \pmod{n}$ . Then there exists some integer  $k$  such that  $a - b = nk$ . Let  $Z_n = \langle x \rangle$ . For any  $x^m \in Z_n$ , we have

$$\sigma_a(x^m) = x^{am} = x^{(b+nk)m} = x^{bm} (x^n)^{km} = x^{bm} \cdot 1 = \sigma_b(x^m)$$

so that  $\sigma_a = \sigma_b$ .

- (c) Let  $Z_n = \langle x \rangle$  and let  $\varphi \in \text{Aut}(Z_n)$ . Since  $\varphi$  is surjective, then there exists some integer  $a$  such that  $\varphi(x) = x^a$ . For any  $x^m \in Z_n$ , we have

$$\varphi(x^m) = \varphi(x)^m = (x^a)^m = x^{am} = \sigma_a(x^m)$$

so that  $\varphi = \sigma_a$ .

- (d) Let  $Z_n = \langle x \rangle$ . For any  $x^m \in Z_n$ , we have

$$\sigma_a \circ \sigma_b(x^m) = \sigma_a(x^{bm}) = x^{abm} = \sigma_{ab}(x^m)$$

so that  $\sigma_a \circ \sigma_b = \sigma_{ab}$ . Consider the map

$$\varphi : (\mathbb{Z}/n\mathbb{Z})^\times \rightarrow \text{Aut}(Z_n) \quad \text{by} \quad \varphi(\bar{a}) = \sigma_a$$

By part (b),  $\varphi$  is well defined. Part (a) shows that  $\varphi$  is injective, while part (c) shows that  $\varphi$  is surjective. The first part of (d) shows that  $\varphi$  is a homomorphism. Hence,  $\varphi$  is an isomorphism of  $(\mathbb{Z}/n\mathbb{Z})^\times$  onto  $\text{Aut}(Z_n)$ . ■

## 2.4 Subgroups Generated by Subsets of a Group

### Exercise 2.4.1

Prove that if  $H$  is a subgroup of  $G$  then  $\langle H \rangle = H$ .

**Solution.** It is clear that  $H \subseteq \langle H \rangle$ . Recall that  $\langle H \rangle$  is the intersection of all subgroups of  $G$  that contain  $H$ . Since  $H$  is itself a subgroup of  $G$  that contains  $H$ , then  $\langle H \rangle \subseteq H$ . Hence,  $\langle H \rangle = H$ . ■

### Exercise 2.4.2

Prove that if  $A$  is a subset of  $B$  then  $\langle A \rangle \leq \langle B \rangle$ . Give an example where  $A \subseteq B$  with  $A \neq B$  but  $\langle A \rangle = \langle B \rangle$ .

**Solution.** Since  $\langle B \rangle$  is the smallest subgroup of  $G$  that contains  $B$ , and  $A \subseteq B$ , then  $A \subseteq \langle B \rangle$ . Since  $\langle A \rangle$  is the smallest subgroup of  $G$  that contains  $A$ , then  $\langle A \rangle \subseteq \langle B \rangle$ , so that  $\langle A \rangle \leq \langle B \rangle$ .

For an example, consider  $G = \mathbb{Z}$ ,  $A = \langle 1 \rangle$ , and  $B = \{1, 2\}$ . Then  $A \subset B$  but  $A \neq B$ , and  $\langle A \rangle = \langle B \rangle = \mathbb{Z}$ . ■

### Exercise 2.4.3

Prove that if  $H$  is an abelian subgroup of a group  $G$  then  $\langle H, Z(G) \rangle$  is abelian. Give an explicit example of an abelian subgroup  $H$  of a group  $G$  such that  $\langle H, C_G(H) \rangle$  is not abelian.

**Solution.** Let  $g, h \in \langle H, Z(G) \rangle$ . Using Proposition 2.9, we may put them as follows:

$$g = g_1^{\epsilon_1} g_2^{\epsilon_2} \dots g_m^{\epsilon_m}, \quad h = h_1^{\delta_1} h_2^{\delta_2} \dots h_n^{\delta_n}$$

where  $\epsilon_i = \delta_i = \pm 1$ , and  $g_i, h_i \in H \cup Z(G)$  for all  $i$ . Since both  $H$  and  $Z(G)$  are abelian, then elements of  $H$  and  $Z(G)$  commute with each other. Then

$$gh = g_1^{\epsilon_1} \dots g_m^{\epsilon_m} h_1^{\delta_1} \dots h_n^{\delta_n} = h_1^{\delta_1} \dots h_n^{\delta_n} g_1^{\epsilon_1} \dots g_m^{\epsilon_m} = hg$$

so that  $\langle H, Z(G) \rangle$  is abelian.

To obtain an example, consider groups that are non-abelian and their centers. Let  $G = D_8$  and  $H = Z(G) = \{1, r^2\}$ . Then  $C_G(H) = G$  since  $H \leq Z(G)$ . However,  $\langle H, C_G(H) \rangle = \langle Z(G), G \rangle = G$  which is non-abelian. ■

### Exercise 2.4.4

Prove that if  $H$  is a subgroup of  $G$  then  $H$  is generated by the set  $H - \{1\}$ .

**Solution.** If  $H$  is trivial, then  $H - \{1\} = \emptyset$  so that  $\langle \emptyset \rangle = 1 = H$ .

Suppose  $H$  is non-trivial, and let  $h \in H$ . If  $h = 1$ , then  $h \in \langle H - \{1\} \rangle \leq G$ . If  $h \neq 1$ , then  $h \in H - \{1\}$  so that  $h \in \langle H - \{1\} \rangle$ . Hence,  $H \subseteq \langle H - \{1\} \rangle$ . Moreover,  $\langle H - \{1\} \rangle$  is the smallest subgroup containing  $H - \{1\}$  which is a subset of  $H$  so that  $\langle H - \{1\} \rangle \subseteq H$ . Therefore,  $\langle H - \{1\} \rangle = H$ . ■

### Exercise 2.4.5

Prove that the subgroup generated by any two distinct elements of order 2 in  $S_3$  is all of  $S_3$ .

**Solution.** Consider the 2-cycles  $(1\ 2)$  and  $(1\ 3)$ . Then

$$\begin{aligned} (1\ 2)(1\ 3) &= (1\ 3\ 2) \\ (1\ 3)(1\ 2) &= (1\ 2\ 3) \\ (1\ 3\ 2)(1\ 2) &= (2\ 3) \end{aligned}$$

so that  $\langle (1\ 2), (1\ 3) \rangle = S_3$ . One can also do similar calculations to ensure  $\langle (1\ 2), (2\ 3) \rangle = \langle (1\ 3), (2\ 3) \rangle = S_3$ . ■

**Exercise 2.4.6**

Prove that the subgroup of  $S_4$  generated by  $(1\ 2)$  and  $(1\ 2)(3\ 4)$  is a noncyclic group of order 4.

**Solution.** Let  $a = (1\ 2)$  and  $b = (3\ 4)$ . Then  $\langle a, ab \rangle = \{1, a, b, ab\}$  where  $ab = ba$  since they are disjoint cycles. Moreover,  $|a| = |b| = |ab| = 2$  so that it is noncyclic. ■

**Exercise 2.4.7**

Prove that the subgroup of  $S_4$  generated by  $(1\ 2)$  and  $(1\ 3)(2\ 4)$  is isomorphic to the dihedral group of order 8.

**Solution.** Let  $\alpha = (1\ 2)$  and  $\beta = (1\ 3)(2\ 4)$ . Let  $\gamma = \alpha\beta = (1\ 3\ 2\ 4)$ . Note that  $\gamma^4 = \alpha^2 = 1$ , and  $\gamma\alpha = \alpha\gamma^{-1}$  so that  $\alpha$  and  $\gamma$  satisfy the relations for  $D_8$ . Moreover, any element of  $\langle \alpha, \beta \rangle$  can be written as  $\alpha^i \gamma^j$  for some  $0 \leq i \leq 1$  and  $0 \leq j \leq 3$  using the relations above. We may then extend the map

$$\varphi : D_8 \rightarrow \langle \alpha, \beta \rangle \quad \text{by} \quad r^j s^i \mapsto \gamma^j \alpha^i$$

to an injective homomorphism, since distinct elements in  $D_8$  map to distinct elements in  $\langle \alpha, \beta \rangle$ . It is easy to see that  $\langle \alpha, \beta \rangle$  has order 8, since each of  $\gamma, \gamma^2, \gamma^3$  are distinct from each other and from  $\alpha, \alpha\gamma, \alpha\gamma^2, \alpha\gamma^3$ , along with the identity. Hence,  $\varphi$  is an isomorphism. ■

**Exercise 2.4.8**

Prove that  $S_4 = \langle (1\ 2\ 3\ 4), (1\ 2\ 4\ 3) \rangle$ .

**Solution.** Let  $A = \langle (1\ 2\ 3\ 4), (1\ 2\ 4\ 3) \rangle$ . Let  $\alpha = (1\ 2\ 3\ 4)$  and  $\beta = (1\ 2\ 4\ 3)$ . To show that  $A = S_4$ , we will show that  $A$  must be of order 12 or 24, then show that it contains both  $(1\ 2)$  and  $(1\ 3)(2\ 4)$  so that by the previous exercise, it has a subgroup of order 8, hence  $A$  must be of order 24.

Note that  $|\alpha| = 4$  so that  $A$  has a subgroup of order 4. Moreover,  $\alpha\beta = (1\ 3\ 2)$  has order 3 so that  $A$  has a subgroup of order 3 as well. Since  $A \leq S_4$ , we may use Lagrange's Theorem to conclude that  $|A|$  is a multiple of 3 and 4 that divides 24, so that  $|A| = 12$  or 24.

To obtain the elements  $(1\ 2)$  and  $(1\ 3)(2\ 4)$ , observe that  $\alpha^2 = (1\ 3)(2\ 4)$ . To obtain  $(1\ 2)$ , we use  $\beta$  to “disrupt” the symmetry of  $\alpha^2$  to potentially obtain a smaller cycle. By computation, we get  $\beta\alpha^2 = (2\ 3)$ . Conjugating  $(2\ 3)$  by  $\alpha^{-1} = \alpha^3$  sends 2 to 1 and 3 to 2, so that  $\alpha^3(2\ 3)\alpha = (1\ 2)$ . Hence,  $(1\ 2) \in A$ . Then by the previous exercise,  $A$  has a subgroup of order 8, so that  $|A| = 24$ . Therefore,  $A = S_4$ . ■

**Exercise 2.4.9**

Prove that  $\text{SL}_2(\mathbb{F}_3)$  is the subgroup of  $\text{GL}_2(\mathbb{F}_3)$  generated by  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ . [Recall from [Exercise 2.1.9](#) that  $\text{SL}_2(\mathbb{F}_3)$  is the subgroup of matrices of determinant 1. You may assume this subgroup has order 24—this will be an exercise in Section 3.2.]

**Solution.** Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Since we know that  $\text{SL}_2(\mathbb{F}_3)$  has order 24, we need to exhibit at least 13 distinct matrices using  $A$  and  $B$ , since



$\langle A, B \rangle$  divides 24. Since  $I, A$ , and  $B$  are 3 distinct matrices, we compute 10 more:

$$\begin{array}{ll} A^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} & B^2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \\ AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} & BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \\ (AB)^2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} & ABA^2 = \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix} \\ A^2B = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix} & B^2A = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} \\ ABA = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} & BAB = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \end{array}$$

Hence  $\langle A, B \rangle = 24$  so that  $\langle A, B \rangle = \text{SL}_2(\mathbb{F}_3)$ . ■

#### Exercise 2.4.10

Prove that the subgroup of  $\text{SL}_2(\mathbb{F}_3)$  generated by  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  is isomorphic to the quaternion group of order 8. [Use a presentation for  $Q_8$ .]

**Solution.** Let

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

By computation, it is easy to see that  $A^2 = B^2 = -I_2$ . Let  $C = AB$ . Then

$$C = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}, \quad C^2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = -I_2$$

By definition,  $I_2$  and  $-I_2$  commute with all matrices in  $\text{SL}_2(\mathbb{F}_3)$ . Note that all the relations we have gathered satisfy the relations in the presentation of  $Q_8$  given in [Exercise 1.4.3](#). The mapping

$$\varphi : Q_8 \rightarrow \langle A, B \rangle \quad \text{by} \quad i \mapsto A, \quad j \mapsto B$$

extends to a homomorphism since the relations are satisfied. Moreover,  $\varphi$  maps generators of  $Q_8$  to generators of  $\langle A, B \rangle$  so that  $\varphi$  is surjective. Since  $|Q_8| = |\langle A, B \rangle| = 8$ , then  $\varphi$  is injective as well. Hence,  $\langle A, B \rangle \cong Q_8$ . ■

#### Exercise 2.4.11

Show that  $\text{SL}_2(\mathbb{F}_3)$  and  $S_4$  are two nonisomorphic groups of order 24.

**Solution.** Recall that  $Q_8$  and  $S_4$  both have 6 elements of order 4. However, no two elements in  $Q_8$  can generate the entirety of  $\text{SL}_2(\mathbb{F}_3)$  since  $|Q_8| = 8$ , while  $S_4 = \langle (1\ 2\ 3\ 4), (1\ 2\ 4\ 3) \rangle$ , hence they cannot be isomorphic. ■

(\*) **Exercise 2.4.12**

Prove that the subgroup of upper triangular matrices in  $\text{GL}_3(\mathbb{F}_2)$  is isomorphic to the dihedral group of order 8. (First find the order of this subgroup.)

**Solution.** Let  $\text{UT}_3(\mathbb{F}_2)$  denote the subgroup of upper triangular matrices in  $\text{GL}_3(\mathbb{F}_2)$ . Let each matrix in  $\text{UT}_3(\mathbb{F}_2)$  be represented as  $(a, b, c)$ , which refer to the elements  $a_{12}, a_{13}, a_{23}$  in a  $3 \times 3$  upper triangular matrix with 1's on the diagonal. Moreover, multiplication of two matrices  $A = (a, b, c)$  and  $B = (d, e, f)$  is written as follows:

$$AB = (a, b, c)(d, e, f) = (a + d, b + e + af, c + f)$$

Since each of  $a, b, c$  can be either 0 or 1, then there are  $2^3 = 8$  elements in  $\text{UT}_3(\mathbb{F}_2)$ .

We first need to find matrices  $A, B \in \text{UT}_3(\mathbb{F}_2)$  such that  $A^4 = B^2 = I_3$ , and  $AB = BA^3$ . Note that

$$(a, b, c)^2 = (2a, ac + 2b, 2c) = (0, ac, 0)$$

so that for  $A = (a, b, c)$  to have order 4, it must be that  $a = c = 1$ . Since  $b$  may vary, choose  $A = (1, 0, 1)$ . For  $B = (d, e, f)$  to have order 2, then  $B^2 = (0, df, 0) = I_3$ , so  $df = 0$ . Then either  $d = 0$  or  $f = 0$ . To satisfy  $AB = BA^3$ , we compute both sides to obtain

$$(1, 0, 1)(d, e, f) = (1 + d, e + f, 1 + f) = (1 + d, 1 + d + e, 1 + f) = (d, e, f)(1, 1, 1)$$

Checking each coordinate, we see that the first and third coordinates are equal to each other, while equality of the second coordinates imply that

$$e + f = 1 + d + e$$

which implies that  $f = 1 + d$ . Since  $d$  and  $f$  must be different, we may set without loss of generality  $d = 1$  and  $f = 0$ . Since  $e$  may vary, we may choose  $B = (1, 0, 0)$ . Since  $A^4 = B^2 = I_3$ , then we have a homomorphism  $\varphi : D_8 \rightarrow \text{UT}_3(\mathbb{F}_2)$  given by

$$\varphi(r) = A, \quad \varphi(s) = B$$

It is easy to see that  $\langle A, B \rangle = \text{UT}_3(\mathbb{F}_2)$  since  $A$  and  $B$  generate all 8 elements of  $\text{UT}_3(\mathbb{F}_2)$ . Since  $\varphi(D_8) = \text{UT}_3(\mathbb{F}_2)$ , then  $\varphi$  is surjective. Moreover,  $|D_8| = |\text{UT}_3(\mathbb{F}_2)| = 8$ , so that  $\varphi$  is injective as well. Therefore,  $\text{UT}_3(\mathbb{F}_2) \cong D_8$ . ■

**Exercise 2.4.13**

Prove that the multiplicative group of positive rational numbers is generated by the set  $\{1/p \mid p \text{ is prime}\}$ .

**Solution.** Let  $P$  denote the set of the prime reciprocals, and let  $\mathbb{Q}_+^\times$  denote the multiplicative group of positive rational numbers. Observe that  $(1/p)^{-1} = p \in \langle P \rangle$  for any  $1/p \in P$ , since  $\langle P \rangle \leq \mathbb{Q}_+^\times$ . It then follows that  $p^k \in \langle P \rangle$  as well for any  $k \in \mathbb{Z}$ .

Let  $q \in \mathbb{Q}_+^\times$ . Then  $q = m/n$  for some  $m \in \mathbb{Z}^+ \cup \{0\}$  and  $n \in \mathbb{Z}^+$ . By the Fundamental Theorem of Arithmetic, both  $m$  and  $n$  can be expressed as a product of primes. Then  $m/n = q \in \langle P \rangle$  since  $\langle P \rangle$  is closed under multiplication. Hence,  $\mathbb{Q}_+^\times \subseteq \langle P \rangle$ . Since  $P \subseteq \mathbb{Q}_+^\times$ , then  $\langle P \rangle \subseteq \mathbb{Q}_+^\times$ . Therefore,  $\langle P \rangle = \mathbb{Q}_+^\times$ . ■

(\*) **Exercise 2.4.14**

A group  $H$  is called *finitely generated* if there is a finite set  $A$  such that  $H = \langle A \rangle$ .

- (a) Prove that every finite group is finitely generated.
- (b) Prove that  $\mathbb{Z}$  is finitely generated.
- (c) Prove that every finitely generated subgroup of the additive group  $\mathbb{Q}$  is cyclic. [If  $H$  is a finitely generated subgroup of  $\mathbb{Q}$ , show that  $H \leq \langle 1/k \rangle$ , where  $k$  is the product of all denominators appearing in a set of generators for  $H$ .]
- (d) Prove that  $\mathbb{Q}$  is not finitely generated.

**Solution.**

- (a) Any finite group  $G$  is generated by  $\langle G \rangle$ .
- (b)  $\mathbb{Z} = \langle 1 \rangle$ .
- (c) Let  $H = \langle A \rangle$ , where

$$A = \left\{ \frac{p_1}{q_1}, \frac{p_2}{q_2}, \dots, \frac{p_n}{q_n} \right\}$$

where  $(p_i, q_i) = 1$  for all  $1 \leq i \leq n$ . Then every element  $h \in H$  is of the form

$$h = \sum_{i=1}^n a_i \frac{p_i}{q_i}, \quad a_i \in \mathbb{Z}$$

since  $\mathbb{Q}$  is abelian. Define the quantities

$$k = \prod_{i=1}^n q_i, \quad k_j = k/q_j$$

so that  $k$  is the product of all denominators in  $A$ , and  $k_j$  is the product of all denominators except the denominator in the  $j$ -th fraction of  $A$ . We may then rewrite  $h$  as

$$h = \frac{1}{k} \sum_{i=1}^n a_i p_i k_i$$

so that  $h \in \langle 1/k \rangle$ , which is a cyclic subgroup of  $\mathbb{Q}$ . Then  $H \leq \langle 1/k \rangle$ , hence it is cyclic.

- (d) Suppose  $\mathbb{Q}$  was finitely generated. Then  $\mathbb{Q} = \langle p/q \rangle$  where  $(p, q) = 1$ , by the previous part. Let  $r \in \mathbb{Z}$  such that  $r$  does not divide  $q$ . Then there is some  $n \in \mathbb{Z}$  such that

$$n \frac{p}{q} = \frac{1}{r}$$

Then  $q = npr$ , contradicting that  $r$  doesn't divide  $q$ . ■

**Exercise 2.4.15**

Exhibit a proper subgroup of  $\mathbb{Q}$  which is not cyclic.

**Solution.** By the previous exercise, such a subgroup cannot be finitely generated. Consider the set

$$A = \left\{ \frac{1}{2^k} \mid k \in \mathbb{Z}^+ \cup \{0\} \right\}$$

where  $\langle A \rangle \leq \mathbb{Q}$ . Note that  $\langle A \rangle < \mathbb{Q}$  since  $1/3 \notin \langle A \rangle$ . Moreover, if  $\langle A \rangle = \langle p/q \rangle$ , then  $q = 2^m$  since every element of  $A$  has a power of 2. Then  $2^{m+1} \notin \langle p/q \rangle$ , but  $2^{m+1} \in \langle A \rangle$ , hence  $\langle A \rangle$  cannot be cyclic. ■

(\*) **Exercise 2.4.16**

A subgroup  $M$  of a group  $G$  is called a *maximal subgroup* if  $M \neq G$  and the only subgroups of  $G$  which contain  $M$  are  $M$  and  $G$ .

- Prove that if  $H$  is a proper subgroup of the finite group  $G$  then there is a maximal subgroup of  $G$  containing  $H$ .
- Show that the subgroup of all rotations in a dihedral group is a maximal subgroup.
- Show that if  $G = \langle x \rangle$  is a cyclic group of order  $n \geq 1$  then a subgroup  $H$  is maximal if and only if  $H = \langle x^p \rangle$  for some prime  $p$  dividing  $n$ .

**Solution.**

- If  $H$  is maximal, then we are done. If not, there exists a subgroup  $H_1 < G$  such that  $H < H_1$ . If  $H_1$  is maximal, we are done, but if not, then there must be another subgroup such that  $H_1 < H_2$ . Continuing on, we can create a chain of subgroups  $H_i$  such that  $H_i < H_{i+1}$ . Since  $G$  is finite, the process of creating subgroups must terminate at some  $k$ . Then  $H_k < G$  is a maximal subgroup that contains  $H$ .
- Note that  $s \notin \langle r \rangle$  so that  $\langle r \rangle$  is a proper subgroup of  $D_{2n}$ . If  $\langle r \rangle$  was not maximal, there must exist some  $H$  such that  $\langle r \rangle < H$  so that some reflection  $sr^k \in H$  for some  $1 \leq k < n$ . Note that  $r^{n-k} \in H$  so that  $sr^k r^{n-k} = s \in H$ . But then  $H = D_{2n}$ , contradicting that  $H < D_{2n}$ . It must be that  $\langle r \rangle$  is maximal.
- Suppose  $H$  is maximal, and put  $H = \langle x^k \rangle$  and  $d = (n, k)$ . Note that  $d > 1$  for the subgroup to be proper. Let  $p$  be a prime that divides  $n$ . If  $k = p$ , then we are done. If  $k \neq p$ , consider  $\langle x^p \rangle$ . Then  $H < \langle x^p \rangle$  since  $p \mid d$ . Since  $H$  is maximal, then  $\langle x^p \rangle = G$ . Then  $(p, n) = 1$ , but this contradicts that  $p \mid n$ . Then  $k = p$ .

Suppose that  $H = \langle x^p \rangle$  for prime  $p \mid n$ . If  $H$  is not maximal, there exists  $\langle x^d \rangle$  such that  $d \mid p$ . Since  $p$  is prime, then either  $d = 1$  or  $p$ . If  $d = 1$ , then  $\langle x^d \rangle = G$  which shows that  $\langle x^d \rangle$  is not a proper subgroup of  $G$ . If  $d = p$ , then  $\langle x^d \rangle = \langle x^p \rangle$  so that  $H$  is not a proper subgroup of  $\langle x^d \rangle$ . It follows that  $H$  is maximal. ■

(\*) **Exercise 2.4.17**

This is an exercise involving Zorn's Lemma. Prove that every nontrivial finitely generated group possesses maximal subgroups. Let  $G$  be a finitely generated group, say  $G = \langle g_1, g_2, \dots, g_n \rangle$ , and let  $\mathcal{S}$  be the set of all proper subgroups of  $G$ . Then  $\mathcal{S}$  is partially ordered by inclusion. Let  $C$  be a chain in  $\mathcal{S}$ .

- Prove that the union  $H$  of all the subgroups in  $C$  is a subgroup of  $G$ .
- Prove that  $H$  is a proper subgroup.
- Use Zorn's Lemma to show that  $\mathcal{S}$  has a maximal element (which is, by definition, a maximal subgroup).

**Solution.**

- Let  $C \subseteq \mathcal{S}$  be a chain. Put

$$H = \bigcup_{K \in C} K$$

Since at least one subgroup is in  $C$  and subgroups are nonempty, then  $C$  is nonempty. Suppose  $g, h \in H$ . Then  $g \in K_1$  and  $h \in K_2$  for some  $K_1, K_2 \in C$  such that  $K_1 \leq K_2$  or  $K_2 \leq K_1$ , or both. Without loss of generality, suppose  $K_1 \leq K_2$ . Then  $g \in K_2$  as well so that  $gh^{-1} \in K_2 \subseteq H$ . Then  $H \leq G$ .

- If  $H$  was not a proper subgroup, then  $H$  must contain all generators  $g_i$ . Associate each generator with a (not necessarily distinct) subgroup  $K_i$  so that  $g_i \in K_i$  for every  $1 \leq i \leq n$ . Since  $C$  is a chain, then we may order the subgroups such that  $K_j \leq K_{j+1}$  for all  $1 \leq j \leq n-1$ . It follows that  $K_n$  contains every generator  $g_i$  so that  $K_n = G$ , contradicting that  $C$  is a chain of proper subgroups of  $G$ .
- Because  $G$  is nontrivial, then  $\{1\} \in \mathcal{S}$  so that  $\mathcal{S}$  is nonempty. Moreover,  $H \in \mathcal{S}$  by the previous part, and for any  $K \in C$ , we have  $K \leq H$  so that  $H$  is an upper bound for  $C$ . Then every chain in  $\mathcal{S}$  has an upper bound, so by Zorn's Lemma,  $\mathcal{S}$  must have a maximal element. ■

**Exercise 2.4.18**

Let  $p$  be a prime and let  $Z = \{z \in \mathbb{C} \mid z^{p^n} = 1 \text{ for some } n \in \mathbb{Z}^+\}$  (so  $Z$  is the multiplicative group of all  $p$ -power roots of unity in  $\mathbb{C}$ ). For each  $k \in \mathbb{Z}^+$  let  $H_k = \{z \in Z \mid z^{p^k} = 1\}$  (the group of  $p^k$ th roots of unity). Prove:

- (a)  $H_k \leq H_m$  if and only if  $k \leq m$ .
- (b)  $H_k$  is cyclic for all  $k$ .
- (c) Every proper subgroup of  $Z$  equals  $H_k$  for some  $k$ . In particular, every proper subgroup of  $Z$  is finite and cyclic.
- (d)  $Z$  is not finitely generated.

**Solution.**

- (a) Note that  $|H_k| = p^k$  for any  $k$  as there are exactly  $p^k$  roots of unity. Now, if  $H_k \leq H_m$ , then  $p^k \mid p^m$  by Lagrange's Theorem. Then  $p^k \leq p^m$ , or  $k \leq m$  since  $p > 0$ .  
If  $k \leq m$ , then  $p^k \leq p^m$ . In particular,  $p^k \mid p^m$ . Then for any  $z \in H_k$ , we have

$$z^{p^m} = (z^{p^k})^{p^{m-k}} = 1$$

so that  $z \in H_m$ . Then  $H_k \leq H_m$ .

- (b) Let  $\theta = e^{2\pi i/p^k}$ . Note that  $\langle \theta \rangle$  has order  $p^k$ , and  $\langle \theta \rangle \subseteq H_k$ . Now for some  $z \in H_k$ , we have  $z = e^{2\pi i x/p^k}$  for some  $0 \leq x < p^k$ . Then  $z = (e^{2\pi i/p^k})^x \in \langle \theta \rangle$  so that  $\langle \theta \rangle = H_k$ .  
(c) Let  $H$  be a proper subgroup of  $Z$ . Note that every element in  $Z$  has order  $p^i$  for some  $i \in \mathbb{Z}^+$ , so elements of  $H$  will have similar orders. Define the set

$$S = \{n \in \mathbb{Z}^+ \mid |h| = p^n \text{ for some } h \in H\}$$

so that  $S$  is the set of integers  $n$  such that  $H$  contains a  $p^n$ -th root of unity. Moreover,  $H$  is trivially nonempty if we define  $H_0 = \{z \in Z \mid z^{p^0} = z = 1\}$  so that the trivial proper subgroup  $\{1\}$  of  $Z$  allows  $1 \in H$ .

Suppose now that  $S$  is infinite. For any  $n \in \mathbb{Z}^+$ , there exists  $h \in H$  and  $m \in \mathbb{Z}^+$  such that  $|h| = p^m > p^n$ , for otherwise it would be that every  $h \in H$  has order  $|h| \leq p^n$ , meaning that  $n$  is an upper bound for  $S$ . Then  $|h| = p^m$  so that  $H_m = \langle h \rangle$ , and  $H_m \leq H$ . Since  $H_n \leq H_m$  for every  $n \in \mathbb{Z}^+$ , then

$$Z = \bigcup_{n \in \mathbb{Z}^+} H_n \subseteq H$$

which shows  $Z \leq H$ . But  $H \leq Z$ , hence  $Z = H$ , which contradicts that  $Z \neq H$ . It must be that  $S$  is finite, so it has some maximal element  $s$ . Because  $s \in S$ , then there is  $h_0 \in H$  where  $|h_0| = p^s$  so that  $H_s = \langle h_0 \rangle \leq H$ . Suppose  $h \in H$  with  $|h| = p^k$  for some  $k \in \mathbb{Z}^+$ . Then  $k \in S$  where  $k \leq s$  so that  $H_k \leq H_s$ . Since  $h \in H_k$ , then  $h \in H_s$  so  $H \leq H_s$ . Hence,  $H = H_s$ .

- (d) Put  $Z = \langle z_1, z_2, \dots, z_n \rangle$  for some  $n \in \mathbb{Z}^+$  such that  $|z_i| = p^{x_i}$ . Let  $x = \max(x_1, x_2, \dots, x_n)$ . Then  $z_i \in H_x$  for every  $1 \leq i \leq n$  so that  $Z \leq H_x$ . But recall that  $Z$  comprises every  $p$ -power roots of unity so that  $H_x \leq H_x$ , contradicting part (a). It must be that  $Z$  is infinitely generated. ■

**(\*) Exercise 2.4.19**

A nontrivial abelian group  $A$  (written multiplicatively) is called *divisible* if for each element  $a \in A$  and each nonzero integer  $k$  there is an element  $x \in A$  such that  $x^k = a$ .

- (a) Prove that the additive group of rational numbers  $\mathbb{Q}$  is divisible.
- (b) Prove that no finite abelian group is divisible.

**Solution.**

- (a) Suppose  $p/q \in \mathbb{Q}$ . Then  $(p/qn)n = p/q$ , where  $p/qn \in \mathbb{Q}$ .  
(b) Suppose we have a finite abelian group  $G$  with  $|G| = n$ . Pick some nonidentity  $g \in G$ . Then there is no such  $h \in G$  where  $h^n = g$ , since  $h^n = 1$ . ■

**Exercise 2.4.20**

Prove that if  $A$  and  $B$  are nontrivial abelian groups, then  $A \times B$  is divisible if and only if both  $A$  and  $B$  are divisible.

**Solution.** Suppose  $A \times B$  is divisible, and let  $a \in A$ ,  $b \in B$ , and  $k \in \mathbb{Z}^+$ . Then there exists  $(c, d) \in A \times B$  such that  $(c, d)^k = (c^k, d^k) = (a, b)$ . But then  $c^k = a$  and  $d^k = b$  so that  $A$  and  $B$  are divisible.

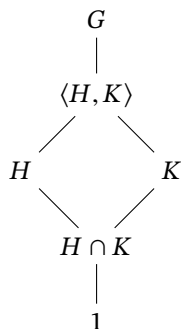
If  $A$  and  $B$  are both divisible, pick  $(a, b) \in A \times B$  and let  $k \in \mathbb{Z}^+$ . Then there exists  $c \in A$  and  $d \in B$  such that  $c^k = a$  and  $d^k = b$ . Then  $(c, d)^k = (c^k, d^k) = (a, b)$  so that  $A \times B$  is divisible. ■

## 2.5 The Lattice of Subgroups of a Group

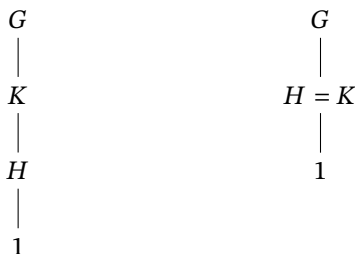
### Exercise 2.5.1

Let  $H$  and  $K$  be subgroups of  $G$ . Exhibit all possible sublattices which show only  $G, 1, H, K$  and their joins and intersections. What distinguishes the different drawings?

**Solution.** If  $H$  and  $K$  are distinct and neither is contained in the other, then the sublattice is a diamond shape:



If  $H \leq K$ , then the sublattice is a chain, noting that  $\langle H, K \rangle = K$  and  $H \cap K = H$ . If  $H = K$ , then the sublattice is also a chain, noting that  $\langle H, K \rangle = H = K$  and  $H \cap K = H = K$ :



Remaining scenarios include more trivial cases where one of  $H$  or  $K$  is either  $G$  or  $1$ . ■

### Exercise 2.5.2

In each of (a) to (d) list all subgroups of  $D_{16}$  that satisfy the given condition.

- (a) Subgroups that are contained in  $\langle sr^2, r^4 \rangle$
- (b) Subgroups that are contained in  $\langle sr^7, r^4 \rangle$
- (c) Subgroups that contain  $\langle r^4 \rangle$
- (d) Subgroups that contain  $\langle s \rangle$ .

**Solution.** Using the subgroup lattice of  $D_{16}$ , we find:

- (a)  $\langle sr^2, r^4 \rangle, \langle sr^6 \rangle, \langle sr^2 \rangle, \langle r^4 \rangle, 1$ .
- (b) Note  $sr^7 = sr^3$ , so  $\langle sr^3, r^4 \rangle, \langle r^4 \rangle, \langle sr^3 \rangle, \langle sr^7 \rangle, 1$ .
- (c)  $\langle r^4 \rangle, \langle sr^2, r^4 \rangle, \langle s, r^4 \rangle, \langle r^2 \rangle, \langle sr^3, r^4 \rangle, \langle sr^5, r^4 \rangle, \langle s, r^2 \rangle, \langle r \rangle, \langle sr, r^2 \rangle, D_{16}$ .
- (d)  $\langle s \rangle, \langle s, r^4 \rangle, \langle s, r^2 \rangle, D_{16}$ . ■

**Exercise 2.5.3**

Show that the subgroup  $\langle s, r^2 \rangle$  of  $D_8$  is isomorphic to  $V_4$ .

**Solution.** Recall that  $V_4 = \{1, a, b, c\}$  where  $a^2 = b^2 = c^2 = 1$  and  $ab = c$ ,  $bc = a$ , and  $ca = b$ . Note that every element in  $\langle s, r^2 \rangle$  has order 2 since  $(r^2)^2 = s^2 = 1$  and  $(sr^2)^2 = sr^2sr^2 = 1$ . Then we may define a mapping  $\varphi : V_4 \rightarrow \langle s, r^2 \rangle$  by

$$\varphi(1) = 1, \quad \varphi(a) = s, \quad \varphi(b) = r^2, \quad \varphi(c) = sr^2$$

We show that  $\varphi$  is a homomorphism by checking the products:

$$\begin{aligned} \varphi(a^2) &= \varphi(1) = 1 = s^2 = \varphi(a)^2 \\ \varphi(b^2) &= \varphi(1) = 1 = (r^2)^2 = \varphi(b)^2 \\ \varphi(ab) &= \varphi(c) = sr^2 = \varphi(a)\varphi(b) \end{aligned}$$

Similar calculations show that  $\varphi(bc) = \varphi(b)\varphi(c)$  and  $\varphi(ca) = \varphi(c)\varphi(a)$ . Since  $\varphi$  is defined on all elements of  $V_4$ , then  $\varphi$  is a homomorphism. Moreover,  $\varphi$  is surjective since  $\varphi(V_4) = \langle s, r^2 \rangle$ . Since  $|V_4| = |\langle s, r^2 \rangle| = 4$ , then  $\varphi$  is injective as well. Therefore,  $\langle s, r^2 \rangle \cong V_4$ . ■

**Exercise 2.5.4**

Use the given lattice to find all pairs of elements that generate  $D_8$  (there are 12 pairs).

**Solution.** Note that  $D_8 = \langle s, r \rangle$ . Moreover,  $s \neq rs$ , and  $r \in \langle s, rs \rangle$  so that  $\langle s, rs \rangle = D_8$ . Since  $r = r^3$ , we also have  $\langle s, r^3 \rangle = D_8$ . We may also replace the reflection  $s$  with  $sr^2$ , since combinations of a reflection with an odd rotation and a reflection with an even rotation contain  $r$  and thus generates  $D_8$ . It follows that the pairs of elements that generate  $D_8$  are:

$$\langle s, r \rangle, \langle s, r^3 \rangle, \langle s, rs \rangle, \langle s, r^3s \rangle, \langle r^2s, r \rangle, \langle r^2s, r^3 \rangle, \langle r^2s, rs \rangle, \langle r^2s, r^3s \rangle, \langle r, rs \rangle, \langle r^3, rs \rangle, \langle r, r^3s \rangle, \langle r^3, r^3s \rangle$$

**Exercise 2.5.5**

Use the given lattice to find all elements  $x \in D_{16}$  such that  $D_{16} = \langle x, s \rangle$  (there are 8 such elements  $x$ ).

**Solution.** Note that  $\langle r \rangle = \langle r^3 \rangle = \langle r^5 \rangle = \langle r^7 \rangle$ . We then pair each generator with an  $s$  so that we obtain just the rotation to then generate  $D_{16}$ :  $x = r, r^3, r^5, r^7, sr, sr^3, sr^5, sr^7$ . ■

**Exercise 2.5.6**

Use the given lattices to help find the centralizers of every element in the following groups:

(a)  $D_8$  (b)  $Q_8$  (c)  $S_3$  (d)  $D_{16}$ .

**Solution.**

(a) To calculate the centralizer of an element  $a$ , start with the cyclic subgroup that contains  $a$  and see if any other elements are contained in  $\langle a \rangle$ . For example, to calculate  $C_{D_8}(rs)$ , start with  $\langle rs \rangle$  in the subgroup lattice. Since  $r^4, sr^5 \in C_{D_8}(rs)$ , then  $C_{D_8}(rs) \leq \langle sr^5, r^4 \rangle$ . Checking the next subgroup, it follows that  $r^2 \in C_{D_8}(rs)$  so that  $C_{D_8}(rs) \leq \langle rs, r^2 \rangle$ . Since  $r \notin C_{D_8}(rs)$ , then  $C_{D_8}(rs) \neq D_{16}$  so that  $C_{D_8}(rs) = \langle rs, r^2 \rangle$ . We use similar reasoning to deduce the other centralizers:

$$\begin{aligned} C_{D_8}(1) &= D_8 & C_{D_8}(s) &= \langle s, r^2 \rangle \\ C_{D_8}(r) &= \langle r \rangle & C_{D_8}(rs) &= \langle rs, r^2 \rangle \\ C_{D_8}(r^2) &= D_8 & C_{D_8}(r^2s) &= \langle s, r^2 \rangle \\ C_{D_8}(r^3) &= \langle r \rangle & C_{D_8}(r^3s) &= \langle sr, r^2 \rangle \end{aligned}$$



(b) Note that  $1, -1 \in Z(Q_8)$ , while none of  $i, j$ , and  $k$  commute with anything but themselves. Then:

$$C_{Q_8}(1) = C_{Q_8}(-1) = Q_8$$

$$C_{Q_8}(i) = C_{Q_8}(-i) = \langle i \rangle$$

$$C_{Q_8}(j) = C_{Q_8}(-j) = \langle j \rangle$$

$$C_{Q_8}(k) = C_{Q_8}(-k) = \langle k \rangle$$

(c) Since no element in  $S_3$  commutes with each other, then  $C_{S_3}(a) = \langle a \rangle$  for all  $a \in S_3 - \{1\}$ , where  $C_{S_3}(1) = S_3$ .

(d) Use similar reasoning as in part (a) to obtain the centralizers:

$$C_{D_{16}}(1) = C_{D_{16}}(r^4) = D_{16}$$

$$C_{D_{16}}(r^k) = \langle r \rangle \text{ for all } k = 1, 2, 3, 5, 6,$$

$$C_{D_{16}}(s) = C_{D_{16}}(sr^4) = \langle s, r^4 \rangle$$

$$C_{D_{16}}(sr) = C_{D_{16}}(sr^5) = \langle sr^5, r^4 \rangle$$

$$C_{D_{16}}(sr^2) = C_{D_{16}}(sr^6) = \langle sr^2, r^4 \rangle$$

$$C_{D_{16}}(sr^3) = C_{D_{16}}(sr^7) = \langle sr^3, r^4 \rangle$$

■

### Exercise 2.5.7

Find the center of  $D_{16}$ .

**Solution.**  $Z(D_{16}) = \{1, r^4\}$  by Exercise 2.2.7. ■

### Exercise 2.5.8

In each of the following groups find the normalizer of each subgroup:

(a)  $S_3$  (b)  $Q_8$ .

**Solution.**

(a) From the subgroup lattice of  $S_3$ , we can deduce that each subgroup is cyclic and maximal except for  $\{id\}$ , where  $N_{S_3}(id) = S_3$ . Then for any  $\alpha \in S_3$ , we observe that  $N_{S_3}(\langle \alpha \rangle) = \langle \alpha \rangle$  or  $S_3$ .

Consider  $\alpha = (1\ 2)$ . Since  $(1\ 3)(1\ 2)(1\ 3) = (2\ 3) \notin \langle (1\ 2) \rangle$ , then  $N_{S_3}(\langle (1\ 2) \rangle) \neq S_3$  so that  $N_{S_3}(\langle (1\ 2) \rangle) = \langle (1\ 2) \rangle$ . By symmetry, the same holds for  $\langle (1\ 3) \rangle$  and  $\langle (2\ 3) \rangle$ .

Consider  $\alpha = (1\ 2\ 3)$ . Observe that

$$(1\ 2)(1\ 2\ 3)(1\ 2) = (1\ 3\ 2) \quad \text{and} \quad (1\ 2)(1\ 3\ 2)(1\ 2) = (1\ 2\ 3)$$

so that  $(1\ 2) \in N_{S_3}(\langle (1\ 2\ 3) \rangle)$ . Then  $N_{S_3}(\langle (1\ 2\ 3) \rangle) = S_3$ .

(b) Note that the subgroups  $\langle -1 \rangle$  and  $\{1\}$  are normal in  $Q_8$  so that  $N_{Q_8}(\langle -1 \rangle) = N_{Q_8}(\{1\}) = Q_8$ . Consider the subgroups of order 4 in  $Q_8$ . In particular, observe for  $i$ :

$$jij^{-1} = -i, j(-i)j^{-1} = i$$

so that  $j \in N_{Q_8}(\langle i \rangle)$ . Then  $N_{Q_8}(\langle i \rangle) = Q_8$ . By symmetry, the same holds for  $\langle j \rangle$  and  $\langle k \rangle$ . ■

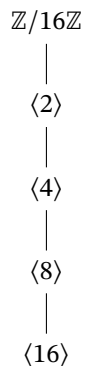
**Exercise 2.5.9**

Draw the lattices of subgroups of the following groups:

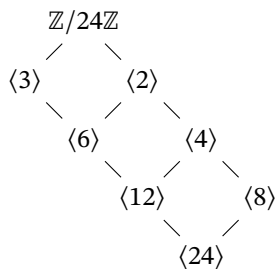
(a)  $\mathbb{Z}/16\mathbb{Z}$  (b)  $\mathbb{Z}/24\mathbb{Z}$  (c)  $\mathbb{Z}/48\mathbb{Z}$ .

**Solution.**

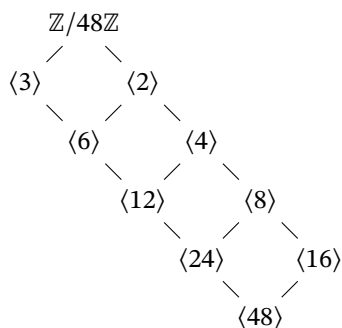
(a)  $\mathbb{Z}/16\mathbb{Z}$



(b)  $\mathbb{Z}/24\mathbb{Z}$



(c)  $\mathbb{Z}/48\mathbb{Z}$



■

**Exercise 2.5.10**

Classify groups of order 4 by proving that if  $|G| = 4$ , then  $G \cong \mathbb{Z}_4$  or  $G \cong V_4$ . [See [Exercise 1.1.36](#).]

**Solution.** Let  $G = \{1, a, b, c\}$ . If  $G$  has an element of order 4, then  $G \cong \mathbb{Z}_4$  by Theorem 2.4.

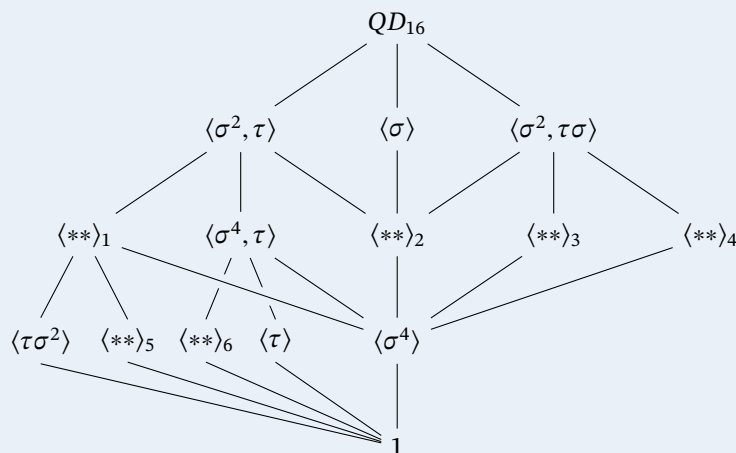
Suppose  $G$  is not cyclic. Then every nonidentity element has order 2 by Lagrange's Theorem. In particular,  $a^2 = b^2 = c^2 = 1$ . Note that  $ab \neq 1$ , since otherwise  $b = a^{-1} = a$ . Similarly,  $ab \neq a$  and  $ab \neq b$ , so that  $ab = c$ . By similar reasoning, we have  $bc = a$  and  $ca = b$ . This is precisely the definition of  $V_4$ , so that  $G \cong V_4$ . ■

**Exercise 2.5.11**

Consider the group of order 16 with the following presentation:

$$QD_{16} = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1, \sigma\tau = \tau\sigma^3 \rangle$$

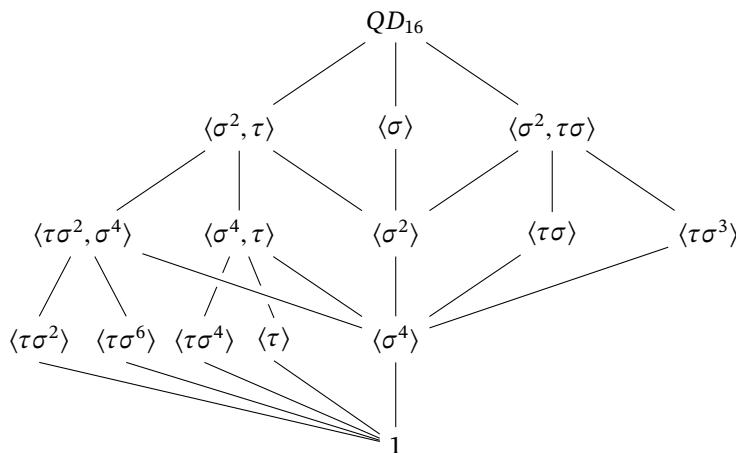
(called the *quasidihedral* or *semidihedral* group of order 16). This group has three subgroups of order 8:  $\langle \tau, \sigma^2 \rangle \cong D_8$ ,  $\langle \sigma \rangle \cong Z_8$ , and  $\langle \sigma^2, \tau\sigma \rangle \cong Q_8$ , and every proper subgroup is contained in one of these three subgroups. Fill in the missing subgroups in the lattice of all subgroups of the quasidihedral group, exhibiting each subgroup with at most two generators.



**Solution.** The above has had subscripts added to distinguish the unknown subgroups. It is clear that  $\langle ** \rangle_2$  must be  $\langle \sigma^2 \rangle$  since it is the only subgroup contained in both  $\langle \sigma \rangle$  and  $\langle \sigma^2, \tau \rangle$ . To find the other subgroups, we calculate the cyclic subgroups of the form  $\langle \tau\sigma^k \rangle$ :

$$\begin{aligned} \langle \tau\sigma \rangle &= \{1, \tau\sigma, \sigma^4, \tau\sigma^5\} & \langle \tau\sigma^3 \rangle &= \{1, \tau\sigma^3, \sigma^4, \tau\sigma^7\} \\ \langle \tau\sigma^4 \rangle &= \{1, \tau\sigma^4\} & \langle \tau\sigma^6 \rangle &= \{1, \tau\sigma^6\} \end{aligned}$$

It becomes clear that 6 must be  $\langle \tau\sigma^4 \rangle$ , since the subgroup above contains both  $\tau$  and  $\sigma^4$ . 3 and 4 must be  $\langle \tau\sigma \rangle$  and  $\langle \tau\sigma^3 \rangle$  respectively, since  $\langle \sigma^2, \tau\sigma \rangle$  both contain  $\tau\sigma$  and  $\sigma^2$  which multiply to  $\tau\sigma^3$ . Certainly, 1 cannot be  $\langle \tau\sigma^6 \rangle$  as it does not contain  $\langle \tau\sigma^2 \rangle$ , so it must be 5. Then 1 must be  $\langle \tau\sigma^2, \sigma^4 \rangle$  as it contains both subgroups. Hence, the completed diagram is



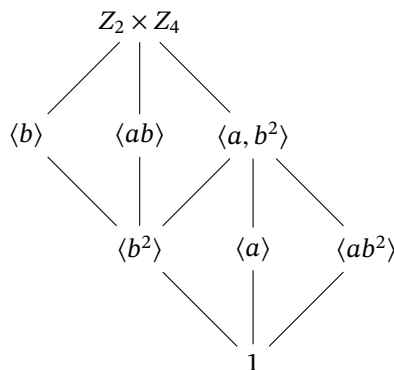
**Exercise 2.5.12**

The group  $A = Z_2 \times Z_4 = \langle a, b \mid a^2 = b^4 = 1, ab = ba \rangle$  has order 8 and has three subgroups of order 4:  $\langle a, b^2 \rangle \cong V_4$ ,  $\langle b \rangle \cong Z_4$ , and  $\langle ab \rangle \cong Z_4$  and every proper subgroup is contained in one of these three. Draw the lattice of all subgroups of  $A$ , giving each subgroup in terms of at most two generators.

**Solution.** Observe that  $A$  is the direct product of two abelian groups, so  $A$  is abelian, and we may write it as

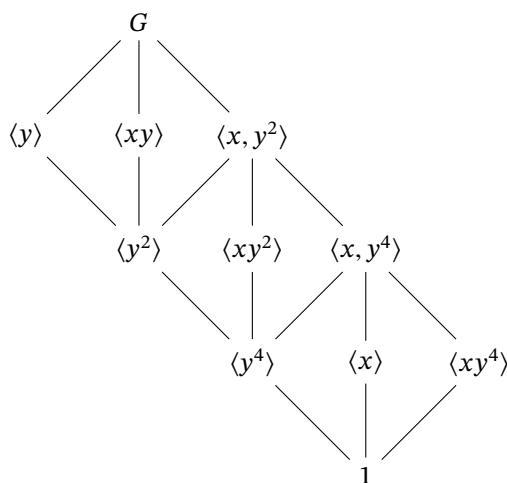
$$A = \{1, a, b, b^2, b^3, ab, ab^2, ab^3\}$$

We know that  $A$  has three subgroups of order 4:  $\langle a, b^2 \rangle$ ,  $\langle b \rangle$ , and  $\langle ab \rangle$ . To find what the containment relations are, observe that  $\langle a, b^2 \rangle$  has three subgroups of order 2:  $\langle b^2 \rangle$ ,  $\langle a \rangle$ , and  $\langle ab^2 \rangle$ .  $\langle b \rangle$  has the single subgroup of order 2:  $\langle b^2 \rangle$ , and  $\langle ab \rangle$  has the single subgroup of order 2:  $\langle ab^2 \rangle$ . Then the subgroup lattice of  $A$  is

**Exercise 2.5.13**

The group  $G = Z_2 \times Z_8 = \langle x, y \mid x^2 = y^8 = 1, xy = yx \rangle$  has order 16 and has three subgroups of order 8:  $\langle x, y^2 \rangle \cong Z_2 \times Z_4$ ,  $\langle y \rangle \cong Z_8$ , and  $\langle xy \rangle \cong Z_8$ , and every proper subgroup is contained in one of those three. Draw the lattice of all subgroups of  $G$ , giving each subgroups in terms of at most two generators (cf. [Exercise 2.5.12](#)).

**Solution.** Let  $x = a$  and  $y^2 = b$ , where  $a$  and  $b$  are the generators from [Exercise 2.5.12](#). It follows that  $G$  contains a copy of  $Z_2 \times Z_4$  as  $\langle x, y^2 \rangle = \langle a, b \rangle$ . Moreover, we may append  $\langle y \rangle$  and  $\langle xy \rangle$  to this lattice as follows:



**Exercise 2.5.14**

Let  $M$  be the group of order 16 with the following presentation:

$$\langle u, v \mid u^2 = v^8 = 1, vu = uv^5 \rangle$$

(sometimes called the *modular* group of order 16). It has three subgroups of order 8:  $\langle u, v^2 \rangle$ ,  $\langle v \rangle$ , and  $\langle uv \rangle$ , and every proper subgroup is contained in one of those three. Prove that  $\langle u, v^2 \rangle \cong Z_2 \times Z_4$ ,  $\langle v \rangle \cong Z_8$ , and  $\langle uv \rangle \cong Z_8$ . Show that the lattice of subgroups of  $M$  is the same as the lattice of subgroups of  $Z_2 \times Z_8$  (cf. Exercise 13) but that these two groups are not isomorphic.

**Solution.** It is clear that  $|v| = 8$  implies that  $\langle v \rangle \cong Z_8$ . Similarly, since  $|uv| = 8$ , then  $\langle uv \rangle \cong Z_8$ . To see that  $\langle u, v^2 \rangle \cong Z_2 \times Z_4$ , we verify the relations. Note that  $u^2 = (v^2)^4 = 1$ , and

$$v^2u = vu v^5 = uv^{10} = uv^2$$

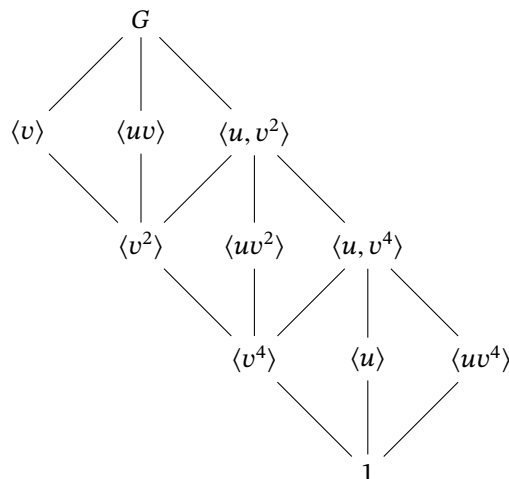
so that  $u$  and  $v^2$  commute, hence it is abelian. We may then list out the elements of  $\langle u, v^2 \rangle$ :

$$\langle u, v^2 \rangle = \{1, v^2, v^4, v^6, u, uv^2, uv^4, uv^6\}$$

Then the mapping  $\varphi : Z_2 \times Z_4 \rightarrow \langle u, v^2 \rangle$  defined by

$$\varphi(a) = u, \quad \varphi(b) = v^2$$

extends to a homomorphism. Moreover, it is clear that distinct elements in  $Z_2 \times Z_4$  map to distinct elements in  $\langle u, v^2 \rangle$ , so that  $\varphi$  is injective. As shown above,  $|\langle u, v^2 \rangle| = 8 = |Z_2 \times Z_4|$ , so that  $\varphi$  is surjective. Hence,  $\varphi$  is an isomorphism and  $Z_2 \times Z_4 \cong \langle u, v^2 \rangle$ . The subgroup lattice of  $M$  is then the same as that of  $Z_2 \times Z_8$  as shown in Exercise 2.5.13, but with  $u = x$  and  $v = y$ :



Finally,  $M$  is not isomorphic to  $Z_2 \times Z_8$  since  $M$  is not abelian. If it were, then  $vu = uv$ , but  $vu = uv^5 = uv$  would imply that  $v^4 = 1$ , contradicting that  $|v| = 8$ . ■

**Exercise 2.5.15**

Describe the isomorphism type of each of the three subgroups of  $D_{16}$  of order 8.

**Solution.** Since  $|r| = 8$ , then  $\langle r \rangle \cong Z_8$ . For the remaining subgroups, note that the lattice for  $D_{16}$  shows a striking similarity to the lattice for  $D_8$ ; in fact, these subgroups *are* isomorphic to  $D_8$  as follows:

For the subgroup  $\langle s, r^2 \rangle$ , observe that  $(r^2)^4 = s^2 = 1$ , and  $sr^2 = r^6s = (r^2)^{-1}s$ . Then the mapping  $\varphi : D_8 \rightarrow \langle s, r^2 \rangle$  given by

$$\varphi(r) = r^2, \quad \varphi(s) = s$$

extends to a homomorphism. Moreover, this mapping is surjective by construction. Since  $\langle s, r^2 \rangle$  contains a subgroup of order 4 and is a subgroup of  $D_{16}$ , it must be 8 so that  $\varphi$  is an isomorphism, and  $D_8 \cong \langle s, r^2 \rangle$ .

For the subgroup  $\langle sr, r^2 \rangle$ , we again observe that  $(r^2)^4 = (sr)^2 = 1$ , and  $(sr)r^2 = sr^3 = r^5s = r^6r^7s = (r^2)^{-1}(sr)$ . The mapping  $\psi : D_8 \rightarrow \langle sr, r^2 \rangle$  given by

$$\psi(r) = r^2, \quad \psi(s) = sr$$

extends to a homomorphism, surjective by construction, and is an isomorphism because  $\langle sr, r^2 \rangle$  has order 8. Then  $D_8 \cong \langle sr, r^2 \rangle$ . ■

**Exercise 2.5.16**

Use the lattice of subgroups of the quasidihedral of order 16 to show that every element of order 2 is contained in the proper subgroup  $\langle \tau, \sigma^2 \rangle$  (cf. [Exercise 2.5.11](#)).

**Solution.** Every element of order 2 generates a cyclic subgroup of order 2. Using the lattice,  $\langle \tau, \sigma^2 \rangle$  properly contains all cyclic subgroups, except  $\langle \tau\sigma \rangle$  and  $\langle \tau\sigma^3 \rangle$ , both of which are order 4. Then  $\langle \tau, \sigma^2 \rangle$  contains all cyclic subgroups of order 2, hence contains all elements of order 2. ■

**Exercise 2.5.17**

Use the lattice of subgroups of the modular group  $M$  of order 16 to show that the set  $\{x \in M \mid x^2 = 1\}$  is a subgroup of  $M$  isomorphic to the Klein 4-group (cf. [Exercise 2.5.14](#)).

**Solution.** Using the lattice in Exercise 14, we see that we have 3 candidates to be isomorphic to  $V_4$ , namely  $\langle v^2 \rangle$ ,  $\langle uv^2 \rangle$ , and  $\langle u, v^4 \rangle$ . The first and second subgroups are cyclic, while  $\langle u, v^4 \rangle = \{1, u, v^4, uv^4\}$ . Since it is not generated by one element, each of these elements are of order 2, and  $v^4u = v^3uv^5 = \dots = uv^{20} = uv^4$  so that it is abelian, then  $\langle u, v^4 \rangle \cong V_4$ . ■

**Exercise 2.5.18**

Use the lattice to help find the centralizer of every element of  $QD_{16}$  (cf. [Exercise 2.5.11](#)).

**Solution.** Note that  $\sigma^4\tau = \sigma^3\tau\sigma^3 = \dots = \tau\sigma^{12} = \tau\sigma^4$  so that  $\sigma^4 \in Z(QD_{16})$  as  $\sigma^4$  already commutes with powers of  $\sigma$ . Moreover, any power of  $\sigma$  does not commute with  $\tau$  except for  $\sigma^4$ . The elements  $\tau\sigma$  and  $\tau\sigma^3$  do not commute with  $\sigma^2$  as  $(\tau\sigma)\sigma^2 = \tau\sigma^3 = \sigma\tau \neq \sigma^2(\tau\sigma) = \tau\sigma^2$ , and  $(\tau\sigma^3)\sigma^2 = \sigma^7\tau \neq \sigma^3\tau = \sigma^2(\tau\sigma^3)$ . Next,  $(\tau\sigma^2)\sigma^2 = \tau\sigma^4 \neq \tau = \sigma^2(\tau\sigma^2)$  so that  $\sigma^2$  does not commute with  $\tau\sigma^2$ . Moreover,  $(\tau\sigma^6)(\tau\sigma^2) = \tau\sigma^2\sigma^4\tau\sigma^2 = (\tau\sigma^2)(\tau\sigma^6)$  so that  $\tau\sigma^2$  commutes with  $\tau\sigma^6$ , but  $\sigma^2\tau\sigma^6 = \tau\sigma^4 \neq \tau = (\tau\sigma^6)\sigma^2$  so  $\sigma^2$  does not commute with  $\tau\sigma^6$ . Lastly,  $\sigma^2(\tau\sigma^4) = \tau\sigma^2 \neq (\tau\sigma^4)\sigma^2$ , and  $\sigma^2\tau = \tau\sigma^6 \neq \tau\sigma^2$  so that  $\sigma^2$  does not commute with  $\tau\sigma^4$  nor with  $\tau$ . It follows that the centralizers of the elements of  $QD_{16}$  are

$$\begin{aligned} C_{QD_{16}}(1) &= C_{QD_{16}}(\sigma^4) = QD_{16} \\ C_{QD_{16}}(\langle \sigma^k \rangle) &= \langle \sigma \rangle \text{ for } k = 1, 2, 3, 5, 6, 7 \\ C_{QD_{16}}(\langle \tau\sigma \rangle) &= C_{QD_{16}}(\langle \tau\sigma^5 \rangle) = \langle \tau\sigma \rangle \\ C_{QD_{16}}(\langle \tau\sigma^3 \rangle) &= C_{QD_{16}}(\langle \tau\sigma^7 \rangle) = \langle \tau\sigma^3 \rangle \\ C_{QD_{16}}(\langle \tau \rangle) &= C_{QD_{16}}(\langle \tau\sigma^4 \rangle) = \langle \sigma^4, \tau \rangle \\ C_{QD_{16}}(\langle \tau\sigma^2 \rangle) &= C_{QD_{16}}(\langle \tau\sigma^6 \rangle) = \langle \tau\sigma^2, \sigma^4 \rangle \end{aligned}$$

**Exercise 2.5.19**

Use the lattice to help find  $N_{D_{16}}(\langle s, r^4 \rangle)$ .

**Solution.** Based on the placement of  $\langle s, r^4 \rangle$ , its normalizer may be itself,  $\langle s, r^2 \rangle$ , or  $D_{16}$ . Note that  $\langle s, r^4 \rangle = \{1, s, r^4, sr^4\}$ , and taking  $r^2$  and  $(r^2)^{-1} = r^6$ , we have

$$r^2 \langle s, r^4 \rangle r^6 = \{1, sr^4, r^4, s\} = \langle s, r^4 \rangle$$

so that  $r^2 \in N_{D_{16}}(\langle s, r^4 \rangle)$ , and  $\langle s, r^2 \rangle \leq N_{D_{16}}(\langle s, r^4 \rangle)$ . Since  $rsr^{-1} = r^2s \neq r$ , then  $r \notin N_{D_{16}}(\langle s, r^4 \rangle)$  so that  $N_{D_{16}}(\langle s, r^4 \rangle) = \langle s, r^2 \rangle$ . ■

**Exercise 2.5.20**

Use the lattice of subgroups of  $QD_{16}$  (cf. [Exercise 2.5.11](#)) to help find the normalizers

(a)  $N_{QD_{16}}(\langle \tau\sigma \rangle)$  (b)  $N_{QD_{16}}(\langle \tau, \sigma^4 \rangle)$ .

**Solution.**

(a) Note that  $\langle \tau\sigma \rangle = \{1, \tau\sigma, \sigma^4, \tau\sigma^5\}$ . Moreover,  $(\sigma^2)^{-1} = \sigma^6$  so that

$$\sigma^2 \langle \tau\sigma \rangle \sigma^6 = \{1, \tau\sigma^4, \sigma^4, \tau\sigma\} = \langle \tau\sigma \rangle$$

and  $\sigma(\tau\sigma)\sigma^7 = \tau\sigma^3 \neq \tau\sigma$  so that  $\sigma \notin N_{QD_{16}}(\langle \tau\sigma \rangle)$ . Then  $N_{QD_{16}}(\langle \tau\sigma \rangle) = \langle \sigma^2, \tau\sigma \rangle$ .

(b)  $\langle \tau, \sigma^4 \rangle = \{1, \tau, \sigma^4, \tau\sigma^4\}$ , and

$$\sigma^2 \langle \tau, \sigma^4 \rangle \sigma^6 = \{1, \tau\sigma^4, \sigma^4, \tau\}$$

while  $\sigma\tau\sigma^7 = \tau\sigma^2 \neq \tau$  so that  $\sigma \notin N_{QD_{16}}(\langle \tau, \sigma^4 \rangle)$ . Then  $N_{QD_{16}}(\langle \tau, \sigma^4 \rangle) = \langle \sigma^2, \tau \rangle$ . ■

## 4 Group Actions

### 4.1 Group Actions and Permutation Representations

Let  $G$  be a group and let  $A$  be a nonempty set.

#### Exercise 4.1.1

Let  $G$  act on the set  $A$ . Prove that if  $a, b \in A$  and  $b = g \cdot a$  for some  $g \in G$ , then  $G_b = gG_ag^{-1}$  ( $G_a$  is the stabilizer of  $a$ ). Deduce that if  $G$  acts transitively on  $A$  then the kernel of the action is

$$\bigcap_{g \in G} gG_ag^{-1}.$$

**Solution.** Suppose  $h \in G_b$ . We want to show that  $h \in gG_ag^{-1}$ . Since  $h \in G_b$ , we have  $h \cdot b = b$ . But  $b = g \cdot a$ , so  $h \cdot (g \cdot a) = g \cdot a$ . Applying  $g^{-1}$  to both sides, we get  $g^{-1}hg \cdot a = a$ . This means that  $g^{-1}hg \in G_a$ , so  $h \in gG_ag^{-1}$ . Thus, we have shown that  $G_b \subseteq gG_ag^{-1}$ . For the other direction, suppose  $h \in gG_ag^{-1}$ . Then there exists some  $k \in G_a$  such that  $h = gkg^{-1}$ . We want to show that  $h \in G_b$ . We have  $h \cdot b = (gkg^{-1}) \cdot (g \cdot a) = g \cdot (k \cdot a) = g \cdot a = b$ , since  $k \in G_a$  implies  $k \cdot a = a$ . Thus,  $h \in G_b$ . Therefore, we have shown that  $gG_ag^{-1} \subseteq G_b$ . Combining both inclusions, we conclude that  $G_b = gG_ag^{-1}$ .

Now, if  $G$  acts transitively on  $A$ , then for any  $b \in A$ , there exists some  $g \in G$  such that  $b = g \cdot a$ . From the first part, we have  $G_b = gG_ag^{-1}$ . The kernel of the action is the intersection of all stabilizers  $G_b$  for  $b \in A$ . Therefore, the kernel is

$$\bigcap_{b \in A} G_b = \bigcap_{g \in G} gG_ag^{-1}. \quad \blacksquare$$

#### Exercise 4.1.2

Let  $G$  be a permutation group on the set  $A$  (i.e.,  $G \leq S_A$ ), let  $\sigma \in G$  and let  $a \in A$ . Prove that  $\sigma G_a \sigma^{-1} = G_{\sigma(a)}$ . Deduce that if  $G$  acts transitively on  $A$  then

$$\bigcap_{\sigma \in G} \sigma G_a \sigma^{-1} = 1.$$

**Solution.** Suppose  $\tau \in \sigma G_a \sigma^{-1}$ . We want to show that  $\tau \in G_{\sigma(a)}$ . Since  $\tau \in \sigma G_a \sigma^{-1}$ , there exists some  $\rho \in G_a$  such that  $\tau = \sigma \rho \sigma^{-1}$ . We have  $\tau \cdot \sigma(a) = (\sigma \rho \sigma^{-1}) \cdot \sigma(a) = \sigma \cdot (\rho \cdot a) = \sigma(a)$ , since  $\rho \in G_a$  implies  $\rho \cdot a = a$ . Thus,  $\tau \in G_{\sigma(a)}$ . Therefore, we have shown that  $\sigma G_a \sigma^{-1} \subseteq G_{\sigma(a)}$ . For the other direction, suppose  $\tau \in G_{\sigma(a)}$ . We want to show that  $\tau \in \sigma G_a \sigma^{-1}$ . We have  $\tau \cdot \sigma(a) = \sigma(a)$ . Applying  $\sigma^{-1}$  to both sides, we get  $\sigma^{-1}\tau \sigma \cdot a = a$ . This means that  $\sigma^{-1}\tau \sigma \in G_a$ , so  $\tau \in \sigma G_a \sigma^{-1}$ . Thus, we have shown that  $G_{\sigma(a)} \subseteq \sigma G_a \sigma^{-1}$ . Combining both inclusions, we conclude that  $\sigma G_a \sigma^{-1} = G_{\sigma(a)}$ .

Now, if  $G$  acts transitively on  $A$ , then for any  $b \in A$ , there exists some  $\sigma \in G$  such that  $b = \sigma(a)$ . From the first part, we have  $\sigma G_a \sigma^{-1} = G_b$ . The kernel of the action is the intersection of all stabilizers  $G_b$  for  $b \in A$ . Therefore, the kernel is

$$\bigcap_{b \in A} G_b = \bigcap_{\sigma \in G} \sigma G_a \sigma^{-1} = 1. \quad \blacksquare$$

#### Exercise 4.1.3

Assume that  $G$  is an abelian, transitive subgroup of  $S_A$ . Show that  $\sigma(a) \neq a$  for all  $\sigma \in G - \{1\}$  and all  $a \in A$ . Deduce that  $|G| = |A|$ . [Use the preceding exercise.]

**Solution.** Since  $G$  is abelian, then the conjugate of  $G_a$  is trivial, i.e.,  $\sigma G_a \sigma^{-1} = G_a$  for all  $\sigma \in G$ . From the previous exercise, we know that the kernel of this action is trivial. However, the intersection of all  $\sigma G_a \sigma^{-1}$  is just  $G_a$  so that  $G_a = 1$ . Then  $G_a$  is trivial for every  $a \in A$ , hence  $\sigma(a) \neq a$  for all  $\sigma \in G - \{1\}$  and all  $a \in A$ . It follows by Proposition 4.2 that  $|G|/|G_a| = |A|$ , hence  $|G| = |A|$  because  $G_a$  is trivial.  $\blacksquare$



**Exercise 4.1.4**

Let  $S_3$  act on the set  $\Omega$  of ordered pairs  $\{(i, j) \mid 1 \leq i, j \leq 3\}$  by  $\sigma((i, j)) = (\sigma(i), \sigma(j))$ . Find the orbits of  $S_3$  on  $\Omega$ . For each  $\sigma \in S_3$  find the cycle decomposition of  $\sigma$  under this action (i.e., find its cycle decomposition when  $\sigma$  is considered as an element of  $S_9$ —first fix a labeling of these nine ordered pairs). For each orbit  $O$  of  $S_3$  acting on these nine points, pick some  $a \in O$  and find the stabilizer of  $a$  in  $S_3$ .

**Solution.** Note that in  $\Omega$ , there are two types of ordered pairs: those with identical elements and those with distinct elements. In the former case, it is clear that any  $\sigma \in S_3$  will map such a pair to another pair with identical elements. In the latter case, it is easy to find some  $\sigma \in S_3$  that maps any ordered pair with distinct elements to any other such pair. We now label the elements of  $\Omega$  accordingly:

- |                |                |                |
|----------------|----------------|----------------|
| • $1 = (1, 1)$ | • $4 = (1, 2)$ | • $7 = (3, 1)$ |
| • $2 = (2, 2)$ | • $5 = (2, 1)$ | • $8 = (2, 3)$ |
| • $3 = (3, 3)$ | • $6 = (1, 3)$ | • $9 = (3, 2)$ |

We then have the two orbits

$$O_1 = \{(i, i) \mid i \in \{1, 2, 3\}\} \quad \text{and} \quad O_2 = \{(i, j) \mid i \neq j\}$$

of  $S_3$  acting on  $\Omega$ . Note that  $|O_1| = 3$  and  $|O_2| = 6$ . Now for each  $\sigma \in S_3$ , we find its cycle decomposition as an element of  $S_9$ . We describe how to compute one such cycle decomposition, and the rest follow similarly. Consider  $\sigma = (123) \in S_3$ . Then we have the following mappings:

- $(1, 1) \mapsto (2, 2) \mapsto (3, 3) \mapsto (1, 1)$ , which corresponds to the cycle  $(1\ 2\ 3)$  in  $S_9$ .
- $(1, 2) \mapsto (2, 3) \mapsto (3, 1) \mapsto (1, 2)$ , which corresponds to the cycle  $(4\ 8\ 7)$  in  $S_9$ .
- $(2, 1) \mapsto (3, 2) \mapsto (1, 3) \mapsto (2, 1)$ , which corresponds to the cycle  $(5\ 9\ 6)$  in  $S_9$ .

Combining these cycles, we find that the cycle decomposition of  $\sigma = (123)$  in  $S_9$  is  $(1\ 2\ 3)(4\ 8\ 7)(5\ 9\ 6)$ . The cycle decompositions for all elements of  $S_3$  acting on  $\Omega$  are as follows:

- |                                            |                                    |                                        |
|--------------------------------------------|------------------------------------|----------------------------------------|
| • $1 \in S_3$ is the same as $1 \in S_9$ . | • $(13): (1\ 3)(4\ 6)(5\ 7)(8\ 9)$ | • $(123): (1\ 2\ 3)(4\ 8\ 7)(5\ 9\ 6)$ |
| • $(12): (1\ 2)(4\ 5)(6\ 7)(8\ 9)$         | • $(23): (1\ 2)(2\ 3)(5\ 6)(7\ 8)$ | • $(132): (1\ 3\ 2)(4\ 7\ 8)(5\ 6\ 9)$ |

For  $O_1$ , we pick  $a = (1, 1)$ . The only  $\sigma \in S_3$  that fixes  $(1, 1)$  is the identity permutation and  $(2\ 3)$ , so the stabilizer of  $(1, 1)$  in  $S_3$  is  $\langle (2\ 3) \rangle$ . Moreover, we have  $|G_a||O_1| = 2 \cdot 3 = 6 = |S_3|$  as expected. For  $O_2$ , we pick  $a = (1, 2)$ . The only  $\sigma \in S_3$  that fixes  $(1, 2)$  is the identity permutation, so the stabilizer of  $(1, 2)$  in  $S_3$  is 1. Again,  $|G_a||O_2| = 1 \cdot 6 = 6 = |S_3|$ . ■

**Exercise 4.1.5**

For each of parts (a) and (b) repeat the preceding exercise but with  $S_3$  acting on the specified set:

- (a) the set of 27 triples  $\{(i, j, k) \mid 1 \leq i, j, k \leq 3\}$
- (b) the set  $\mathcal{P}(\{1, 2, 3\}) - \{\emptyset\}$  of all 7 nonempty subsets of  $\{1, 2, 3\}$

**Solution.**

- (a) We begin by classifying these set of triples. There are five types of triples:

- Type 1: Triples with all identical elements.
- Type 2: Triples whose two first coordinates are identical.
- Type 3: Triples whose first and last coordinates are identical.
- Type 4: Triples whose last two coordinates are identical.
- Type 5: Triples with all distinct elements.

Note that this is a correct classification. One may suggest to combine Types 2, 3, and 4 into a singular type. However, observe that no elements may be permuted such that the location of a pair of identical coordinates move to another one, i.e., there is no such permutation  $\sigma$  in  $S_3$  such that  $(1, 1, 2)$  maps to  $(1, 2, 2)$ . We now label the elements of  $\Omega$  lexicographically as follows, i.e., we begin by increasing the last coordinate, then increasing the middle coordinate, and finally increasing the first coordinate:

- |                 |                  |                  |
|-----------------|------------------|------------------|
| • 1 = (1, 1, 1) | • 10 = (2, 1, 1) | • 19 = (3, 1, 1) |
| • 2 = (1, 1, 2) | • 11 = (2, 1, 2) | • 20 = (3, 1, 2) |
| • 3 = (1, 1, 3) | • 12 = (2, 1, 3) | • 21 = (3, 1, 3) |
| • 4 = (1, 2, 1) | • 13 = (2, 2, 1) | • 22 = (3, 2, 1) |
| • 5 = (1, 2, 2) | • 14 = (2, 2, 2) | • 23 = (3, 2, 2) |
| • 6 = (1, 2, 3) | • 15 = (2, 2, 3) | • 24 = (3, 2, 3) |
| • 7 = (1, 3, 1) | • 16 = (2, 3, 1) | • 25 = (3, 3, 1) |
| • 8 = (1, 3, 2) | • 17 = (2, 3, 2) | • 26 = (3, 3, 2) |
| • 9 = (1, 3, 3) | • 18 = (2, 3, 3) | • 27 = (3, 3, 3) |

The classification yields the following orbits of  $S_3$  acting on  $\Omega$ :

- $O_1 = \{(i, i, i) \mid i \in \{1, 2, 3\}\}$  with order 3.
- $O_2 = \{(i, i, j) \mid i \neq j\}$  with order 6.
- $O_3 = \{(i, j, i) \mid i \neq j\}$  with order 6.
- $O_4 = \{(j, i, i) \mid i \neq j\}$  with order 6.
- $O_5 = \{(i, j, k) \mid i, j, k \text{ distinct}\}$  with order 6.

The cycle decompositions for all elements of  $S_3$  acting on  $\Omega$  are as follows:

- $1 \in S_3$  is the same as  $1 \in S_{27}$ .
- (1 2): (1 14)(2 13)(3 15)(4 11)(5 10)(6 12)(7 17)(8 16)(9 18)(19 23)(20 22)(21 24)(25 26)
- (1 3): (1 27)(2 26)(3 25)(4 24)(5 23)(6 22)(7 21)(8 20)(9 19)(10 18)(11 17)(12 16)(13 15)
- (2 3): (2 3)(4 7)(5 9)(6 8)(10 19)(11 21)(12 20)(13 25)(14 27)(15 26)(16 22)(17 24)(18 23)
- (1 2 3): (1 14 27)(2 15 25)(3 13 26)(4 17 21)(5 18 19)(6 16 20)(7 11 24)(8 12 22)(9 10 23)
- (1 3 2): (1 27 14)(2 25 15)(3 26 13)(4 21 17)(5 19 18)(6 20 16)(7 24 11)(8 22 12)(9 23 10)

For  $O_1$ , pick  $a = (1 1 1)$ . The elements that stabilize  $a$  are the identity permutation and (2 3), so the stabilizer of  $a$  in  $S_3$  is  $\langle (2 3) \rangle$ . We have  $|G_a||O_1| = 2 \cdot 3 = 6 = |S_3|$ . For the other orbits, note that  $S_3$  acts transitively on each of them, so the stabilizer of any chosen element in these orbits is trivial. For example, for  $O_2$ , pick  $a = (1 1 2)$ . The only element that stabilizes  $a$  is the identity permutation, so the stabilizer of  $a$  in  $S_3$  is 1. We have  $|G_a||O_2| = 1 \cdot 6 = 6 = |S_3|$ . The same logic applies to  $O_3, O_4$ , and  $O_5$ .

- (b) We begin by classifying the nonempty subsets of  $\{1, 2, 3\}$ . There are three types of subsets: 1-element subsets, 2-element subsets, and the 3-element subset. We now label the elements of  $\mathcal{P}(\{1, 2, 3\}) - \{\emptyset\}$  as follows:

- |           |              |              |                 |
|-----------|--------------|--------------|-----------------|
| • 1 = {1} | • 3 = {3}    | • 5 = {1, 3} | • 7 = {1, 2, 3} |
| • 2 = {2} | • 4 = {1, 2} | • 6 = {2, 3} |                 |

The classification yields the following orbits of  $S_3$  acting on  $\mathcal{P}(\{1, 2, 3\}) - \{\emptyset\}$ :

- $O_1 = \{\{1\}, \{2\}, \{3\}\}$  with order 3.
- $O_2 = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$  with order 3.
- $O_3 = \{\{1, 2, 3\}\}$  with order 1.

The cycle decompositions for all elements of  $S_3$  acting on  $\mathcal{P}(\{1, 2, 3\}) - \{\emptyset\}$  are as follows:

- $1 \in S_3$  is the same as  $1 \in S_7$ .
- (1 2): (1 2)(5 6)
- (1 3): (1 3)(4 6)
- (2 3): (2 3)(4 5)
- (1 2 3): (1 2 3)(4 6 5)
- (1 3 2): (1 3 2)(4 5 6)

For  $O_1$ , pick  $a = \{1\}$ . The only elements that stabilize  $a$  are the identity permutation and (2 3), so the stabilizer of  $a$  in  $S_3$  is  $\langle (2 3) \rangle$ . We have  $|G_a||O_1| = 2 \cdot 3 = 6 = |S_3|$ . For  $O_2$ , pick  $a = \{1, 2\}$ . The only elements that stabilize  $a$  are the identity permutation and (1 2), so the stabilizer of  $a$  in  $S_3$  is  $\langle (1 2) \rangle$ . We have  $|G_a||O_2| = 2 \cdot 3 = 6 = |S_3|$ . For  $O_3$ , note that  $S_3$  acts trivially on this orbit, so the stabilizer of  $\{1, 2, 3\}$  in  $S_3$  is all of  $S_3$ . We have  $|G_a||O_3| = 6 \cdot 1 = 6 = |S_3|$ . ■

**Exercise 4.1.6**

As in [Exercise 2.2.12](#), let  $R$  be the set of all polynomials with integer coefficients in the independent variables  $x_1, x_2, x_3, x_4$  and let  $S_4$  act on  $R$  by permuting the indices of the four variables:  $\sigma \cdot p(x_1, x_2, x_3, x_4) = p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)})$  for all  $\sigma \in S_4$ .

- Find the polynomials in the orbit of  $S_4$  on  $R$  containing  $x_1 + x_2$  (recall from [Exercise 2.2.12](#) that the stabilizer of this polynomial has order 4).
- Find the polynomials in the orbit of  $S_4$  on  $R$  containing  $x_1x_2 + x_3x_4$  (recall from [Exercise 2.2.12](#) that the stabilizer of this polynomial has order 8).
- Find the polynomials in the orbit of  $S_4$  on  $R$  containing  $(x_1 + x_2)(x_3 + x_4)$ .

**Solution.**

- Note that the size of the orbit is given by  $|S_4|/|G_{x_1+x_2}| = 24/4 = 6$ . The polynomials in the orbit of  $S_4$  on  $R$  containing  $x_1 + x_2$  are

$$x_1 + x_2, x_1 + x_3, x_1 + x_4, x_2 + x_3, x_2 + x_4, x_3 + x_4$$

- The size of the orbit is 3. The polynomials in the orbit of  $S_4$  on  $R$  containing  $x_1x_2 + x_3x_4$  are

$$x_1x_2 + x_3x_4, x_1x_3 + x_2x_4, x_1x_4 + x_2x_3$$

- Note that the stabilizer of  $(x_1 + x_2)(x_3 + x_4)$  has order 8 (it is the same as that of  $x_1x_2 + x_3x_4$ ). Thus, the size of the orbit is 3. The polynomials in the orbit of  $S_4$  on  $R$  containing  $(x_1 + x_2)(x_3 + x_4)$  are

$$(x_1 + x_2)(x_3 + x_4), (x_1 + x_3)(x_2 + x_4), (x_1 + x_4)(x_2 + x_3)$$

■

**(\*) Exercise 4.1.7**

Let  $G$  be a transitive permutation group on the finite set  $A$ . A *block* is a nonempty subset  $B$  of  $A$  such that for all  $\sigma \in G$  either  $\sigma(B) = B$  or  $\sigma(B) \cap B = \emptyset$  (here  $\sigma(B) = \{\sigma(b) \mid b \in B\}$ ).

- Prove that if  $B$  is a block containing the element  $a$  of  $A$ , then the set  $G_B$  defined by  $G_B = \{\sigma \in G \mid \sigma(B) = B\}$  is a subgroup of  $G$  containing  $G_a$ .
- Show that if  $B$  is a block and  $\sigma_1(B), \sigma_2(B), \dots, \sigma_n(B)$  are all the distinct images of  $B$  under the elements of  $G$ , then these form a partition of  $A$ .
- A (transitive) group  $G$  on a set  $A$  is said to be *primitive* if the only blocks in  $A$  are the trivial ones: the sets of size 1 and  $A$  itself. Show that  $S_4$  is primitive on  $A = \{1, 2, 3, 4\}$ . Show that  $D_8$  is not primitive as a permutation group on the four vertices of a square.
- Prove that the transitive group  $G$  is primitive on  $A$  if and only if for each  $a \in A$ , the only subgroups of  $G$  containing  $G_a$  are  $G_a$  and  $G$  (i.e.,  $G_a$  is a *maximal* subgroup of  $G$ , cf. [Exercise 2.4.16](#)). [Use part (a).]

**Solution.**

- We first show that  $G_B$  is a subgroup of  $G$ . Since  $1(B) = B$ , then  $1 \in G_B$ , hence it is nonempty. If  $\sigma, \tau \in G$  such that  $\sigma(B) = B$  and  $\tau(B) = B$ , we note that  $\tau^{-1}(B) = B$  as well, hence  $\sigma, \tau^{-1} \in G_B$ . Then  $(\sigma\tau^{-1})(B) = \sigma(\tau^{-1}(B)) = \sigma(B) = B$ , so  $\sigma\tau^{-1} \in G_B$ . Hence  $G_B \leq G$ .

Now, let  $a \in B$ . If  $\sigma \in G_a$ , then  $\sigma(a) = a$ . Since  $a \in B$ , then  $\sigma(a) \in \sigma(B)$ . But  $\sigma(a) = a \in B$ , so  $\sigma(B) \cap B \neq \emptyset$ . By the definition of a block, this implies that  $\sigma(B) = B$ , hence  $\sigma \in G_B$ . Therefore, we have shown that  $G_a \leq G_B$ .

- Let  $B$  be a block and let  $\sigma_1(B), \sigma_2(B), \dots, \sigma_n(B)$  be all the distinct images of  $B$  under the elements of  $G$ . By transitivity of  $G$  on  $A$ , then for some  $a \in A$ , there exists  $b \in B$  such that  $\sigma(b) = a$ . Then  $a \in \sigma(B)$ , and since  $\sigma(B)$  is one of the sets  $\sigma_1(B), \sigma_2(B), \dots, \sigma_n(B)$ , it follows that

$$\bigcup_{i=1}^n \sigma_i(B) = A.$$

Suppose there existed  $i \neq j$  where  $\sigma_i(B) \cap \sigma_j(B) \neq \emptyset$ . Then there are  $b, b' \in B$  such that  $\sigma_i(b) = \sigma_j(b')$ , or  $(\sigma_j^{-1}\sigma_i)(b) = b'$ , hence  $\sigma_j^{-1}\sigma_i(B) \cap B \neq \emptyset$ . Since  $B$  is a block, then  $\sigma_j^{-1}\sigma_i(B) = B$ , hence  $\sigma_i(B) = \sigma_j(B)$ ,

contradicting the assumption that they are distinct. Therefore, the sets  $\sigma_1(B), \sigma_2(B), \dots, \sigma_n(B)$  are disjoint, and we conclude that they form a partition of  $A$ .

- (c) Let  $B \subset A$ , i.e., a proper subset of  $A$ . If  $|B| = 1$ , then it is trivial. Suppose  $|B| > 1$ , and without loss of generality, let  $1 \in B$  and some  $1 \neq i \in B$  as well. Then there exists some  $\sigma \in S_4$  such that  $\sigma(1) = 1$  and  $\sigma(i) = j$  for  $j \notin B$ . Then  $\sigma(B) \cap B \neq \emptyset$  so that  $\sigma(B) = B$ , hence  $j \in B$ . We may repeat this to show that  $B = A$ , contradicting the assumption that  $B$  is a proper subset of  $A$ . Therefore, the only blocks in  $A$  are the trivial ones, and  $S_4$  is primitive on  $A$ .

Now, consider  $D_8$  acting on the four vertices of a square. Let  $B$  be the set containing the two opposite vertices. Then for any  $\sigma \in D_8$ , either  $\sigma(B) = B$  or  $\sigma(B) \cap B = \emptyset$ . Thus,  $B$  is a nontrivial block, and  $D_8$  is not primitive in its action on the four vertices of a square.

- (d) ( $\Rightarrow$ ) If  $G$  is primitive on  $A$ . Let  $H \leq G$  such that  $G_a \leq H \leq G$ , and let  $B$  be the orbit of  $H$  so that  $B = H \cdot a$ . Since  $1 \in H$  because  $H \leq G$ , then  $1 \cdot a = a \in B$ . By part (a), we know that  $a \in B$  implies  $G_a \leq G_B$  so that  $B$  is now a block containing  $a$ . Primitivity of  $G$  implies that  $B$  is either  $\{a\}$  or  $A$  itself. If  $B = \{a\}$ , then for any  $\sigma \in H$ , we have  $\sigma \cdot a \in B$ , hence  $\sigma \cdot a = a$ , so  $\sigma \in G_a$ . Therefore,  $H \leq G_a$ , and since  $G_a \leq H$ , then  $H = G_a$ . If  $B = A$ , then for any  $b \in A$ , there exists some  $\sigma \in H$  such that  $\sigma \cdot a = b$ . Thus, for any  $b \in A$ , there exists some  $\sigma \in H$  such that  $\sigma(b) = a$ , hence  $H$  is transitive on  $A$ . Therefore,  $H = G$ . We have shown that the only subgroups of  $G$  containing  $G_a$  are  $G_a$  and  $G$ .

( $\Leftarrow$ ) Suppose now that  $G_a$  is maximal in  $G$ . Let  $B$  be a block of  $A$  that contains some  $a \in A$ . By part (a), we know that  $G_a \leq G_B \leq G$ . Maximality of  $G_a$  implies that either  $G_B = G_a$  or  $G_B = G$ . If  $G_B = G_a$ , then for any  $\sigma \in G_B$ , we have  $\sigma \in G_a$ , hence  $\sigma(a) = a$ . Since  $a \in B$ , then  $\sigma(a) \in \sigma(B)$ , so  $\sigma(a) \in B$ . Therefore,  $\sigma(a) = a \in B$ , and it follows that  $\sigma(B) \cap B \neq \emptyset$ . By the definition of a block, this implies that  $\sigma(B) = B$ . Thus, for any  $\sigma \in G_B$ , we have  $\sigma(B) = B$ , so  $B = \{a\}$ . If  $G_B = G$ , then for any  $b \in A$ , there exists some  $\sigma \in G$  such that  $\sigma(a) = b$ . Since  $\sigma \in G_B$ , then  $\sigma(B) = B$ , hence  $b \in B$ . Therefore,  $B = A$ . We have shown that the only blocks in  $A$  are the trivial ones, so  $G$  is primitive on  $A$ . ■

#### Exercise 4.1.8

A transitive permutation group  $G$  on a set  $A$  is called *doubly transitive* if for any (hence all)  $a \in A$  the subgroup  $G_a$  is transitive on the set  $A - \{a\}$ .

- (a) Prove that  $S_n$  is doubly transitive on  $\{1, 2, \dots, n\}$  for all  $n \geq 2$ .  
 (b) Prove that a doubly transitive group is primitive. Deduce that  $D_8$  is not doubly transitive in its action on the 4 vertices of a square.

#### Solution.

- (a) Let  $G_a$  be the stabilizer of  $a \in \{1, 2, \dots, n\}$ . Note that  $G_a$  is the set of all permutations that fix  $a$  and permute the remaining  $n - 1$  elements so that  $G_a \cong S_{n-1}$ . Since  $S_{n-1}$  is transitive on the set  $\{1, 2, \dots, n\} - \{a\}$  for all  $n - 1 \geq 1$ , then  $G_a$  is transitive on  $A - \{a\}$  for all  $n \geq 2$ . Therefore,  $S_n$  is doubly transitive on  $\{1, 2, \dots, n\}$  for all  $n \geq 2$ .  
 (b) Let  $G$  be double transitive and let  $B$  be a block containing some  $a \in A$ . Take  $b \in B$  such that  $b \neq a$ . Double transitivity of  $G$  implies that there exists some  $\sigma \in G_a$  such that  $\sigma(b) = c$  for any  $c \in A - \{a\}$ . Since  $\sigma \in G_a$ , then  $\sigma(B) = B$ , hence  $c \in B$ . Therefore,  $B = A$ , and we conclude that the only blocks in  $A$  are the trivial ones. Thus,  $G$  is primitive on  $A$ . Since  $D_8$  is not primitive in its action on the four vertices of a square (as shown in part (c) of the previous exercise), then it is not doubly transitive. ■

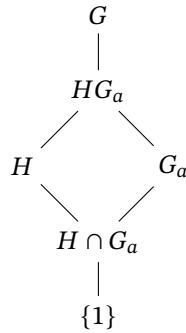
(\*) **Exercise 4.1.9**

Assume  $G$  acts transitively on the finite set  $A$  and let  $H$  be a normal subgroup of  $G$ . Let  $O_1, O_2, \dots, O_r$  be the distinct orbits of  $H$  on  $A$ .

- (a) Prove that  $G$  permutes the sets  $O_1, O_2, \dots, O_r$ , in the sense that for each  $g \in G$  and each  $i \in \{1, \dots, r\}$  there is a  $j$  such that  $gO_i = O_j$ , where  $gO = \{g \cdot a \mid a \in O\}$  (i.e., in the notation of Exercise 7 the sets  $O_1, \dots, O_r$  are blocks). Prove that  $G$  is transitive on  $\{O_1, \dots, O_r\}$ . Deduce that all orbits of  $H$  on  $A$  have the same cardinality.
- (b) Prove that if  $a \in O_1$  then  $|O_1| = |H : H \cap G_a|$  and prove that  $r = |G : HG_a|$ . [Draw the sublattice describing the Second Isomorphism Theorem for the subgroups  $H$  and  $G_a$  of  $G$ . Note that  $H \cap G_a = H_a$ .]

**Solution.**

- (a) Let  $g \in G$  and  $O_i$  be an orbit of  $H$  on  $A$ . Let  $a \in O_i$  so that  $O_i = H \cdot a = \{h \cdot a \mid h \in H\}$ . Consider the set  $gO_i = \{g \cdot (h \cdot a) \mid h \in H\}$ . Since  $H$  is normal in  $G$ , then for any  $h \in H$ , we have  $ghg^{-1} \in H$ . Then  $gO_i = H \cdot (g \cdot a)$ , which is the orbit of  $H$  containing  $g \cdot a$ . Therefore, there exists some  $j$  such that  $gO_i = O_j$ .
- To show that  $G$  is transitive on  $\{O_1, \dots, O_r\}$ , let  $O_i$  and  $O_j$  be any two orbits of  $H$  on  $A$ . Since  $G$  is transitive on  $A$ , then for some  $a \in O_i$  and  $b \in O_j$ , there exists some  $g \in G$  such that  $g \cdot a = b$ . Then  $gO_i$  is the orbit of  $H$  containing  $b$ , hence  $gO_i = O_j$ . Therefore,  $G$  is transitive on  $\{O_1, \dots, O_r\}$ . Since  $G$  is transitive on  $\{O_1, \dots, O_r\}$ , then all orbits of  $H$  on  $A$  have the same cardinality.
- (b) Suppose  $a \in O_1$ . By Proposition 4.2, we have  $|O_1| = |H : H_a|$ . Since  $H_a = H \cap G_a$ , then  $|O_1| = |H : H \cap G_a|$ . Moreover, the orbits of  $H$  partition  $A$  and have the same size by the previous solution, so we have  $|A| = r|O_1|$ . By applying Proposition 4.2 again, we have  $|A| = |G : G_a|$ . Therefore,  $|G : G_a| = r|O_1| = r|H : H \cap G_a|$ . By the Second Isomorphism Theorem, we also have that  $H/(H \cap G_a) \cong HG_a/G_a$  so that  $|H : H \cap G_a| = |HG_a : G_a|$ . Now recall by [Exercise 3.2.11](#) that  $|G : K| = |G : H||H : K|$  for subgroups  $K \leq H \leq G$ . Applying this to the subgroups  $G_a \leq HG_a \leq G$ , we have  $|G : G_a| = |G : HG_a||HG_a : G_a| = |G : HG_a||H : H \cap G_a|$ . We then have  $r|H : H \cap G_a| = |G : HG_a||H : H \cap G_a|$ , or  $r = |G : HG_a|$ . Moreover, the sublattice is given as follows:



■

(\*) **Exercise 4.1.10**

Let  $H$  and  $K$  be subgroups of the group  $G$ . For each  $x \in G$  define the  $HK$  double coset of  $x$  in  $G$  to be the set  $HxK = \{h x k \mid h \in H, k \in K\}$ .

- (a) Prove that  $HxK$  is the union of the left cosets  $x_1K, \dots, x_nK$  where  $\{x_1K, \dots, x_nK\}$  is the orbit containing  $xK$  of  $H$  acting by left multiplication on the set of left cosets of  $K$ .
- (b) Prove that  $HxK$  is a union of right cosets of  $H$ .
- (c) Show that  $HxK$  and  $HyK$  are either the same set or are disjoint for all  $x, y \in G$ . Show that the set of  $HK$  double cosets partitions  $G$ .
- (d) Prove that  $|HxK| = |K| \cdot |H : H \cap xKx^{-1}|$ .
- (e) Prove that  $|HxK| = |H| \cdot |K : K \cap x^{-1}Hx|$ .

**Solution.**

- (a) Let  $\mathcal{O}$  be the orbit containing  $xK$  of  $H$  acting by left multiplication on the set of left cosets of  $K$ . Then  $\mathcal{O} = \{hxK \mid h \in H\}$ , and let  $\{x_1K, \dots, x_nK\}$  be the distinct left cosets in  $\mathcal{O}$ . Let  $\mathcal{K}$  be the union of these left cosets, i.e.,  $\mathcal{K} = \bigcup_1^n x_iK$ .

Now pick  $y \in HxK$ . Then there exist  $h \in H$  and  $k \in K$  such that  $y = h x k$ . Note that  $h x K \in \mathcal{O}$ , so there exists some  $i$  such that  $h x K = x_iK$ . Therefore,  $y = h x k \in x_iK \subseteq \mathcal{K}$ , hence  $HxK \subseteq \mathcal{K}$ . If  $z \in \mathcal{K}$ , then there exists some  $i$  and some  $k' \in K$  such that  $z = x_i k'$ . Since  $x_iK \in \mathcal{O}$ , then there exists some  $h' \in H$  such that  $x_iK = h' x K$ , hence  $x_i = h' x k''$  for some  $k'' \in K$ . Therefore,  $z = x_i k' = h' x k k'' \in HxK$ , so  $\mathcal{K} \subseteq HxK$ . We have shown that  $HxK = \mathcal{K}$ , i.e.,  $HxK$  is the union of the left cosets  $x_1K, \dots, x_nK$ .

- (b) The proof is similar to that of part (a).
- (c) Let  $x, y \in G$ . Suppose  $HxK \cap HyK \neq \emptyset$ . Then there exist  $h_1, h_2 \in H$  and  $k_1, k_2 \in K$  such that  $h_1 x k_1 = h_2 y k_2$ . Rearranging, we have  $y = h_2^{-1} h_1 x k_1 k_2^{-1}$ , hence  $y \in HxK$ . Therefore,  $HyK \subseteq HxK$ . By symmetry, we also have  $HxK \subseteq HyK$ , so  $HxK = HyK$ . We have shown that  $HxK$  and  $HyK$  are either the same set or are disjoint for all  $x, y \in G$ . Since every element of  $G$  is in some double coset of the form  $HxK$ , then the set of  $HK$  double cosets partitions  $G$ .
- (d) Recall that the orbit of  $xK$  under this action is  $\mathcal{O} = \{hxK \mid h \in H\}$ . By Proposition 4.2, we have  $|\mathcal{O}| = |H : H_{xK}|$ , where  $H_{xK}$  is the stabilizer of  $xK$  in  $H$ . Observe that  $H_{xK} = \{h \in H \mid hxK = xK\}$ . Then  $hxK = xK$  implies that there exists some  $k \in K$  such that  $h x k = x$ , or equivalently,  $h = x k x^{-1}$ . Therefore,  $H_{xK} = H \cap xKx^{-1}$ . By part (a), we have  $|HxK| = |\mathcal{O}| \cdot |K| = |H : H \cap xKx^{-1}| \cdot |K|$ .
- (e) Similar to part (d). ■

## 4.2 Groups Acting on Themselves by Left Multiplication—Cayley's Theorem

### Exercise 4.2.1

Let  $G$  be  $\{1, a, b, c\}$ , the Klein 4-group whose group table is written out in Section 2.5.

- (a) Label  $1, a, b, c$  with the integers  $1, 2, 4, 3$ , respectively, and prove that under the left regular representation of  $G$  into  $S_4$  the nonidentity elements are mapped as follows:

$$a \mapsto (1\ 2)(3\ 4), \quad b \mapsto (1\ 4)(2\ 3), \quad c \mapsto (1\ 3)(2\ 4).$$

- (b) Relabel  $1, a, b, c$  as  $1, 4, 2, 3$ , respectively, and compute the image of each element of  $G$  under the left regular representation of  $G$  into  $S_4$ . Show that the image of  $G$  in  $S_4$  under this labelling is the same *subgroup* as the image of  $G$  in part (a) (even though the nonidentity elements individually map to different permutations under the two different labellings).

### Solution.

- (a) We have  $a \cdot 1 = a$  so that  $\sigma_a(1) = 2$ . Similarly,  $\sigma_a(2) = 1$ . We also have  $a \cdot b = c$  so that  $\sigma_a(4) = 3$ , and  $\sigma_a(3) = 4$ , and we have  $a \mapsto (1\ 2)(3\ 4)$ . The others are computed similarly.
- (b) Again, we have  $a \cdot 1 = a$  so that  $\sigma_a(1) = 4$ . Similarly,  $\sigma_a(4) = 1$ . We also have  $a \cdot b = c$  so that  $\sigma_a(2) = 3$ , and  $\sigma_a(3) = 2$ , and we have  $a \mapsto (1\ 4)(2\ 3)$ . For  $b$ , we have  $b \mapsto (1\ 2)(3\ 4)$ , and for  $c$ , we have  $c \mapsto (1\ 3)(2\ 4)$ . The image of  $G$  in  $S_4$  under this labelling is  $\{1, (1\ 4)(2\ 3), (1\ 2)(3\ 4), (1\ 3)(2\ 4)\}$ , which is the same subgroup as in part (a). ■

### Exercise 4.2.2

List the elements of  $S_3$  as  $1, (1\ 2), (2\ 3), (1\ 3), (1\ 2\ 3), (1\ 3\ 2)$  and label these with the integers  $1, 2, 3, 4, 5, 6$  respectively. Exhibit the image of each element of  $S_3$  under the left regular representation of  $S_3$  into  $S_6$ .

**Solution.** Consider  $(1\ 2)$ . Then we have the following computations:

|                                   |                          |
|-----------------------------------|--------------------------|
| $(1\ 2) \cdot 1 = (1\ 2)$         | $\sigma_{(1\ 2)}(1) = 2$ |
| $(1\ 2) \cdot (1\ 2) = 1$         | $\sigma_{(1\ 2)}(2) = 1$ |
| $(1\ 2) \cdot (2\ 3) = (1\ 3\ 2)$ | $\sigma_{(1\ 2)}(4) = 5$ |
| $(1\ 2) \cdot (1\ 3) = (1\ 2\ 3)$ | $\sigma_{(1\ 2)}(3) = 6$ |
| $(1\ 2) \cdot (1\ 2\ 3) = (2\ 3)$ | $\sigma_{(1\ 2)}(5) = 3$ |
| $(1\ 2) \cdot (1\ 3\ 2) = (1\ 3)$ | $\sigma_{(1\ 2)}(6) = 4$ |

Thus, we have  $(1\ 2) \mapsto (1\ 2)(3\ 6\ 4\ 5)$ . We calculate that  $(2\ 3) \mapsto (1\ 3\ 5\ 2)(4\ 6)$ . Since  $(1\ 2)(2\ 3) = (1\ 2\ 3)$ , then

$$(1\ 2\ 3) \mapsto (1\ 2)(3\ 6\ 4\ 5)(1\ 3\ 5\ 2)(4\ 6) = (1\ 5\ 6)(2\ 4\ 3)$$

Continuing in this way, we find that the images of the elements of  $S_3$  under the left regular representation are as follows:

$$\begin{aligned} 1 &\mapsto 1 \\ (1\ 2) &\mapsto (1\ 2)(3\ 6\ 4\ 5) \\ (2\ 3) &\mapsto (1\ 3\ 5\ 2)(4\ 6) \\ (1\ 3) &\mapsto (1\ 4)(2\ 6)(3\ 5) \\ (1\ 2\ 3) &\mapsto (1\ 5\ 6)(2\ 4\ 3) \\ (1\ 3\ 2) &\mapsto (1\ 6\ 5)(2\ 3\ 4) \end{aligned}$$

■

**Exercise 4.2.3**

Let  $r$  and  $s$  be the usual generators for the dihedral group of order 8.

- (a) List the elements of  $D_8$  as  $1, r, r^2, r^3, s, sr, sr^2, sr^3$  and label these with the integers  $1, 2, \dots, 8$  respectively. Exhibit the image of each element of  $D_8$  under the left regular representation of  $D_8$  into  $S_8$ .
- (b) Relabel this same list of elements of  $D_8$  with the integers  $1, 3, 5, 7, 2, 4, 6, 8$  respectively and recompute the image of each element of  $D_8$  under the left regular representation with respect to this new labelling. Show that the two subgroups of  $S_8$  obtained in parts (a) and (b) are different.

**Solution.**

- (a) We calculate  $\sigma_r = (1\ 2\ 3\ 4)(5\ 8\ 7\ 6)$  and  $\sigma_s = (1\ 5)(2\ 6)(3\ 7)(4\ 8)$ . Continuing in this way, we find that the images of the elements of  $D_8$  under the left regular representation are as follows:

$$\begin{array}{ll} 1 \mapsto 1 & s \mapsto (1\ 5)(2\ 6)(3\ 7)(4\ 8) \\ r \mapsto (1\ 2\ 3\ 4)(5\ 8\ 7\ 6) & sr \mapsto (1\ 6)(2\ 7)(3\ 8)(4\ 5) \\ r^2 \mapsto (1\ 3)(2\ 4)(5\ 7)(6\ 8) & sr^2 \mapsto (1\ 7)(2\ 8)(3\ 5)(4\ 6) \\ r^3 \mapsto (1\ 4\ 3\ 2)(5\ 6\ 7\ 8) & sr^3 \mapsto (1\ 8)(2\ 5)(3\ 6)(4\ 7) \end{array}$$

- (b) The images of  $D_8$  under the left regular representation with respect to this new labelling are as follows:

$$\begin{array}{ll} 1 \mapsto 1 & s \mapsto (1\ 2)(3\ 4)(5\ 6)(7\ 8) \\ r \mapsto (1\ 3\ 5\ 7)(2\ 8\ 6\ 4) & sr \mapsto (1\ 4)(2\ 7)(3\ 6)(5\ 8) \\ r^2 \mapsto (1\ 5)(3\ 7)(2\ 6)(4\ 8) & sr^2 \mapsto (1\ 6)(2\ 5)(3\ 8)(4\ 7) \\ r^3 \mapsto (1\ 7\ 5\ 3)(2\ 4\ 6\ 8) & sr^3 \mapsto (1\ 8)(2\ 3)(4\ 5)(6\ 7) \end{array}$$

Clearly, the two subgroups of  $S_8$  obtained in parts (a) and (b) are different since, for example, the image of  $r$  in part (a) is a product of two 4-cycles while the image of  $r$  in part (b) is a product of two 4-cycles but with different elements. ■

**Exercise 4.2.4**

Use the left regular representation of  $Q_8$  to produce two elements of  $S_8$  which generate a subgroup of  $S_8$  isomorphic to the quaternion group  $Q_8$ .

**Solution.** At minimum, we have that  $Q_8 = \langle i, j \rangle$  (any 2 elements of order 4 can be used), so we compute the images of  $i$  and  $j$  under the left regular representation. We label  $Q_8$  as  $1, -1, i, -i, j, -j, k, -k$  and label these with the integers  $1, 2, \dots, 8$  respectively. We find that

$$\begin{aligned} i &\mapsto \sigma_i = (1\ 3\ 4\ 2)(5\ 7)(6\ 8) \\ j &\mapsto \sigma_j = (1\ 5\ 2\ 6)(3\ 8)(4\ 7) \end{aligned}$$

Hence  $Q_8 \cong \langle \sigma_i, \sigma_j \rangle \leq S_8$ . ■



**Exercise 4.2.5**

Let  $r$  and  $s$  be the usual generators for the dihedral group of order 8 and let  $H = \langle s \rangle$ . List the left cosets of  $H$  in  $D_8$  as  $1H, rH, r^2H$  and  $r^3H$ .

- Label these cosets with the integers 1, 2, 3, 4, respectively. Exhibit the image of each element of  $D_8$  under the representation  $\pi_H$  of  $D_8$  into  $S_4$  obtained from the action of  $D_8$  by left multiplication on the set of 4 left cosets of  $H$  in  $D_8$ . Deduce that this representation is faithful (i.e., the elements of  $S_4$  obtained form a subgroup isomorphic to  $D_8$ ).
- Repeat part (a) with the list of cosets relabelled by the integers 1, 3, 2, 4, respectively. Show that the permutations obtained from this labelling form a subgroup of  $S_4$  that is different from the subgroup obtained in part (a).
- Let  $K = \langle sr \rangle$ , list the cosets of  $K$  in  $D_8$  as  $1K, rK, r^2K$  and  $r^3K$ , and label these with the integers 1, 2, 3, 4. Prove that, with respect to this labelling, the image of  $D_8$  under the representation  $\pi_K$  obtained from left multiplication on the cosets of  $K$  is the same *subgroup* of  $S_4$  as in part (a) (even though the subgroups  $H$  and  $K$  are different and some of the elements of  $D_8$  map to different permutations under the two homomorphisms).

**Solution.**

- (a) We have  $\pi_H(r) = (1\ 2\ 3\ 4)$  and  $\pi_H(s) = (2\ 4)$ . We continue in this way to find that the images of the elements of  $D_8$  under  $\pi_H$  are:

$$\begin{array}{ll} \pi_H(1) = 1 & \pi_H(s) = (2\ 4) \\ \pi_H(r) = (1\ 2\ 3\ 4) & \pi_H(sr) = (1\ 2)(3\ 4) \\ \pi_H(r^2) = (1\ 3)(2\ 4) & \pi_H(sr^2) = (1\ 3) \\ \pi_H(r^3) = (1\ 4\ 3\ 2) & \pi_H(sr^3) = (1\ 4)(2\ 3) \end{array}$$

so that the subgroup  $\langle \pi_H(r), \pi_H(s) \rangle \leq S_4$  is isomorphic to  $D_8$ . Since the kernel of  $\pi_H$  is trivial, then the representation is faithful.

- (b) The new labelling affords the permutations  $\pi_H(r) = (1\ 3\ 2\ 4)$  and  $\pi_H(s) = (3\ 4)$ . We have the following images:

$$\begin{array}{ll} \pi_H(1) = 1 & \pi_H(s) = (3\ 4) \\ \pi_H(r) = (1\ 3\ 2\ 4) & \pi_H(sr) = (1\ 3)(2\ 4) \\ \pi_H(r^2) = (1\ 2)(3\ 4) & \pi_H(sr^2) = (1\ 2) \\ \pi_H(r^3) = (1\ 4\ 2\ 3) & \pi_H(sr^3) = (1\ 4)(2\ 3) \end{array}$$

Moreover, the subgroup is different from that in part (a) since, for example, the image of  $r$  in part (a) is  $(1\ 2\ 3\ 4)$  while the image of  $r$  in this part is  $(1\ 3\ 2\ 4)$ .

- (c) Under  $\pi_K$ , we have  $\pi_K(r) = (1\ 2\ 3\ 4)$  and  $\pi_K(s) = (1\ 2)(3\ 4)$ . With respect to this labeling, we have the subgroup  $\widehat{K} = \langle (1\ 2\ 3\ 4), (1\ 2)(3\ 4) \rangle$ . Let  $\widehat{H}$  be the subgroup obtained in part (a). Note that  $(1\ 2\ 3\ 4)$  is contained in both  $\widehat{H}$  and  $\widehat{K}$ . Moreover, we have following:

$$(1\ 2\ 3\ 4)(2\ 4)(1\ 4\ 3\ 2) = (1\ 2)(3\ 4) \in \widehat{H} \quad \text{and} \quad (1\ 2\ 3\ 4)^2(1\ 2)(3\ 4) = (2\ 4) \in \widehat{K}$$

so that both subgroups contain the other generator of the other subgroup. Therefore,  $\widehat{H} = \widehat{K}$ . ■

**Exercise 4.2.6**

Let  $r$  and  $s$  be the usual generators for the dihedral group of order 8 and let  $N = \langle r^2 \rangle$ . List the left cosets of  $N$  in  $D_8$  as  $1N, rN, sN$  and  $srN$ . Label these cosets with the integers 1, 2, 3, 4 respectively. Exhibit the image of each element of  $D_8$  under the representation  $\pi_N$  of  $D_8$  into  $S_4$  obtained from the action of  $D_8$  by left multiplication on the set of 4 left cosets of  $N$  in  $D_8$ . Deduce that this representation is not faithful and prove that  $\pi_N(D_8)$  is isomorphic to the Klein 4-group.

**Solution.** We have  $\pi_N(r) = (1\ 2)(3\ 4)$  and  $\pi_N(s) = (1\ 3)(2\ 4)$ . Continuing in this way, we find that the images of the elements of  $D_8$  under  $\pi_N$  are:

$$\begin{array}{ll} \pi_N(1) = 1 & \pi_N(s) = (1\ 3)(2\ 4) \\ \pi_N(r) = (1\ 2)(3\ 4) & \pi_N(sr) = (1\ 4)(2\ 3) \\ \pi_N(r^2) = 1 & \pi_N(sr^2) = (1\ 3)(2\ 4) \\ \pi_N(r^3) = (1\ 2)(3\ 4) & \pi_N(sr^3) = (1\ 4)(2\ 3) \end{array}$$

Since  $\ker(\pi_N)$  contains the nontrivial element  $r^2$ , then the representation is not faithful. Moreover, we have  $\pi_N(D_8) = \{1, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$ , which is isomorphic to the Klein 4-group. ■

**Exercise 4.2.7**

Let  $Q_8$  be the quaternion group of order 8.

- Prove that  $Q_8$  is isomorphic to a subgroup of  $S_8$ .
- Prove that  $Q_8$  is not isomorphic to a subgroup of  $S_n$  for any  $n \leq 7$ . [If  $Q_8$  acts on any set  $A$  of order  $\leq 7$  show that the stabilizer of any point  $a \in A$  must contain the subgroup  $\langle -1 \rangle$ .]

**Solution.**

- This is done in [Exercise 4.2.4](#).
- Let  $Q_8$  act on a set  $A$  of order  $n \leq 7$ . For any  $a \in A$ , consider the orbit  $O_a$  of  $a$  under this action. By Proposition 4.2, we have  $|O_a| = |Q_8 : (Q_8)_a|$ , where  $(Q_8)_a$  is the stabilizer of  $a$  in  $Q_8$ . Since  $|O_a|$  divides  $|A| \leq 7$ , then  $|O_a|$  must be 1, 2, 4, or 7. However, since  $Q_8$  has no subgroup of index 7, then  $|O_a|$  cannot be 7. If  $|O_a| = 4$ , then  $(Q_8)_a$  has order 2, but the only subgroup of order 2 in  $Q_8$  is  $\langle -1 \rangle$ . If  $|O_a| = 2$ , then  $(Q_8)_a$  has order 4, and the only subgroups of order 4 in  $Q_8$  are  $\langle i \rangle$ ,  $\langle j \rangle$ , and  $\langle k \rangle$ , all of which contain  $\langle -1 \rangle$ . If  $|O_a| = 1$ , then  $(Q_8)_a = Q_8$ , which also contains  $\langle -1 \rangle$ . Therefore, in all cases, the stabilizer  $(Q_8)_a$  contains  $\langle -1 \rangle$ . Since this holds for all  $a \in A$ , then  $\langle -1 \rangle$  is contained in the kernel of the action. The action is not faithful, hence  $Q_8$  cannot be isomorphic to a subgroup of  $S_n$  for any  $n \leq 7$ . ■

**Exercise 4.2.8**

Prove that if  $H$  has finite index  $n$  then there is a normal subgroup  $K$  of  $G$  with  $K \leq H$  and  $|G : K| \leq n!$ .

**Solution.** Let  $G$  act by left multiplication on the set of left cosets of  $H$  in  $G$ . Then we have the homomorphism  $\varphi : G \rightarrow S_n$ , which is the permutation representation of  $G$  on  $G/H$ . Let  $K = \ker(\varphi) \subseteq H$ , since  $g \in K$  if and only if  $gH = H$ . By the First Isomorphism Theorem, we have  $G/K \cong \varphi(G) \leq S_n$ , so that  $|G : K| = |G/K| = |\varphi(G)|$  divides  $n!$ . Therefore,  $|G : K| \leq n!$ . ■

**Exercise 4.2.9**

Prove that if  $p$  is a prime and  $G$  is a group of order  $p^\alpha$  for some  $\alpha \in \mathbb{Z}^+$ , then every subgroup of index  $p$  is normal in  $G$ . Deduce that every group of order  $p^2$  has a normal subgroup of order  $p$ .

**Solution.** Let  $H$  be a subgroup of  $G$  with index  $p$ . Then  $|G : H| = p$ , so the action of  $G$  on the left cosets of  $H$  in  $G$  gives a homomorphism  $\varphi : G \rightarrow S_p$ . Since  $|G| = p^\alpha$ , then by Lagrange's Theorem, the order of  $\varphi(G)$  divides  $p^\alpha$ . However, the only subgroups of  $S_p$  whose order divides  $p^\alpha$  are the trivial group and groups of order  $p$ . Since  $\varphi(G)$  acts transitively on the  $p$  cosets of  $H$ , then  $\varphi(G)$  cannot be trivial. Therefore,  $|\varphi(G)| = p$ , which is prime, so  $\varphi(G)$  is cyclic and hence abelian. The kernel of  $\varphi$  is a normal subgroup of  $G$  contained in  $H$ . Since the image  $\varphi(G)$  has order  $p$ , then by the First Isomorphism Theorem, we have  $|G : \ker(\varphi)| = p$ . But since  $\ker(\varphi) \subseteq H$  and both have index  $p$ , then  $\ker(\varphi) = H$ . Therefore,  $H$  is normal in  $G$ .

To deduce that every group of order  $p^2$  has a normal subgroup of order  $p$ , let  $G$  be a group of order  $p^2$ . By Cauchy's Theorem, there exists an element of order  $p$  in  $G$ , which generates a subgroup  $H$  of order  $p$ . Since the index of  $H$  in  $G$  is  $p$ , by the previous result,  $H$  is normal in  $G$ . ■

**Exercise 4.2.10**

Prove that every non-abelian group of order 6 has a nonnormal subgroup of order 2. Use this to classify groups of order 6. [Produce an injective homomorphism into  $S_3$ .]

**Solution.** Note that  $2 \mid 6$  and  $3 \mid 6$ . By Cauchy's Theorem, there exists subgroups  $H$  and  $K$  of orders 2 and 3 respectively. Let  $H = \{1, h\}$ , suppose it is normal, and let  $g \in G - H$ . Since  $H \trianglelefteq G$ , then  $gHg^{-1} = H$ . In particular,  $ghg^{-1} \in H$ , so either  $ghg^{-1} = 1$  or  $ghg^{-1} = h$ . Since the first case implies  $h = 1$ , it must be that  $ghg^{-1} = h$ , or  $gh = hg$ . Then  $h$  commutes with every element. Moreover, we may use Exercise 3.3.3 to conclude that  $G = HK$  since  $H \trianglelefteq G$ . Since  $K$  is also abelian as it is cyclic, then  $G = hk$  for  $h \in H$  and  $k \in K$ , implying that  $G$  is abelian, a contradiction. Therefore,  $H$  is not normal in  $G$ .

To classify groups of order 6, let  $G$  be a group of order 6. If  $G$  is abelian, then  $G \cong \mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$ . If  $G$  is non-abelian, then by the above result,  $G$  has a nonnormal subgroup  $H$  of order 2. Let  $K$  be the subgroup of order 3. Then  $G = HK$ . Let  $G$  act by left multiplication on the left cosets of  $K$  in  $G$ . This action gives a homomorphism  $\varphi : G \rightarrow S_3$ . Since  $H$  is not normal in  $G$ , then the action is nontrivial, so  $\varphi$  is injective. Therefore,  $G \cong \varphi(G) \leq S_3$ . Since  $|G| = 6 = |S_3|$ , then  $\varphi(G) = S_3$ , and hence  $G \cong S_3$ . ■

**Exercise 4.2.11**

Let  $G$  be a finite group and let  $\pi : G \rightarrow S_G$  be the left regular representation. Prove that if  $x$  is an element of  $G$  of order  $n$  and  $|G| = mn$ , then  $\pi(x)$  is a product of  $m$   $n$ -cycles. Deduce that  $\pi(x)$  is an odd permutation if and only if  $|x|$  is even and  $|G|/|x|$  is odd.

**Solution.** Let  $x \in G$  with  $|x| = n$ . Consider the action of  $\langle x \rangle$  on  $G$  by left multiplication. The orbit of any  $g \in G$  under this action is  $O_g = \{x^k g \mid k = 0, 1, \dots, n-1\}$ . Since  $|x| = n$ , then  $|O_g| = n$  for all  $g \in G$ . By Proposition 4.2, we have  $|O_g| = |\langle x \rangle : (\langle x \rangle)_g|$ , where  $(\langle x \rangle)_g$  is the stabilizer of  $g$  in  $\langle x \rangle$ . Since  $|O_g| = n$ , then  $(\langle x \rangle)_g$  is trivial for all  $g \in G$ . Therefore, the orbits of this action partition  $G$  into subsets of size  $n$ . Since  $|G| = mn$ , there are exactly  $m$  such orbits. Each orbit corresponds to an  $n$ -cycle in the permutation  $\pi(x)$ . Therefore,  $\pi(x)$  is a product of  $m$   $n$ -cycles.

To determine when  $\pi(x)$  is an odd permutation, we note that an  $n$ -cycle is an odd permutation if and only if  $n$  is even. Since  $\pi(x)$  is a product of  $m$   $n$ -cycles, the parity of  $\pi(x)$  is determined by the parity of  $m$  and  $n$ . Specifically,  $\pi(x)$  is odd if and only if  $n$  is even and  $m$  is odd. Thus,  $\pi(x)$  is an odd permutation if and only if  $|x|$  is even and  $|G|/|x|$  is odd. ■

**Exercise 4.2.12**

Let  $G$  and  $\pi$  be as in the preceding exercise. Prove that if  $\pi(G)$  contains an odd permutation then  $G$  has a subgroup of index 2. [Use [Exercise 3.3.3](#).]

**Solution.** Recall the  $\epsilon$  homomorphism from Proposition 3.23. Moreover,  $\pi$  is a homomorphism from  $G$  to  $S_G$ . Consider the composition  $\epsilon \circ \pi : G \rightarrow \{\pm 1\}$ . Since  $\pi(G)$  contains an odd permutation, then  $\epsilon \circ \pi$  is surjective. By the First Isomorphism Theorem, we have  $G/\ker(\epsilon \circ \pi) \cong \{\pm 1\}$ , so that  $|\ker(\epsilon \circ \pi)| = |G|/2$ . Therefore,  $\ker(\epsilon \circ \pi)$  is a subgroup of  $G$  of index 2. ■

**Exercise 4.2.13**

Prove that if  $|G| = 2k$  where  $k$  is odd then  $G$  has a subgroup of index 2. [Use Cauchy's Theorem to produce an element of order 2 and then use the preceding two exercises.]

**Solution.** By Cauchy's Theorem, there exists  $x \in G$  such that  $|x| = 2$ . Then  $\pi(x)$  is of order 2 and is hence a product of disjoint transpositions. Since  $|G| = 2k$  with  $k$  odd, then use  $m = k$  and  $n = 2$  in [Exercise 4.2.11](#) along with even  $|x|$  to conclude that  $\pi(x)$  is an odd permutation. By the [Exercise 4.2.12](#),  $G$  has a subgroup of index 2. ■

**Exercise 4.2.14**

Let  $G$  be a finite group of composite order  $n$  with the property that  $G$  has a subgroup of order  $k$  for each positive integer  $k$  dividing  $n$ . Prove that  $G$  is not simple.

**Solution.** Let  $S$  be the set of all prime factors of  $n$ . By the Well Ordering Principle, this has a minimal element  $p$ . By hypothesis,  $G$  has a subgroup  $P$  of order  $n/p$  with index  $p$ . By Corollary 4.5,  $P$  is normal in  $G$  since  $p$  is the smallest prime dividing  $n$ . Therefore,  $G$  is not simple. ■

### 4.3 Groups Acting on Themselves by Conjugation—The Class Equation

Let  $G$  be a group.

**(\*) Exercise 4.3.1**

Suppose  $G$  has a left action on a set  $A$ , denoted by  $g \cdot a$  for all  $g \in G$  and  $a \in A$ . Denote the corresponding right action on  $A$  by  $a \cdot g$ . Prove that the (equivalence) relations  $\sim$  and  $\sim'$  defined by

$$a \sim b \quad \text{if and only if} \quad a = g \cdot b \quad \text{for some } g \in G$$

and

$$a \sim' b \quad \text{if and only if} \quad a = b \cdot g \quad \text{for some } g \in G$$

are the same relation (i.e.,  $a \sim b$  if and only if  $a \sim' b$ ).

**Solution.** If  $a \sim b$ , there exists  $g \in G$  such that  $a = g \cdot b$ . Let  $h = g^{-1}$ . Then  $b = h \cdot a$ , so  $a = b \cdot g$ . Thus,  $a \sim' b$ . Conversely, if  $a \sim' b$ , there exists  $g \in G$  such that  $a = b \cdot g$ . Let  $h = g^{-1}$ . Then  $b = h \cdot a$ , so  $a = g \cdot b$ . Thus,  $a \sim b$ . Therefore, the relations  $\sim$  and  $\sim'$  are the same. ■

**Exercise 4.3.2**

Find all conjugacy classes and their sizes in the following groups:

- (a)  $D_8$    (b)  $Q_8$    (c)  $A_4$

**Solution.**

- (a) Discussed in the text, the conjugacy classes of  $D_8$  are  $\{1\}$ ,  $\{r^2\}$ ,  $\{r, r^3\}$ ,  $\{s, sr^2\}$ , and  $\{sr, sr^3\}$  with sizes 1, 1, 2, 2, and 2 respectively.
- (b) Again discussed in the text, the conjugacy classes of  $Q_8$  are  $\{1\}$ ,  $\{-1\}$ ,  $\{i, -i\}$ ,  $\{j, -j\}$ , and  $\{k, -k\}$  with sizes 1, 1, 2, 2, and 2 respectively.
- (c) We proceed similarly as the text. The possible cycle types are 1, (1 2 3), and (1 2)(3 4).

For (1 2 3), we have  $C_{A_4}((1 2 3)) = \langle (1 2 3) \rangle$  since the only permutation that fixes 1, 2, and 3 is the identity. Therefore, the conjugacy class of (1 2 3) has size  $12/3 = 4$ . However, there are 8 3-cycles in  $A_4$ , so there exists a 3-cycle  $\sigma \in A_4$  not in the conjugacy class of (1 2 3). By similar reasoning, the conjugacy class of  $\sigma$  also has size 4. There are then 2 conjugacy classes of size 4 corresponding to the 3-cycles.

For (1 2)(3 4), it is trivial to calculate that the remaining double transpositions are in its conjugacy class. Therefore, the conjugacy class of (1 2)(3 4) has size 3. ■

**Exercise 4.3.3**

Find all conjugacy classes and their sizes in the following groups:

- (a)  $Z_2 \times S_3$  (b)  $S_3 \times S_3$  (c)  $Z_3 \times A_4$

**Solution.** We first prove a (somewhat) trivial idea: if  $G$  and  $H$  are groups, let  $g \in G$  and  $h \in H$ . Let  $\mathcal{K}_g$  and  $\mathcal{K}_h$  be the conjugacy classes of  $g$  in  $G$  and  $h$  in  $H$  respectively. Then  $\mathcal{K}_{(g,h)} = \mathcal{K}_g \times \mathcal{K}_h$  is the conjugacy class of  $(g, h)$  in  $G \times H$ . To see this, note that for any  $(x, y) \in G \times H$ , we have

$$(x, y)(g, h)(x, y)^{-1} = (xgx^{-1}, yhy^{-1}) \in \mathcal{K}_g \times \mathcal{K}_h.$$

Conversely, for any  $(g', h') \in \mathcal{K}_g \times \mathcal{K}_h$ , there exist  $x \in G$  and  $y \in H$  such that  $g' = xgx^{-1}$  and  $h' = yhy^{-1}$ . Therefore,  $(g', h') = (x, y)(g, h)(x, y)^{-1}$ , so  $(g', h')$  is in the conjugacy class of  $(g, h)$  in  $G \times H$ . This result shows two important things: the conjugacy classes of  $G \times H$  are precisely the products of the conjugacy classes of  $G$  and  $H$ , and the size of the conjugacy class of  $(g, h)$  in  $G \times H$  is the product of the sizes of the conjugacy classes of  $g$  in  $G$  and  $h$  in  $H$ . We now use this to answer the question.

- (a) Let  $Z_2 = \langle x \rangle$ . The conjugacy classes  $\mathcal{K}_1$  and  $\mathcal{K}_x$  of  $Z_2$  have sizes 1 and 1 respectively. For  $S_3$ , the partitions of 3 are 3 1-cycles, 2-cycle and 1-cycle, and a 3-cycle with representatives 1,  $(1\ 2)$ , and  $(1\ 2\ 3)$  respectively. Since  $C_{S_3}((1\ 2)) = \langle (1\ 2) \rangle$  and  $C_{S_3}((1\ 2\ 3)) = \langle (1\ 2\ 3) \rangle$ , then the conjugacy classes  $\mathcal{K}_1$ ,  $\mathcal{K}_{(1\ 2)}$ , and  $\mathcal{K}_{(1\ 2\ 3)}$  of  $S_3$  have sizes 1, 3, and 2 respectively. Therefore,  $Z_2 \times S_3$  has 6 conjugacy classes of size 1, 3, 2, 1, 3, and 2 respectively.
- (b) By the above,  $S_3$  has 3 conjugacy classes of sizes 1, 3, and 2 respectively. Then  $S_3 \times S_3$  has 9 conjugacy classes with sizes 1, 3, 2, 3, 9, 6, 2, 6, and 4 respectively.
- (c)  $Z_3$  has 3 conjugacy classes of size 1 each. From [Exercise 4.3.2](#), we know that  $A_4$  has 4 conjugacy classes of sizes 1, 4, 4, and 3. Therefore,  $Z_3 \times A_4$  has 12 conjugacy classes with sizes 1, 4, 4, 3, 1, 4, 4, 3, 1, 4, 4, and 3 respectively. ■

**Exercise 4.3.4**

Prove that if  $S \subseteq G$  and  $g \in G$  then  $gN_G(S)g^{-1} = N_G(gSg^{-1})$  and  $gC_G(S)g^{-1} = C_G(gSg^{-1})$ .

**Solution.** Suppose  $x \in gN_G(S)g^{-1}$ , and consider  $xgSg^{-1}x^{-1}$ . Since  $x = gng^{-1}$  for some  $n \in N_G(S)$ , then we have  $gng^{-1}gSg^{-1}gn^{-1}g^{-1} = gnSn^{-1}g^{-1} = gSg^{-1}$ , so that  $x \in N_G(gSg^{-1})$ . Conversely, suppose  $x \in N_G(gSg^{-1})$ . Then  $xgSg^{-1}x^{-1} = gSg^{-1}$ . Letting  $n = g^{-1}xg$ , we have  $nSn^{-1} = S$ , so  $n \in N_G(S)$  and hence  $x \in gN_G(S)g^{-1}$ . Therefore,  $gN_G(S)g^{-1} = N_G(gSg^{-1})$ .

Suppose  $x \in gC_G(S)g^{-1}$ , and consider  $xsx^{-1}$  for some  $s \in gSg^{-1}$ . Since  $x = gng^{-1}$  for some  $n \in C_G(S)$ , then we have  $gng^{-1}sgn^{-1}g^{-1} = gn(g^{-1}sg)n^{-1}g^{-1} = g(g^{-1}sg)g^{-1} = s$ , so that  $x \in C_G(gSg^{-1})$ . Conversely, suppose  $x \in C_G(gSg^{-1})$ . Then  $xsx^{-1} = s$  for all  $s \in gSg^{-1}$ . Letting  $n = g^{-1}xg$ , we have  $ntn^{-1} = t$  for all  $t \in S$ , so  $n \in C_G(S)$  and hence  $x \in gC_G(S)g^{-1}$ . Therefore,  $gC_G(S)g^{-1} = C_G(gSg^{-1})$ . ■

**Exercise 4.3.5**

If the center of  $G$  is of index  $n$ , prove that every conjugacy class has at most  $n$  elements.

**Solution.** Let  $\mathcal{K}_g$  be the conjugacy class of some  $g \in G$ . By Proposition 4.6, we have  $|\mathcal{K}_g| = |G : C_G(g)|$ . Since  $Z(G) \leq C_G(g)$  for all  $g \in G$ , Lagrange's Theorem shows that  $|Z(G)|$  divides  $|C_G(g)|$ . Therefore,  $|G : C_G(g)|$  divides  $|G : Z(G)| = n$ . Hence,  $|\mathcal{K}_g| \leq n$ . ■

(\*) **Exercise 4.3.6**

Assume  $G$  is a non-abelian group of order 15. Prove that  $Z(G) = 1$ . Use the fact that  $\langle g \rangle \leq C_G(g)$  for all  $g \in G$  to show that there is at most one possible class equation for  $G$ . [Use [Exercise 3.1.36](#).]

**Solution.** Recall that  $Z(G) \leq G$ . Since  $G$  is non-abelian, then  $Z(G) \neq G$ . Assume  $1 < Z(G) < 15$ . By Lagrange's Theorem,  $Z(G)$  has order 3 or 5. If  $|Z(G)| = 3$ , then  $|G/Z(G)| = 5$  and  $G/Z(G) \cong Z_5$ . If  $|Z(G)| = 5$ , then  $|G/Z(G)| = 3$  and  $G/Z(G) \cong Z_3$ . In either case,  $G/Z(G)$  is cyclic by Corollary 3.10, and [Exercise 3.1.36](#) implies that  $G$  is abelian, a contradiction. Therefore,  $Z(G) = 1$ .

Now suppose  $g \in G$  is nonidentity. Since  $\langle g \rangle \leq C_G(g)$ , then  $|C_G(g)|$  is 3, 5, or 15 by Lagrange's Theorem. If  $|C_G(g)| = 15$ , then  $C_G(g) = G$ , so  $g \in Z(G)$ , a contradiction. Therefore,  $|C_G(g)|$  is 3 or 5 for all nonidentity  $g \in G$ . By Proposition 4.6, we have  $|\mathcal{K}_g| = |G : C_G(g)|$ , so that  $|\mathcal{K}_g|$  is 5 or 3 respectively. The identity element forms a conjugacy class of size 1, and the class equation must involve a summation of sizes 3 and 5 that adds to 14. The only such combination is three classes of size 3 and one class of size 5. Therefore, the class equation of  $G$  is  $15 = 1 + 3 + 3 + 3 + 5$ . ■

**Exercise 4.3.7**

For  $n = 3, 4, 6$ , and 7 make lists of the partitions of  $n$  and give representatives for the corresponding conjugacy classes of  $S_n$ .

**Solution.**  $n = 3$ :

| Partition of 3 | Representative of Cycle Type |
|----------------|------------------------------|
| 1, 1, 1        | 1                            |
| 2, 1           | (1 2)                        |
| 3              | (1 2 3)                      |

$n = 4$ :

| Partition of 4 | Representative of Cycle Type |
|----------------|------------------------------|
| 1, 1, 1, 1     | 1                            |
| 2, 1, 1        | (1 2)                        |
| 2, 2           | (1 2)(3 4)                   |
| 3, 1           | (1 2 3)                      |
| 4              | (1 2 3 4)                    |

$n = 6$ :

| Partition of 6   | Representative of Cycle Type |
|------------------|------------------------------|
| 1, 1, 1, 1, 1, 1 | 1                            |
| 2, 1, 1, 1, 1    | (1 2)                        |
| 2, 2, 1, 1       | (1 2)(3 4)                   |
| 2, 2, 2          | (1 2)(3 4)(5 6)              |
| 3, 1, 1, 1       | (1 2 3)                      |
| 3, 2, 1          | (1 2 3)(4 5)                 |
| 3, 3             | (1 2 3)(4 5 6)               |
| 4, 1, 1          | (1 2 3 4)                    |
| 4, 2             | (1 2 3 4)(5 6)               |
| 5, 1             | (1 2 3 4 5)                  |
| 6                | (1 2 3 4 5 6)                |

$n = 7$ :

| Partition of 7      | Representative of Cycle Type |
|---------------------|------------------------------|
| 1, 1, 1, 1, 1, 1, 1 | 1                            |
| 2, 1, 1, 1, 1, 1    | (1 2)                        |
| 2, 2, 1, 1, 1       | (1 2)(3 4)                   |
| 2, 2, 2, 1          | (1 2)(3 4)(5 6)              |
| 3, 1, 1, 1, 1       | (1 2 3)                      |
| 3, 2, 1, 1          | (1 2 3)(4 5)                 |
| 3, 2, 2             | (1 2 3)(4 5)(6 7)            |
| 3, 3, 1             | (1 2 3)(4 5 6)               |
| 4, 1, 1, 1          | (1 2 3 4)                    |
| 4, 2, 1             | (1 2 3 4)(5 6)               |
| 4, 3                | (1 2 3 4)(5 6 7)             |
| 5, 1, 1             | (1 2 3 4 5)                  |
| 5, 2                | (1 2 3 4 5)(6 7)             |
| 6, 1                | (1 2 3 4 5 6)                |
| 7                   | (1 2 3 4 5 6 7)              |

#### Exercise 4.3.8

Prove that  $Z(S_n) = 1$  for all  $n \geq 3$ .

**Solution.** Suppose  $\sigma \in Z(S_n)$  for some  $n \geq 3$ . Then  $\sigma$  commutes with every element of  $S_n$ . In particular,  $\sigma$  commutes with all transpositions (the set of which generate  $S_n$ ). Let  $(a b)$  be any transposition in  $S_n$ . Then we have  $\sigma(a b)\sigma^{-1} = (a b)$ . This implies that  $(\sigma(a) \sigma(b)) = (a b)$ , so  $\sigma(a) = a$  and  $\sigma(b) = b$ . Since  $a$  and  $b$  were arbitrary,  $\sigma$  fixes every element of  $\{1, 2, \dots, n\}$ . Therefore,  $\sigma$  is the identity permutation. Hence,  $Z(S_n) = \{1\}$  for all  $n \geq 3$ . ■

#### Exercise 4.3.9

Show that  $|C_{S_n}((1 2)(3 4))| = 8(n - 4)!$  for all  $n \geq 4$ . Determine the elements in this centralizer explicitly.

**Solution.** The  $(n - 4)!$  factor arises from the fact that permutations on the  $n - 4$  integers not involved in the cycle  $(1 2)(3 4)$  commute with it. Therefore, we need only consider the permutations of  $\{1, 2, 3, 4\}$  that commute with  $(1 2)(3 4)$ . Computing  $C_{S_4}((1 2)(3 4))$  directly, the permutations that fall in this set need to leave  $(1 2)(3 4)$  unchanged, or swap the two transpositions. The permutations that leave  $(1 2)(3 4)$  unchanged are 1,  $(1 2)$ ,  $(3 4)$ , and  $(1 2)(3 4)$ . The permutations that swap the two transpositions are  $(1 3)(2 4)$ ,  $(1 4)(2 3)$ ,  $(1 3 2 4)$ , and  $(1 4 2 3)$ . Therefore,

$$C_{S_4}((1 2)(3 4)) = \{1, (1 2), (3 4), (1 2)(3 4), (1 3)(2 4), (1 4)(2 3), (1 3 2 4), (1 4 2 3)\},$$

which has size 8. Combining this with the  $(n - 4)!$  factor, we have  $|C_{S_n}((1 2)(3 4))| = 8(n - 4)!$  for all  $n \geq 4$ . ■

#### Exercise 4.3.10

Let  $\sigma$  be the 5-cycle  $(1 2 3 4 5)$  in  $S_5$ . In each of (a) to (c) find an explicit element  $\tau \in S_5$  which accomplishes the specified conjugation:

- (a)  $\tau\sigma\tau^{-1} = \sigma^2$
- (b)  $\tau\sigma\tau^{-1} = \sigma^{-1}$
- (c)  $\tau\sigma\tau^{-1} = \sigma^{-2}$

**Solution.**

- (a)  $\sigma^2 = (1 3 5 2 4)$ . Then  $\tau = (2 3 5 4)$ .
- (b)  $\sigma^{-1} = (1 5 4 3 2)$ . Then  $\tau = (1 5)(2 4)$ .
- (c)  $\sigma^{-2} = (1 4 2 5 3)$ . Then  $\tau = (2 4 5 3)$ . ■



**Exercise 4.3.11**

In each of (a) – (d) determine whether  $\sigma_1$  and  $\sigma_2$  are conjugate. If they are, given an explicit permutation  $\tau$  such that  $\tau\sigma_1\tau^{-1} = \sigma_2$ .

- (a)  $\sigma_1 = (1\ 2)(3\ 4\ 5)$  and  $\sigma_2 = (1\ 2\ 3)(4\ 5)$   
 (b)  $\sigma_1 = (1\ 5)(3\ 7\ 2)(10\ 6\ 8\ 11)$  and  $\sigma_2 = (3\ 7\ 5\ 10)(4\ 9)(13\ 11\ 2)$   
 (c)  $\sigma_1 = (1\ 5)(3\ 7\ 2)(10\ 6\ 8\ 11)$  and  $\sigma_2 = \sigma_1^3$   
 (d)  $\sigma_1 = (1\ 3)(2\ 4\ 6)$  and  $\sigma_2 = (3\ 5)(2\ 4)(5\ 6)$

**Solution.**

- (a) Rewrite  $\sigma_2$  as  $(4\ 5)(1\ 2\ 3)$  so that both have the same cycle type. Then  $\tau = (1\ 4\ 2\ 5\ 3)$ .  
 (b) Rewrite  $\sigma_2$  as  $(4\ 9)(13\ 11\ 2)(3\ 7\ 5\ 10)$ . Then  $\tau = (1\ 4)(8\ 5\ 9)(6\ 7\ 11\ 10\ 3\ 13)$ .  
 (c) Note that  $\sigma_2 = (1\ 5)(6\ 10\ 11\ 8)$ , which does not have the same cycle type as  $\sigma_1$ . Therefore, they are not conjugate.  
 (d)  $\sigma_2 = (2\ 4)(3\ 5\ 6)$ . Then  $\tau = (1\ 2\ 3\ 4\ 5)$ . ■

**Exercise 4.3.12**

Find a representative for each conjugacy class of elements of order 4 in  $S_8$  and  $S_{12}$ .

**Solution.**  $S_8$ :

| Cycle Type    | Representative             |
|---------------|----------------------------|
| 4, 4          | $(1\ 2\ 3\ 4)(5\ 6\ 7\ 8)$ |
| 4, 2, 2       | $(1\ 2\ 3\ 4)(5\ 6)(7\ 8)$ |
| 4, 2, 1, 1    | $(1\ 2\ 3\ 4)(5\ 6)$       |
| 4, 1, 1, 1, 1 | $(1\ 2\ 3\ 4)$             |

$S_{12}$ :

| Cycle Type                | Representative                            |
|---------------------------|-------------------------------------------|
| 4, 4, 4                   | $(1\ 2\ 3\ 4)(5\ 6\ 7\ 8)(9\ 10\ 11\ 12)$ |
| 4, 4, 2, 2                | $(1\ 2\ 3\ 4)(5\ 6\ 7\ 8)(9\ 10)(11\ 12)$ |
| 4, 4, 2, 1, 1             | $(1\ 2\ 3\ 4)(5\ 6\ 7\ 8)(9\ 10)$         |
| 4, 4, 1, 1, 1, 1          | $(1\ 2\ 3\ 4)(5\ 6\ 7\ 8)$                |
| 4, 2, 2, 2, 2             | $(1\ 2\ 3\ 4)(5\ 6)(7\ 8)(9\ 10)(11\ 12)$ |
| 4, 2, 2, 2, 1, 1          | $(1\ 2\ 3\ 4)(5\ 6)(7\ 8)(9\ 10)$         |
| 4, 2, 2, 1, 1, 1, 1       | $(1\ 2\ 3\ 4)(5\ 6)(7\ 8)$                |
| 4, 2, 1, 1, 1, 1, 1, 1    | $(1\ 2\ 3\ 4)(5\ 6)$                      |
| 4, 1, 1, 1, 1, 1, 1, 1, 1 | $(1\ 2\ 3\ 4)$                            |

**(\*) Exercise 4.3.13**

Find all finite groups which have exactly two conjugacy classes.

**Solution.** Let  $G$  be a finite group with exactly two conjugacy classes. One of these classes must be  $\{1\}$ , the identity element. Let  $\mathcal{K}$  be the other conjugacy class, and let  $g \in \mathcal{K}$ . By Proposition 4.6, we have  $|\mathcal{K}| = |G : C_G(g)|$ . Since there are only two conjugacy classes, then  $|\mathcal{K}| = |G| - 1$ . Therefore,  $|G : C_G(g)| = |G| - 1$ , which implies that  $|C_G(g)| = |G|/(|G| - 1)$ . Since  $|C_G(g)|$  is an integer, then  $|G| - 1$  divides  $|G|$ . This is only possible if  $|G| - 1 = 1$ , so that  $|G| = 2$ . Therefore, the only finite group with exactly two conjugacy classes is the group of order 2, which is isomorphic to  $Z_2$ . ■

**Exercise 4.3.14**

In [Exercise 4.2.1](#) two labellings of the elements  $\{1, a, b, c\}$  of the Klein 4-group  $V_4$  were chosen to give two versions of the left regular representation of  $V_4$  into  $S_4$ . Let  $\pi_1$  be the version of regular representation obtained in part (a) of that exercise, and let  $\pi_2$  be the version obtained via the labelling in part (b). Let  $\tau = (2\ 4)$ . Show that  $\tau \circ \pi_1(g) \circ \tau^{-1} = \pi_2(g)$  for each  $g \in V_4$  (i.e., conjugation by  $\tau$  sends the image of  $\pi_1$  to the image of  $\pi_2$  elementwise).

**Solution.** We will compute for  $a$ , as the rest follow similarly. We have  $\pi_1(a) = (1\ 2)(3\ 4)$  and  $\pi_2(a) = (1\ 4)(2\ 3)$ . Then  $\tau \circ \pi_1(a) \circ \tau^{-1} = (2\ 4)(1\ 2)(3\ 4)(2\ 4) = (1\ 4)(2\ 3) = \pi_2(a)$ . ■

**Exercise 4.3.15**

Find an element of  $S_8$  which conjugates the subgroup of  $S_8$  obtained in part (a) of [Exercise 4.2.3](#) to the subgroup of  $S_8$  obtained in part (b) of that same exercise (both of these subgroups are isomorphic to  $D_8$ ).

**Solution.** Recall by [Exercise 1.7.17](#) that left conjugation is an automorphism, so it suffices to find an element which sends the generators of the first subgroup to the generators of the second subgroup. The generators of the first subgroup are  $(1\ 2\ 3\ 4)(5\ 8\ 7\ 6)$  and  $(1\ 5)(2\ 6)(3\ 7)(4\ 8)$ , while the generators of the second subgroup are  $(1\ 3\ 5\ 7)(2\ 8\ 6\ 4)$  and  $(1\ 2)(3\ 4)(5\ 6)(7\ 8)$  respectively. It suffices to check some element  $\tau$  which sends the first generator to the second; the second generator will follow automatically. One such element is  $\tau = (5\ 2\ 3)(6\ 4\ 7)$ . ■

**Exercise 4.3.16**

Find an element of  $S_4$  which conjugates the subgroup of  $S_4$  obtained in part (a) of [Exercise 4.2.5](#) to the subgroup of  $S_4$  obtained in part (b) of that same exercise (both of these subgroups are isomorphic to  $D_8$ ).

**Solution.** We proceed similarly to the preceding exercise. The generators of the first subgroup are  $(1\ 2\ 3\ 4)$  and  $(2\ 4)$ , while the generators of the second subgroup are  $(1\ 3\ 2\ 4)$  and  $(3\ 4)$ . We obtain  $\tau = (2\ 3)$ . ■

**(\*) Exercise 4.3.17**

Let  $A$  be a nonempty set and let  $X$  be any subset of  $S_A$ . Let

$$F(X) = \{a \in A \mid \sigma(a) = a \text{ for all } \sigma \in X\} \quad \text{—the fixed set of } X$$

Let  $M(X) = A - F(X)$  be the elements which are *moved* by some element of  $X$ . Let  $D = \{\sigma \in S_A \mid |M(\sigma)| < \infty\}$ . Prove that  $D$  is a normal subgroup of  $S_A$ .

**Solution.** Note that  $1 \in D$ , since  $M(1) = \emptyset$  as it fixes all elements of  $A$ . Let  $\sigma, \tau \in D$ . Then  $|M(\sigma)| < \infty$  and  $|M(\tau)| < \infty$ . Suppose  $a \in F(\sigma) \cap F(\tau)$ . Clearly,  $a \in F(\sigma\tau)$ . By contrapositive, then  $a \in M(\sigma\tau)$  implies  $a \in M(\sigma) \cup M(\tau)$ , which are both finite, hence  $M(\sigma\tau)$  is finite. Therefore,  $\sigma\tau \in D$ .

Now consider  $\sigma \in D$ . If  $a \in A$  is fixed by  $\sigma$ , then it is fixed by  $\sigma^{-1}$ . Therefore, any element moved by  $\sigma^{-1}$  must be moved by  $\sigma$ , and vice-versa. Hence,  $M(\sigma^{-1}) = M(\sigma)$ , which is finite. Therefore,  $\sigma^{-1} \in D$ .

Finally, consider  $\sigma \in D$  and  $\tau \in S_A$ . Suppose  $a \in F(\tau\sigma\tau^{-1})$  so that  $\tau\sigma\tau^{-1}(a) = a$ . Then  $\sigma(\tau^{-1}(a)) = \tau^{-1}(a)$ , hence  $\tau^{-1}(a) \in F(\sigma)$ . By contrapositive, if  $a \in M(\tau\sigma\tau^{-1})$ , then  $\tau^{-1}(a) \in M(\sigma)$ . Since  $M(\sigma)$  is finite, then  $M(\tau\sigma\tau^{-1})$  is finite. Therefore,  $\tau\sigma\tau^{-1} \in D$ . ■

**Exercise 4.3.18**

Let  $A$  be a set, let  $H$  be a subgroup of  $S_A$ , and let  $F(H)$  be the fixed points of  $H$  on  $A$  as defined in the preceding exercise. Prove that if  $\tau \in N_{S_A}(H)$ , then  $\tau$  stabilizes the set  $F(H)$  and its complement  $A - F(H)$ .

**Solution.** Suppose  $\sigma \in H$ . Since  $\tau \in N_{S_A}(H)$ , we have  $\tau^{-1} \in N_{S_A}(H)$ , hence  $\tau^{-1}\sigma\tau \in H$ . Suppose  $a \in F(H)$ . Then  $\tau^{-1}\sigma\tau(a) = a$ , so that  $\sigma\tau(a) = \tau(a)$ . Therefore,  $\tau(a) \in F(H)$ , and  $\tau(F(H)) \subseteq F(H)$ . Bijectivity of  $\tau$  also implies that  $\tau^{-1}(F(H)) \subseteq F(H)$ , which shows that  $F(H) \subseteq \tau(F(H))$ , hence  $\tau(F(H)) = F(H)$ . Therefore,  $\tau$  stabilizes  $F(H)$ . Moreover, since  $\tau$  is a bijection, it also stabilizes the complement  $A - F(H)$ . ■

**Exercise 4.3.19**

Assume  $H$  is a normal subgroup of  $G$ ,  $\mathcal{K}$  is a conjugacy class of  $G$  contained in  $H$  and  $x \in \mathcal{K}$ . Prove that  $\mathcal{K}$  is a union of  $k$  conjugacy classes of equal size in  $H$ , where  $k = |G : HC_G(x)|$ . Deduce that a conjugacy class in  $S_n$  which consists of even permutations is either a single conjugacy class under the action of  $A_n$  or is a union of two classes of the same size in  $A_n$ . [Let  $A = C_G(x)$  and  $B = H$  so  $A \cap B = C_H(x)$ . Draw the lattice diagram associated to the Second Isomorphism Theorem and interpret the appropriate indices. See also [Exercise 4.1.9](#).]

**Solution.** Let  $G$  act on  $\mathcal{K}$  by conjugation. Since  $\mathcal{K}$  is a conjugacy class of  $G$ , this action is transitive. Because  $H \trianglelefteq G$  and  $\mathcal{K} \subseteq H$ , then  $H$  also acts on  $\mathcal{K}$  by conjugation. Let  $O$  be one such orbit of this action, and suppose  $y \in O$ . By Proposition 4.6, we have  $|O| = |H : C_H(y)|$ . By normality of  $H$  in  $G$ , we have  $C_H(y) = C_G(y) \cap H$ . Using the Second Isomorphism Theorem, we have

$$HC_G(y)/C_G(y) \cong H/C_H(y)$$

so that  $|H : C_H(y)| = |HC_G(y) : C_G(y)|$ . Therefore,  $|O| = |HC_G(y) : C_G(y)|$ . Using [Exercise 4.1.9](#), we know that each orbit of  $H$  on  $\mathcal{K}$  has the same size, so every orbit has size  $|HC_G(y) : C_G(y)|$ . Suppose there are  $k$  orbits. Then

$$|\mathcal{K}| = k|HC_G(y) : C_G(y)|.$$

By Proposition 4.6, we also have  $|\mathcal{K}| = |G : C_G(y)|$ . Therefore,

$$|G : C_G(y)| = k|HC_G(y) : C_G(y)|.$$

Dividing both sides by  $|HC_G(y) : C_G(y)|$ , we obtain  $k = |G : HC_G(y)|$ . Hence,  $\mathcal{K}$  is a union of  $k$  conjugacy classes of equal size in  $H$ .

Now suppose  $\mathcal{K} \subseteq A_n$  is a conjugacy class of  $S_n$  consisting of even permutations. Let  $x \in \mathcal{K}$ . Applying the preceding result with  $H = A_n$ , we have that  $\mathcal{K}$  is a union of  $k$  conjugacy classes of equal size in  $A_n$ , where  $k = |S_n : A_n C_{S_n}(x)|$ . Since  $|S_n : A_n| = 2$ , then  $k$  is either 1 or 2. Therefore,  $\mathcal{K}$  is either a single conjugacy class under the action of  $A_n$  or is a union of two classes of the same size in  $A_n$ . ■

**Exercise 4.3.20**

Let  $\sigma \in A_n$ . Show that all elements in the conjugacy class of  $\sigma$  in  $S_n$  (i.e., all elements of the same cycle type as  $\sigma$ ) are conjugate in  $A_n$  if and only if  $\sigma$  commutes with an odd permutation. [Use the preceding exercise.]

**Solution.** ( $\Rightarrow$ ) Recall by the previous exercise that the conjugacy class of  $\sigma$  in  $S_n$  is either a single conjugacy class in  $A_n$  or a union of two conjugacy classes of equal size in  $A_n$ . Suppose all elements in the conjugacy class of  $\sigma$  in  $S_n$  are conjugate in  $A_n$ . Then the conjugacy class of  $\sigma$  in  $S_n$  is a single conjugacy class in  $A_n$ , so that  $k = |S_n : A_n C_{S_n}(\sigma)| = 1$ . Therefore,  $S_n = A_n C_{S_n}(\sigma)$ , so that there exists some odd permutation  $\tau \in S_n$  such that  $\tau \in C_{S_n}(\sigma)$ . Hence,  $\sigma$  commutes with an odd permutation.

( $\Leftarrow$ ) Suppose  $\sigma$  commutes with an odd permutation  $\tau \in S_n$ . Then  $\tau \in C_{S_n}(\sigma)$ , so that  $A_n C_{S_n}(\sigma)$  contains odd permutations. Therefore,  $|S_n : A_n C_{S_n}(\sigma)| = 1$ , so that the conjugacy class of  $\sigma$  in  $S_n$  is a single conjugacy class in  $A_n$ . Hence, all elements in the conjugacy class of  $\sigma$  in  $S_n$  are conjugate in  $A_n$ . ■

**(\*) Exercise 4.3.21**

Let  $\mathcal{K}$  be a conjugacy class in  $S_n$  and assume that  $\mathcal{K} \subseteq A_n$ . Show  $\sigma \in S_n$  does *not* commute with any odd permutation if and only if the cycle type of  $\sigma$  consists of distinct odd integers. Deduce that  $\mathcal{K}$  consists of two conjugacy classes in  $A_n$  if and only if the cycle type of an element of  $\mathcal{K}$  consists of distinct odd integers. [Assume first that  $\sigma \in \mathcal{K}$  does not commute with any odd permutation. Observe that  $\sigma$  commutes with each individual cycle in its cycle decomposition—use this to show that all its cycles must be of odd length. If two cycles have the same odd length,  $k$ , find a product of  $k$  transpositions which interchanges them and commutes with  $\sigma$ . Conversely, if the cycle type of  $\sigma$  consists of distinct integers, prove that  $\sigma$  commutes *only* with the group generated by the cycles in its cycle decomposition.]

**Solution.** ( $\Rightarrow$ ) Suppose  $\sigma \in S_n$  does not commute with any odd permutation. Let the cycle decomposition of  $\sigma$  be  $\sigma = \tau_1 \tau_2 \cdots \tau_k$ , where each  $\tau_i$  is a disjoint cycle. Since  $\sigma$  commutes with each  $\tau_i$ , then each  $\tau_i$  must be of odd length; otherwise,  $\tau_i$  would be an odd permutation commuting with  $\sigma$ . Now suppose two cycles  $\tau_i$  and  $\tau_j$  have the same odd length  $m$ . Then the product of  $m$  transpositions which interchanges the elements of  $\tau_i$  and  $\tau_j$  is an odd permutation which commutes with  $\sigma$ , contradicting our assumption. Therefore, all cycles in the cycle decomposition of  $\sigma$  must have distinct odd lengths.

( $\Leftarrow$ ) Suppose the cycle type of  $\sigma$  consists of distinct odd integers. Let the cycle decomposition of  $\sigma$  be  $\sigma = \tau_1 \tau_2 \cdots \tau_k$ , where each  $\tau_i$  is a disjoint cycle of odd length. Any permutation that commutes with  $\sigma$  must permute the cycles  $\tau_i$  among themselves. However, since the lengths of the cycles are distinct, the only way to permute them while preserving their lengths is to leave them unchanged. Therefore, any permutation that commutes with  $\sigma$  must be a product of powers of the individual cycles  $\tau_i$ . Since each  $\tau_i$  has odd length, any such product will be an even permutation. Hence,  $\sigma$  does not commute with any odd permutation. ■

**Exercise 4.3.22**

Show that if  $n$  is odd, then the set of all  $n$ -cycles consists of two conjugacy classes of equal size in  $A_n$ .

**Solution.** Let  $\sigma \in S_n$  be an  $n$ -cycle. Since  $n$  is odd, the cycle type of  $\sigma$  consists of the single odd integer  $n$ , which is distinct. By the previous exercise,  $\sigma$  does not commute with any odd permutation. Therefore, by the exercise before that, the conjugacy class of  $\sigma$  in  $S_n$  consists of two conjugacy classes of equal size in  $A_n$ . Since this holds for any  $n$ -cycle  $\sigma$ , the set of all  $n$ -cycles consists of two conjugacy classes of equal size in  $A_n$ . ■

**Exercise 4.3.23**

Recall (cf. [Exercise 2.4.16](#)) that a proper subgroup  $M$  of  $G$  is called *maximal* if whenever  $M \leq H \leq G$ , either  $H = M$  or  $H = G$ . Prove that if  $M$  is a maximal subgroup of  $G$  then either  $N_G(M) = M$  or  $N_G(M) = G$ . Deduce that if  $M$  is a maximal subgroup of  $G$  that is not normal in  $G$ , then the number of nonidentity elements of  $G$  that are contained in conjugates of  $M$  is at most  $(|M| - 1)|G : M|$ .

**Solution.** Suppose  $M$  is a maximal subgroup of  $G$ . By definition,  $N_G(M)$  is a subgroup of  $G$  containing  $M$ . Therefore, by maximality of  $M$ , either  $N_G(M) = M$  or  $N_G(M) = G$ .

Now suppose  $M$  is a maximal subgroup of  $G$  that is not normal in  $G$ . Then by the previous result, we have  $N_G(M) = M$ . By Proposition 4.6, the size of the conjugacy class of  $M$  in  $G$  is  $|G : N_G(M)| = |G : M|$ . Each conjugate of  $M$  contains  $|M| - 1$  nonidentity elements. Therefore, the total number of nonidentity elements of  $G$  contained in conjugates of  $M$  is at most  $(|M| - 1)|G : M|$ . ■

**Exercise 4.3.24**

Assume  $H$  is a proper subgroup of the finite group  $G$ . Prove

$$G \neq \bigcup_{g \in G} gHg^{-1}$$

i.e.,  $G$  is not the union of the conjugates of any proper subgroup. [Put  $H$  in some maximal subgroup and use the preceding exercise.]

**Solution.** Suppose  $H$  is a proper subgroup of the finite group  $G$ . Then there exists a maximal subgroup  $M$  of  $G$  such that  $H \leq M < G$ . By the previous exercise, either  $N_G(M) = M$  or  $N_G(M) = G$ , so that we have two cases: either  $M \trianglelefteq G$  or  $M$  is not normal in  $G$ .

If  $M$  is normal in  $G$ , then all conjugates of  $M$  are equal to  $M$ . Therefore, the union of the conjugates of  $H$  is contained in  $M$ , which is a proper subset of  $G$ . Hence,

$$G \neq \bigcup_{g \in G} gHg^{-1}.$$

Now suppose  $M$  is not normal in  $G$ . Then by the previous exercise, the number of nonidentity elements of  $G$  contained in conjugates of  $M$  is at most  $(|M| - 1)|G : M|$ . Moreover,  $H \leq M$  implies that any conjugate of  $H$  is contained in a conjugate of  $M$ . Therefore, the number of nonidentity elements of  $G$  contained in conjugates of  $H$  is also at most  $(|M| - 1)|G : M|$ . Since  $M$  is a proper subgroup of  $G$ , then  $|G : M| \geq 2$ , so that  $(|M| - 1)|G : M| < |G| - 1$ . Therefore, there exists at least one nonidentity element of  $G$  that is not contained in any conjugate of  $H$ , hence

$$G \neq \bigcup_{g \in G} gHg^{-1}. \quad \blacksquare$$

**Exercise 4.3.25**

Let  $G = \text{GL}_2(\mathbb{C})$  and let

$$H = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in \mathbb{C}, ac \neq 0 \right\}.$$

Prove that every element of  $G$  is conjugate to some element of the subgroup  $H$  and deduce that  $G$  is the union of conjugates of  $H$ . [Show that every element of  $\text{GL}_2(\mathbb{C})$  has an eigenvector.]

**Solution.** Suppose  $X \in G$ , where  $X$  is the matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

and consider the characteristic polynomial  $\chi_X(t) = \det(X - tI) = t^2 - (a + d)t + (ad - bc)$ . Since  $\mathbb{C}$  is closed under complex multiplication,  $\chi_X(t)$  has at least one root  $\lambda \in \mathbb{C}$ . Therefore, there exists a nonzero vector  $\mathbf{v} \in \mathbb{C}^2$  such that  $X\mathbf{v} = \lambda\mathbf{v}$ , so that  $\mathbf{v}$  is an eigenvector of  $X$  corresponding to the eigenvalue  $\lambda$ . We may extend  $\mathbf{v}$  to a basis  $\{\mathbf{v}, \mathbf{w}\}$  of  $\mathbb{C}^2$ . Let  $P$  be the matrix whose columns are  $\mathbf{v}$  and  $\mathbf{w}$ . Then  $P$  is invertible, and we have

$$P^{-1}XP = \begin{pmatrix} \lambda & \alpha \\ 0 & \beta \end{pmatrix} \in H.$$

for some  $\alpha, \beta \in \mathbb{C}$  such that  $X\mathbf{w} = \alpha\mathbf{v} + \beta\mathbf{w}$ . Therefore, every element of  $G$  is conjugate to some element of the subgroup  $H$ , hence every element of  $G$  is contained in some conjugate of  $H$ . Thus,

$$G = \bigcup_{g \in G} gHg^{-1}. \quad \blacksquare$$

**Exercise 4.3.26**

Let  $G$  be a transitive permutation group on the finite set  $A$  with  $|A| > 1$ . Show that there is some  $\sigma \in G$  such that  $\sigma(a) \neq a$  for all  $a \in A$  (such an element  $\sigma$  is called *fixed point free*).

**Solution.** For each  $a \in A$ , consider  $G_a$ . By transitivity of  $G$  on  $A$ , we have  $|G : G_a| = |A| > 1$ , so that  $G_a$  is a proper subgroup of  $G$ . By Exercise 4.3.24, we have

$$G \neq \bigcup_{g \in G} gG_ag^{-1}.$$

Therefore, there exists some  $\sigma \in G$  such that  $\sigma \notin gG_ag^{-1}$  for all  $g \in G$ . Moreover, if  $\sigma(a) = a$  for some  $a \in A$ , then  $\sigma \in G_a$ , hence  $\sigma \in gG_ag^{-1}$  for  $g = 1$ , a contradiction. Therefore,  $\sigma(a) \neq a$  for all  $a \in A$ . ■

**Exercise 4.3.27**

Let  $g_1, g_2, \dots, g_r$  be representatives of the conjugacy classes of the finite group  $G$  and assume these elements pairwise commute. Prove that  $G$  is abelian.

**Solution.** Pick some  $g_i$  and consider its conjugacy class  $\mathcal{K}_i$ . Since  $g_i$  commutes with all  $g_j$  for  $1 \leq j \leq r$ , then for any  $x \in G$ , we have  $xg_ix^{-1} = g_i$  so that  $x \in C_G(g_i)$ . Therefore,  $C_G(g_i) = G$ , so that  $\mathcal{K}_i = \{g_i\}$ . Since this holds for all  $1 \leq i \leq r$ , then every conjugacy class of  $G$  is a singleton set. Hence, for any  $x, y \in G$ , we have  $xyx^{-1} = y$ , so that  $xy = yx$ . Therefore,  $G$  is abelian. ■

**Exercise 4.3.28**

Let  $p$  and  $q$  be primes with  $p < q$ . Prove that a non-abelian group  $G$  of order  $pq$  has a nonnormal subgroup of index  $q$  so that there exists an injective homomorphism into  $S_q$ . Deduce that  $G$  is isomorphic to a subgroup of the normalizer in  $S_q$  of the cyclic group generated by the  $q$ -cycle  $(1\ 2\ \cdots\ q)$ .

**Solution.** By Cauchy's Theorem, there exists  $g \in G$  such that  $|g| = q$ , hence  $\langle g \rangle$  has order  $q$  and is of index  $p$ . Moreover,  $\langle g \rangle$  is not normal in  $G$ , for otherwise  $G/\langle g \rangle$  would be cyclic and isomorphic to  $Z_p$ , contradicting that  $G$  is non-abelian. Similarly,  $G$  also contains a subgroup  $P$  of order  $p$  and index  $q$  and is similarly not normal in  $G$ .

Let  $G$  act on the left cosets of  $P$  in  $G$  by left multiplication. Then there is the associated permutation representation  $\pi : G \rightarrow S_q$ . In particular,  $\pi$  is injective: If  $\pi(x) = 1$  for some  $x \in G$ , then  $xgP = gP$  for every  $g \in G$ , or  $g^{-1}xg \in P$  for every  $g \in G$ . But  $\ker \pi \trianglelefteq G$  and is contained in  $P$  so that  $\ker \pi$  has order 1 or  $p$ . If  $|\ker \pi| = p$ , then  $\ker \pi = P$  is normal in  $G$ , a contradiction. Therefore,  $\ker \pi = 1$  and  $\pi$  is injective.

Recall that  $|g| = q$  so that  $|\pi(g)| = q$ , so  $\pi(g)$  acts transitively on  $G/P$  since its powers generate  $q$  distinct cosets. Now pick  $h \in G$ . From Exercise 1.1.22, it follows that  $|hgh^{-1}| = |g| = q$  so that  $hgh^{-1} = g^k$  for some integer  $k$  with  $1 \leq k < q$ . Therefore,

$$\pi(h)\pi(g)\pi(h)^{-1} = \pi(hgh^{-1}) = \pi(g^k) = (\pi(g))^k \in \langle \pi(g) \rangle.$$

Hence,  $\pi(h) \in N_{S_q}(\langle \pi(g) \rangle)$ . Since  $h$  was arbitrary, we have  $G \cong \pi(G) \leq N_{S_q}(\langle \pi(g) \rangle)$ . Therefore,  $G$  is isomorphic to a subgroup of the normalizer in  $S_q$  of the cyclic group generated by the  $q$ -cycle  $(1\ 2\ \cdots\ q)$ . ■

**Exercise 4.3.29**

Let  $p$  be a prime and let  $G$  be a group of order  $p^\alpha$ . Prove that  $G$  has a subgroup of order  $p^\beta$ , for every  $\beta$  with  $0 \leq \beta \leq \alpha$ . [Use Theorem 8 and induction on  $\alpha$ .]

**Solution.** We proceed by induction on  $\alpha$ . If  $\alpha = 0$ , then  $G$  is the trivial group, which has a subgroup of order  $p^0 = 1$ . Now suppose the result holds for all groups of order  $p^k$  for some  $k \geq 0$ . Let  $G$  be a group of order  $p^{k+1}$ . By Theorem 8,  $Z(G)$  is nontrivial, so there exists some  $x \in Z(G)$  such that  $|x| = p$ . Then  $\langle x \rangle$  is a normal subgroup of  $G$  of order  $p$ . Consider the quotient group  $G/\langle x \rangle$ , which has order  $p^k$ . By the induction hypothesis, for every  $\beta$  with  $0 \leq \beta \leq k$ , there exists a subgroup  $H/\langle x \rangle$  of  $G/\langle x \rangle$  such that  $|H/\langle x \rangle| = p^\beta$ . By the Lattice

Isomorphism Theorem, there exists a subgroup  $H$  of  $G$  such that  $\langle x \rangle \leq H$  and  $|H| = p^{\beta+1}$ . Therefore, for every  $\beta$  with  $1 \leq \beta \leq k+1$ , there exists a subgroup of  $G$  of order  $p^\beta$ . Since the trivial subgroup has order  $p^0 = 1$ , the result holds for all  $\beta$  with  $0 \leq \beta \leq k+1$ . By induction, the result holds for all  $\alpha \geq 0$ . ■

**Exercise 4.3.30**

If  $G$  is a group of odd order, prove for any nonidentity element  $x \in G$  that  $x$  and  $x^{-1}$  are not conjugate in  $G$ .

**Solution.** Suppose for contradiction that  $x$  and  $x^{-1}$  are conjugate in  $G$ . Then there exists  $g \in G$  such that  $gxg^{-1} = x^{-1}$ . If we conjugate both sides by  $g$  again, we obtain

$$g^2 x g^{-2} = g x^{-1} g^{-1} = (g x g^{-1})^{-1} = (x^{-1})^{-1} = x$$

so that  $g^2 x = x g^2$ . Then  $g^2 \in C_G(x)$ . Consider the quotient group  $G/C_G(x)$ . Since  $g^2 \in C_G(x)$ , we have  $(gC_G(x))^2 = C_G(x)$ , so that the order of  $gC_G(x)$  in  $G/C_G(x)$  is 1 or 2. However, since  $G$  has odd order, then  $G/C_G(x)$  also has odd order, so the order of  $gC_G(x)$  cannot be 2. Therefore, the order of  $gC_G(x)$  is 1, so that  $g \in C_G(x)$ . But this implies that  $gxg^{-1} = x$ , contradicting our assumption. Therefore,  $x$  and  $x^{-1}$  are not conjugate in  $G$ . ■

**Exercise 4.3.31**

Using the usual generators and relations for the dihedral group  $D_{2n}$  (cf. Section 1.2) show that for  $n = 2k$  an even integer the conjugacy classes in  $D_{2n}$  are the following:  $\{1\}$ ,  $\{r^k\}$ ,  $\{r^{\pm 1}\}$ ,  $\{r^{\pm 2}\}$ ,  $\dots$ ,  $\{r^{\pm(k-1)}\}$ ,  $\{sr^{2b} \mid b = 1, \dots, k\}$  and  $\{sr^{2b-1} \mid b = 1, \dots, k\}$ . Give the class equation for  $D_{2n}$ .

**Solution.** We immediately have two conjugacy classes,  $\{1\}$  and  $\{r^n\}$ , since  $r^n \in Z(D_{2n})$  when  $n$  is even. Moreover, recall that the elements of  $D_{2n}$  are of the form  $r^m$  or  $sr^m$  for some integer  $m$ . Consider the conjugacy class of  $r^k$  for some  $1 \leq k < n$  except  $k = n/2$ . For any  $j$ , we have

$$r^j r^k r^{-j} = r^k, \quad \text{and} \quad sr^j r^k r^{-j} s = r^{-k}$$

so that the conjugacy class of  $r^k$  is  $\{r^{\pm k}\}$ . Now consider the conjugacy class of  $sr^m$  for some integer  $m$ . For any  $j$ , we have

$$r^j sr^m r^{-j} = sr^{m-2j}, \quad \text{and} \quad sr^j sr^m r^{-j} s = r^{-(m-2j)}$$

so that the conjugacy class of  $sr^m$  is  $\{sr^{m-2j} \mid j \in \mathbb{Z}\}$ . If  $m$  is even, then this conjugacy class is  $\{sr^{2b} \mid b = 1, \dots, k\}$ ; if  $m$  is odd, then this conjugacy class is  $\{sr^{2b-1} \mid b = 1, \dots, k\}$ . Therefore, the conjugacy classes in  $D_{2n}$  when  $n$  is even are  $\{1\}$ ,  $\{r^k\}$ ,  $\{r^{\pm 1}\}$ ,  $\{r^{\pm 2}\}$ ,  $\dots$ ,  $\{r^{\pm(k-1)}\}$ ,  $\{sr^{2b} \mid b = 1, \dots, k\}$  and  $\{sr^{2b-1} \mid b = 1, \dots, k\}$ . Lastly, the class equation for  $D_{2n}$  is

$$|D_{2n}| = 1 + 1 + 2(k-1) + k + k. \quad \blacksquare$$

**Exercise 4.3.32**

For  $n = 2k + 1$  an odd integer, show that the conjugacy classes in  $D_{2n}$  are  $\{1\}$ ,  $\{r^{\pm 1}\}$ ,  $\{r^{\pm 2}\}$ ,  $\dots$ ,  $\{r^{\pm k}\}$ , and  $\{sr^b \mid b = 1, \dots, n\}$ . Give the class equation for  $D_{2n}$ .

**Solution.** We immediately have the conjugacy class  $\{1\}$ . Moreover, recall that the elements of  $D_{2n}$  are of the form  $r^m$  or  $sr^m$  for some integer  $m$ . Consider the conjugacy class of  $r^k$  for some  $1 \leq k \leq n$  except  $k = n/2$ . For any  $j$ , we have

$$r^j r^k r^{-j} = r^k, \quad \text{and} \quad sr^j r^k r^{-j} s = r^{-k}$$

so that the conjugacy class of  $r^k$  is  $\{r^{\pm k}\}$ . Now consider the conjugacy class of  $sr^m$  for some integer  $m$ . For any  $j$ , we have

$$r^j sr^m r^{-j} = sr^{m-2j}, \quad \text{and} \quad sr^j sr^m r^{-j} s = r^{-(m-2j)}$$

so that the conjugacy class of  $sr^m$  is  $\{sr^{m-2j} \mid j \in \mathbb{Z}\}$ . Since  $n$  is odd, this conjugacy class is  $\{sr^b \mid b = 1, \dots, n\}$ . Therefore, the conjugacy classes in  $D_{2n}$  when  $n$  is odd are  $\{1\}$ ,  $\{r^{\pm 1}\}$ ,  $\{r^{\pm 2}\}$ ,  $\dots$ ,  $\{r^{\pm k}\}$ , and  $\{sr^b \mid b = 1, \dots, n\}$ . Lastly, the class equation for  $D_{2n}$  is

$$|D_{2n}| = 1 + 2k + n. \quad \blacksquare$$



(\*) **Exercise 4.3.33**

This exercise gives a formula for the size of each conjugacy class in  $S_n$ . Let  $\sigma$  be a permutation in  $S_n$  and let  $m_1, m_2, \dots, m_s$  be the *distinct* integers which appear in the cycle type of  $\sigma$  (including 1-cycles). For each  $i \in \{1, 2, \dots, s\}$  assume  $\sigma$  has  $k_i$  cycles of length  $m_i$  (so that  $\sum_1^s k_i m_i = n$ ). Prove that the number of conjugates of  $\sigma$  is

$$\frac{n!}{(k_1! m_1^{k_1})(k_2! m_2^{k_2}) \cdots (k_s! m_s^{k_s})}$$

[See [Exercise 1.3.6](#) and [Exercise 1.3.7](#) where this formula was given in some special cases.]

**Solution.** We start with  $n$  integers. We see that there are  $n!$  ways to arrange these integers. Now consider the  $k_1$  cycles of length  $m_1$  in the cycle decomposition of  $\sigma$ . Clearly, we must use  $k_1 m_1$  of the  $n$  integers to form these cycles. However, there are  $m_1^{k_1}$  ways to arrange the integers within each of these  $k_1$  cycles, and there are  $k_1!$  ways to arrange the  $k_1$  cycles among themselves. Therefore, we must divide  $n!$  by  $(k_1! m_1^{k_1})$  to account for these arrangements. Continuing in this manner for each  $i$  from 1 to  $s$ , we see that the total number of distinct arrangements of the  $n$  integers that correspond to the same cycle type as  $\sigma$  is given by

$$\frac{n!}{(k_1! m_1^{k_1})(k_2! m_2^{k_2}) \cdots (k_s! m_s^{k_s})}.$$

Since the conjugacy class of  $\sigma$  in  $S_n$  consists of all permutations with the same cycle type as  $\sigma$ , the number of conjugates of  $\sigma$  is given by this formula. ■

**Exercise 4.3.34**

Prove that if  $p$  is a prime and  $P$  is a subgroup of  $S_p$  of order  $p$ , then  $|N_{S_p}(P)| = p(p-1)$ . [Argue that every conjugate of  $P$  contains exactly  $p-1$   $p$ -cycles and use the formula for the number of  $p$ -cycles to compute the index of  $N_{S_p}(P)$  in  $S_p$ .]

**Solution.** Note that  $P$  is cyclic of order  $p$ , so  $P = \langle \sigma \rangle$  for some  $p$ -cycle  $\sigma \in S_p$ . Moreover, every nonidentity element of  $P$  is a  $p$ -cycle, so  $P$  contains exactly  $p-1$   $p$ -cycles. Now for some  $\tau \in S_p$ , the conjugate  $\tau P \tau^{-1} = \langle \tau \sigma \tau^{-1} \rangle$  also has order  $p$  and contains exactly  $p-1$   $p$ -cycles. Therefore, each conjugate of  $P$  contains exactly  $p-1$   $p$ -cycles. Note that the number of distinct  $p$ -cycles in  $S_p$  is given by  $p!/p = (p-1)!$ .

Recall by Proposition 4.6 that the size of the conjugacy class of  $P$  in  $S_p$  is given by  $|S_p : N_{S_p}(P)|$ . Let this index be  $k$ . Since each conjugate of  $P$  contains exactly  $p-1$   $p$ -cycles, the total number of distinct  $p$ -cycles contained in all conjugates of  $P$  is  $k(p-1)$ . However, since every  $p$ -cycle in  $S_p$  is contained in some conjugate of  $P$ , we have  $k(p-1) = (p-1)!$ . Therefore,  $k = (p-2)!$ , so that

$$|N_{S_p}(P)| = \frac{|S_p|}{k} = \frac{p!}{(p-2)!} = p(p-1). \quad \blacksquare$$

**Exercise 4.3.35**

Let  $p$  be a prime. Find a formula for the number of conjugacy classes of elements of order  $p$  in  $S_n$  (using the greatest integer function).

**Solution.** Note that elements of order  $p$  in  $S_n$  are products of disjoint  $p$ -cycles. Moreover, each  $p$ -cycle is conjugate to any other  $p$ -cycle in  $S_n$ . Therefore, the conjugacy class of an element of order  $p$  in  $S_n$  is determined by the number of disjoint  $p$ -cycles in its cycle decomposition. Let  $k$  be the number of disjoint  $p$ -cycles in the cycle decomposition of such an element. Then we have  $1 \leq k \leq \lfloor n/p \rfloor$ , hence the number of conjugacy classes of elements of order  $p$  in  $S_n$  is given by  $\lfloor n/p \rfloor$ . ■



(\*) **Exercise 4.3.36**

Let  $\pi : G \rightarrow S_G$  be the left regular representation afforded by the action of  $G$  on itself by left multiplication. For each  $g \in G$  denote the permutation  $\pi(g)$  by  $\sigma_g$  so that  $\sigma_g(x) = gx$  for all  $x \in G$ . Let  $\lambda : G \rightarrow S_G$  be the permutation representation afforded by the corresponding right action of  $G$  on itself, and for each  $h \in G$  denote the permutation  $\lambda(h)$  by  $\tau_h$ . Thus  $\tau_h(x) = xh^{-1}$  for all  $x \in G$  ( $\lambda$  is called the *right regular representation* of  $G$ ).

- (a) Prove that  $\sigma_g$  and  $\tau_h$  commute for all  $g, h \in G$ . (Thus the centralizer in  $S_G$  of  $\pi(G)$  contains the subgroup  $\lambda(G)$  which is isomorphic to  $G$ ).
- (b) Prove that  $\sigma_g = \tau_g$  if and only if  $g$  is an element of order 1 or 2 in the center of  $G$ .
- (c) Prove that  $\sigma_g = \tau_h$  if and only if  $g$  and  $h$  lie in the center of  $G$ . Deduce that  $\pi(G) \cap \lambda(G) = \pi(Z(G)) = \lambda(Z(G))$ .

**Solution.**

- (a) For any  $x \in G$ , we have

$$\sigma_g(\tau_h(x)) = \sigma_g(xh^{-1}) = g(xh^{-1}) = (gx)h^{-1} = \tau_h(gx) = \tau_h(\sigma_g(x)).$$

Therefore,  $\sigma_g$  and  $\tau_h$  commute for all  $g, h \in G$ .

- (b) If  $\sigma_g = \tau_g$ , then  $gx = xg^{-1}$  for all  $x \in G$ . In particular,  $x = 1$  implies  $g = g^{-1}$  so that  $g^2 = 1$ , hence  $|g|$  is 1 or 2. Moreover, for any  $x \in G$ , we have  $gx = xg^{-1} = xg$ , so that  $g \in Z(G)$ .

If  $g$  is an element of order 1 or 2 in the center of  $G$ , then for any  $x \in G$ , we have  $\sigma_g(x) = gx = xg = xg^{-1} = \tau_g(x)$ . Therefore,  $\sigma_g = \tau_g$ .

- (c) If  $\sigma_g = \tau_h$ , then  $gx = xh^{-1}$  for all  $x \in G$ . In particular,  $x = 1$  implies  $g = h^{-1}$ , so that  $h = g^{-1}$ . Therefore,  $gx = xg^{-1}$  for all  $x \in G$ , so that by part (b),  $g$  lies in the center of  $G$ . Since  $h = g^{-1}$ , then  $h$  also lies in the center of  $G$ .

If  $g$  and  $h$  lie in the center of  $G$ , then for any  $x \in G$ , we have  $\sigma_g(x) = gx = xg = xh^{-1} = \tau_h(x)$ . Therefore,  $\sigma_g = \tau_h$ . Moreover, if  $\sigma_g \in \pi(G) \cap \lambda(G)$ , then there exists  $h \in G$  such that  $\sigma_g = \tau_h$ . By the above result, both  $g$  and  $h$  lie in the center of  $G$ , so that  $\sigma_g \in \pi(Z(G))$  and  $\tau_h \in \lambda(Z(G))$ . Conversely, if  $g \in Z(G)$ , then by part (b), we have  $\sigma_g = \tau_g$ , so that  $\sigma_g \in \pi(G) \cap \lambda(G)$ . Therefore,  $\pi(G) \cap \lambda(G) = \pi(Z(G)) = \lambda(Z(G))$ . ■

## 4.4 Automorphisms

### (\*) Exercise 4.4.1

Let  $G$  be a group. If  $\sigma \in \text{Aut}(G)$  and  $\phi_g$  is conjugation by  $g$ , prove that  $\sigma\phi_g\sigma^{-1} = \phi_{\sigma(g)}$ . Deduce that  $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$ . (The group  $\text{Aut}(G)/\text{Inn}(G)$  is called the *outer automorphism group* of  $G$ .)

**Solution.** For any  $x \in G$ , then

$$(\sigma\phi_g\sigma^{-1})(x) = \sigma(\phi_g(\sigma^{-1}(x))) = \sigma(g\sigma^{-1}(x)g^{-1}) = \sigma(g)\sigma(\sigma^{-1}(x))\sigma(g)^{-1} = \sigma(g)x\sigma(g)^{-1} = \phi_{\sigma(g)}(x).$$

Then  $\sigma\phi_g\sigma^{-1} = \phi_{\sigma(g)}$ . Therefore, for any  $\phi_g \in \text{Inn}(G)$  and any  $\sigma \in \text{Aut}(G)$ , we have  $\sigma\phi_g\sigma^{-1} \in \text{Inn}(G)$ , so that  $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$ . ■

### (\*) Exercise 4.4.2

Prove that if  $G$  is an abelian group of order  $pq$ , where  $p$  and  $q$  are distinct primes, then  $G$  is cyclic. (Use Cauchy's Theorem to produce elements of order  $p$  and  $q$  and consider the order of their product.)

**Solution.** By Cauchy's Theorem, there are elements  $g, h \in G$  such that  $|g| = p$  and  $|h| = q$ . Since  $G$  is abelian, we have  $|gh| = \text{lcm}(|g|, |h|) = \text{lcm}(p, q) = pq$ . Therefore,  $G = \langle gh \rangle$  is cyclic. ■

### Exercise 4.4.3

Prove that under any automorphism of  $D_8$ ,  $r$  has at most 2 possible images and  $s$  has at most 4 possible images (where  $r$  and  $s$  are the usual generators). Deduce that  $|\text{Aut}(D_8)| \leq 8$ .

**Solution.** Since  $|r| = 4$ , any automorphism  $\sigma$  of  $D_8$  must send  $r$  to an element of order 4. The only elements of order 4 in  $D_8$  are  $r$  and  $r^3$ , so  $r$  has at most 2 possible images under  $\sigma$ . Moreover,  $|s| = 2$ , so  $\sigma(s)$  must be an element of order 2. The elements of order 2 in  $D_8$  are  $s, sr, sr^2$ , and  $sr^3$ , so  $s$  has at most 4 possible images under  $\sigma$ . Therefore, there are at most  $2 \cdot 4 = 8$  possible automorphisms of  $D_8$ , so that  $|\text{Aut}(D_8)| \leq 8$ . ■

### Exercise 4.4.4

Use arguments similar to those in the preceding exercise to show that  $|\text{Aut}(Q_8)| \leq 24$ .

**Solution.** Note that  $|i| = |j| = |k| = 4$ , so any automorphism  $\sigma$  of  $Q_8$  must send  $i$  to an element of order 4. The only elements of order 4 in  $Q_8$  are  $i, j, k, i^{-1}, j^{-1}, k^{-1}$ , so  $i$  has at most 6 possible images under  $\sigma$ . Moreover, since  $ij = k$ , we have  $\sigma(i)\sigma(j) = \sigma(k)$ . Therefore, once we choose the image of  $i$ , there are 4 possible choices for the image of  $j$  (since it cannot be the inverse of the image of  $i$ ). Hence, there are at most  $6 \cdot 4 = 24$  possible automorphisms of  $Q_8$ , so that  $|\text{Aut}(Q_8)| \leq 24$ . ■

### Exercise 4.4.5

Use the fact that  $D_8 \trianglelefteq D_{16}$  to prove that  $\text{Aut}(D_8) \cong D_8$ .

**Solution.** Consider the subgroup  $H = \langle r^2, s \rangle$  of  $D_{16}$ . Note that  $(r^2)^4 = s^2 = 1$ , so this satisfies the same relations as  $D_8$ , hence  $H \cong D_8$ . Moreover,  $H$  is normal in  $D_{16}$  as it has index 2. Then  $N_{D_{16}}(H) = D_{16}$ . Moreover,  $C_{D_{16}}(H)$  can be computed by considering the centralizers of the generators of  $H$ . Note that  $C_{D_{16}}(r^2) = \langle r \rangle$  since only rotations commute with each other. Consider now the centralizer of  $s$ . For any rotation  $r^k$ , we have  $sr^k = r^k s$  if and only if  $r^{-k} = r^k$ , which holds if and only if  $k = 0$  or  $k = 4$ . For any reflection  $sr^k$ , we obtain the same conclusion so that  $C_{D_{16}}(s) = \{1, r^4, s, sr^4\}$ . Intersecting these, it is easy to see that  $C_{D_{16}}(H) = Z(D_{16})$ .

We now have that  $N_{D_{16}}(H)/C_{D_{16}}(H) \cong D_{16}/Z(D_{16}) \cong D_8$ . Now recall that  $D_{16}/Z(D_{16})$  is still isomorphic to a subgroup of  $\text{Aut}(H)$  when acted on by conjugation. By Exercise 4.4.3, we know that  $|\text{Aut}(D_8)| \leq 8$ , so that  $\text{Aut}(H) \cong D_8$ . Therefore,  $\text{Aut}(D_8) \cong D_8$ . ■

**Exercise 4.4.6**

Prove that characteristic subgroups are normal. Give an example of a normal subgroup that is not characteristic.

**Solution.** If  $H \text{ char } G$ , then  $\varphi(H) = H$  for every  $\varphi \in \text{Aut}(G)$ . In particular, this holds for every  $\varphi_g \in \text{Inn}(G)$  so that  $\varphi_g(H) = gHg^{-1} = H$  for every  $g \in G$ . Therefore,  $H \trianglelefteq G$ .

Consider the abelian group  $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ . Then every subgroup of  $G$  is normal since  $G$  is abelian. However, the subgroup  $H = \{(0, 0), (1, 0)\}$  is not characteristic in  $G$  since the automorphism  $\sigma : G \rightarrow G$  defined by  $\sigma((a, b)) = (b, a)$  sends  $H$  to the distinct subgroup  $\{(0, 0), (0, 1)\}$ . Therefore,  $H$  is a normal subgroup of  $G$  that is not characteristic. ■

**Exercise 4.4.7**

If  $H$  is the unique subgroup of a given order in a group  $G$ , prove that  $H$  is characteristic in  $G$ .

**Solution.** For any  $\sigma \in \text{Aut}(G)$ , the subgroup  $\sigma(H)$  has the same order as  $H$ . Since  $H$  is the unique subgroup of that order in  $G$ , we must have  $\sigma(H) = H$ . Therefore,  $H$  is characteristic in  $G$ . ■

**(\*) Exercise 4.4.8**

Let  $G$  be a group with subgroups  $H$  and  $K$  with  $H \leq K$ .

- Prove that if  $H$  is characteristic in  $K$  and  $K$  is normal in  $G$ , then  $H$  is normal in  $G$ .
- Prove that if  $H$  is characteristic in  $K$  and  $K$  is characteristic in  $G$ , then  $H$  is characteristic in  $G$ . Use this to prove that the Klein 4-group  $V_4$  is characteristic in  $S_4$ .
- Give an example to show that if  $H$  is normal in  $K$  and  $K$  is characteristic in  $G$ , then  $H$  need not be normal in  $G$ .

**Solution.**

- For any  $g \in G$ , consider the inner automorphism  $\varphi_g \in \text{Inn}(G)$ . Since  $K \trianglelefteq G$ , we have  $\varphi_g(K) = K$ . Moreover, since  $H \text{ char } K$ , we have  $\varphi_g(H) = H$ . Therefore,  $gHg^{-1} = H$  for all  $g \in G$ , so that  $H \trianglelefteq G$ .
- For any  $\sigma \in \text{Aut}(G)$ , since  $K \text{ char } G$ , we have  $\sigma(K) = K$ . Moreover, since  $H \text{ char } K$ , we have  $\sigma(H) = H$ . Therefore,  $H \text{ char } G$ .

Consider  $V_4$ . By [Exercise 3.5.8](#),  $V_4$  is the unique subgroup of order 4 in  $A_4$  so that  $V_4 \text{ char } A_4$  by [Exercise 4.4.7](#). Moreover,  $A_4$  is the unique subgroup of order 12 in  $S_4$ , for otherwise if  $H \leq S_4$  such that  $|H| = 12$ , then it would contain only odd permutations, which is impossible since the product of two odd permutations is an even permutation. Therefore,  $A_4 \text{ char } S_4$  by [Exercise 4.4.7](#). By the above result,  $V_4 \text{ char } S_4$ .

- Consider  $A_4$ . Then  $V_4 \text{ char } A_4$  as shown previously. Consider  $H = \langle (1\ 2)(3\ 4) \rangle \leq V_4$ . Since  $V_4$  is abelian, then  $H \trianglelefteq V_4$ . However, taking  $(1\ 2\ 3) \in A_4$ , we have

$$(1\ 2\ 3)(1\ 2)(3\ 4)(3\ 2\ 1) = (1\ 4)(2\ 3) \notin H$$

so that  $H$  is not normal in  $A_4$ . ■

**Exercise 4.4.9**

If  $r, s$  are the usual generators for the dihedral group  $D_{2n}$ , use the preceding two exercises to deduce that every subgroup of  $\langle r \rangle$  is normal in  $D_{2n}$ .

**Solution.** By [Exercise 3.1.34](#), we know  $\langle r \rangle$  is normal in  $D_{2n}$ . Moreover, every subgroup of a cyclic group is characteristic, so every subgroup of  $\langle r \rangle$  is characteristic in  $\langle r \rangle$ . Since  $\langle r \rangle \trianglelefteq D_{2n}$ , then by [Exercise 4.4.8](#), every subgroup of  $\langle r \rangle$  is normal in  $D_{2n}$ . ■

**Exercise 4.4.10**

Let  $G$  be a group, let  $A$  be an abelian normal subgroup of  $G$ , and write  $\bar{G} = G/A$ . Show that  $\bar{G}$  acts (on the left) by conjugation on  $A$  by  $\bar{g} \cdot a = gag^{-1}$ , where  $g$  is any representative of the coset  $\bar{g}$  (in particular, show that this action is well defined). Give an explicit example to show that this action is not well defined if  $A$  is non-abelian.

**Solution.** Let  $\bar{g}$  and  $\bar{h}$  be the same coset in  $\bar{G}$ . Then there exists some  $a_0 \in A$  such that  $h = ga_0$ . Then for any  $a \in A$ , we have

$$\bar{g} \cdot a = gag^{-1} = gaa_0a_0^{-1}g^{-1} = ga_0a(ga_0)^{-1} = hah^{-1} = \bar{h} \cdot a$$

since  $A$  is abelian. Therefore, the action is well defined.

Consider  $G = S_3$  and  $A = S_3$  itself. Then  $\bar{1} = (1\ 2)$  in  $\bar{G}$ . However, for  $a = (1\ 3)$ , we have  $\bar{1} \cdot a = a = (1\ 3)$  but,  $(1\ 2) \cdot a = (1\ 2)(1\ 3)(1\ 2) = (2\ 3) \neq a$ . Therefore, the action is not well defined if  $A$  is non-abelian. ■

**Exercise 4.4.11**

If  $p$  is a prime and  $P$  is a subgroup of  $S_p$  of order  $p$ , prove that  $N_{S_p}(P)/C_{S_p}(P) \cong \text{Aut}(P)$ . (Use [Exercise 4.3.34](#)).

**Solution.** By [Exercise 4.3.34](#), we know that  $|N_{S_p}(P)| = p(p-1)$ . Moreover, since  $P$  is cyclic of order  $p$ , we have  $\text{Aut}(P) \cong (\mathbb{Z}/p\mathbb{Z})^\times$  by Proposition 4.16, so that  $|\text{Aut}(P)| = p-1$ . Let  $N_{S_p}$  act on  $P$  by conjugation. Then we have the map

$$\varphi : N_{S_p}(P) \rightarrow \text{Aut}(P), \quad \varphi(g)(x) = gxg^{-1}$$

for all  $g \in N_{S_p}(P)$  and  $x \in P$ . Recall that  $\ker \varphi = C_{S_p}(P)$  by Proposition 4.13. Therefore, by the First Isomorphism Theorem, we have

$$N_{S_p}(P)/C_{S_p}(P) \cong \text{im } \varphi \leq \text{Aut}(P).$$

Since  $P$  is cyclic, then the only permutations that centralize  $P$  are the elements of  $P$  itself, so that  $C_{S_p}(P) = P$ . Therefore,

$$|N_{S_p}(P)/C_{S_p}(P)| = \frac{|N_{S_p}(P)|}{|C_{S_p}(P)|} = \frac{p(p-1)}{p} = p-1 = |\text{Aut}(P)|.$$

Hence,  $\text{im } \varphi = \text{Aut}(P)$  so that  $N_{S_p}(P)/C_{S_p}(P) \cong \text{Aut}(P)$ . ■

**Exercise 4.4.12**

Let  $G$  be a group of order 3825. Prove that if  $H$  is a normal subgroup of order 17 in  $G$  then  $H \leq Z(G)$ .

**Solution.** Observe that  $3825 = 15^2 \cdot 17$ . Since  $|H| = 17$  a prime, then  $H$  is cyclic. Moreover, we have  $\text{Aut}(H) \cong (\mathbb{Z}/17\mathbb{Z})^\times$  by Proposition 4.16, so that  $|\text{Aut}(H)| = 16$ . Now consider the action of  $G$  on  $H$  by conjugation. This gives the map

$$\varphi : G \rightarrow \text{Aut}(H), \quad \varphi(g)(x) = gxg^{-1}$$

for all  $g \in G$  and  $x \in H$ . Recall that  $\ker \varphi = C_G(H)$  by Proposition 4.13. Therefore, by the First Isomorphism Theorem, we have

$$G/C_G(H) \cong \text{im } \varphi \leq \text{Aut}(H).$$

Hence,  $|G/C_G(H)|$  divides  $|\text{Aut}(H)| = 16$ . However,  $|G| = 3825$  is relatively prime to 16, so that  $|G/C_G(H)| = 1$ . Therefore,  $G = C_G(H)$ , so that  $H \leq Z(G)$ . ■

**Exercise 4.4.13**

Let  $G$  be a group of order 203. Prove that if  $H$  is a normal subgroup of order 7 in  $G$  then  $H \leq Z(G)$ . Deduce that  $G$  is abelian in this case.

**Solution.** Note that  $203 = 7 \cdot 29$ . The proof of showing that  $H \leq Z(G)$  is similar to the previous exercise. To see that  $G$  is abelian, consider the quotient group  $G/H$ . Since  $|G/H| = 29$  is prime, then  $G/H$  is cyclic. By [Exercise 3.1.36](#), we have that  $G$  is abelian. ■

**Exercise 4.4.14**

Let  $G$  be a group of order 1575. Prove that if  $H$  is a normal subgroup of order 9 in  $G$  then  $H \leq Z(G)$ .

**Solution.** This exercise is similar to the previous two. Following the proofs, we will find that  $|G/C_G(H)|$  divides  $|\text{Aut}(H)| = 6$  or 48. However,  $1575 = 3^2 \cdot 5^2 \cdot 7$  results in  $(|G|, |\text{Aut}(H)|) = 1$  or 3. If the latter is true, then  $|C_G(H)| = 525$ . However, this is impossible since  $H \leq C_G(H)$  and  $|H| = 9$  does not divide 525. Therefore, we must have  $|G/C_G(H)| = 1$ , so that  $G = C_G(H)$  and hence  $H \leq Z(G)$ . ■

**Exercise 4.4.15**

Prove that each of the following (multiplicative) groups is cyclic:  $(\mathbb{Z}/5\mathbb{Z})^\times$ ,  $(\mathbb{Z}/9\mathbb{Z})^\times$ , and  $(\mathbb{Z}/18\mathbb{Z})^\times$ .

**Solution.** Since  $|(\mathbb{Z}/5\mathbb{Z})^\times| = 4$ , then  $(\mathbb{Z}/5\mathbb{Z})^\times$  is isomorphic to either  $Z_4$  or  $V_4$ . However,  $\bar{2} \in (\mathbb{Z}/5\mathbb{Z})^\times$  has order 4, so that  $(\mathbb{Z}/5\mathbb{Z})^\times \cong Z_4$  is cyclic.

Observe that  $|(\mathbb{Z}/9\mathbb{Z})^\times| = |(\mathbb{Z}/18\mathbb{Z})^\times| = 6$ . By [Exercise 4.2.10](#), groups of order 6 are isomorphic to either  $Z_6$  or  $S_3$ . However, both groups are abelian, so that  $(\mathbb{Z}/9\mathbb{Z})^\times$  and  $(\mathbb{Z}/18\mathbb{Z})^\times$  must be isomorphic to  $Z_6$ . Therefore, both groups are cyclic. ■

**(\*) Exercise 4.4.16**

Prove that  $(\mathbb{Z}/24\mathbb{Z})^\times$  is an elementary abelian group of order 8. (We shall see later that  $(\mathbb{Z}/n\mathbb{Z})^\times$  is an elementary abelian group if and only if  $n \mid 24$ .)

**Solution.** We first prove the lemma: let  $a, b, m, n \in \mathbb{Z}$  such that  $(m, n) = 1$ . If  $a \equiv b \pmod{m}$  and  $a \equiv b \pmod{n}$ , then  $a \equiv b \pmod{mn}$ . Indeed, we have that

$$a - b = xm \quad \text{and} \quad a - b = yn$$

for some  $x, y \in \mathbb{Z}$ . Therefore,  $xm = yn$ , so that  $m \mid yn$ . Since  $(m, n) = 1$ , then  $m \mid y$ . Letting  $y = km$  for some  $k \in \mathbb{Z}$ , we have

$$a - b = yn = kmn,$$

so that  $a \equiv b \pmod{mn}$ .

Let  $x \in (\mathbb{Z}/24\mathbb{Z})^\times$ . Note that  $x$  is always odd. Then  $\gcd(x, 24) = 1$ , hence  $(x, 3) = (x, 8) = 1$ . Since  $(x, 3) = 1$ , then  $x \equiv 1$  or  $2 \pmod{3}$ . Then  $x^2 \equiv 1 \pmod{3}$ . Since  $(x, 8) = 1$ , let  $x = 2k + 1$  for some  $k \in \mathbb{Z}$ . Then

$$x^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1.$$

Since one of  $k$  or  $k + 1$  is even, then  $4k(k + 1)$  is divisible by 8, so that  $x^2 \equiv 1 \pmod{8}$ . By the lemma above, we have  $x^2 \equiv 1 \pmod{24}$ . Therefore, every element of  $(\mathbb{Z}/24\mathbb{Z})^\times$  has order dividing 2, so that  $(\mathbb{Z}/24\mathbb{Z})^\times$  is an elementary abelian group. ■

**Exercise 4.4.17**

Let  $G = \langle x \rangle$  be a cyclic group of order  $n$ . For  $n = 2, 3, 4, 5, 6$  write out the elements of  $\text{Aut}(G)$  explicitly (by Proposition 16 above we know  $\text{Aut}(G) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ , so for each element  $a \in (\mathbb{Z}/n\mathbb{Z})^\times$ , write out explicitly what the automorphism  $\psi_a$  does to the elements  $\{1, x, x^2, \dots, x^{n-1}\}$  of  $G$ ).

**Solution.**

- $n = 2$ :  $(\mathbb{Z}/2\mathbb{Z})^\times = \{\bar{1}\}$ . Then  $\psi_1 : x^k \mapsto x^{1 \cdot k} = x^k$  for  $k = 0, 1$ . Therefore,  $\text{Aut}(G) = \{\text{id}\}$ .
- $n = 3$ :  $(\mathbb{Z}/3\mathbb{Z})^\times = \{\bar{1}, \bar{2}\}$ . Then

$$\psi_1 : x^k \mapsto x^{1 \cdot k} = x^k, \quad \psi_2 : x^k \mapsto x^{2 \cdot k}$$

for  $k = 0, 1, 2$ . Therefore,  $\text{Aut}(G) = \{\psi_1, \psi_2\}$ .

- $n = 4$ :  $(\mathbb{Z}/4\mathbb{Z})^\times = \{\bar{1}, \bar{3}\}$ . Then

$$\psi_1 : x^k \mapsto x^{1 \cdot k} = x^k, \quad \psi_3 : x^k \mapsto x^{3 \cdot k}$$

for  $k = 0, 1, 2, 3$ . Therefore,  $\text{Aut}(G) = \{\psi_1, \psi_3\}$ .

- $n = 5$ :  $(\mathbb{Z}/5\mathbb{Z})^\times = \{\bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ . Then

$$\psi_1 : x^k \mapsto x^{1 \cdot k}, \quad \psi_2 : x^k \mapsto x^{2 \cdot k}, \quad \psi_3 : x^k \mapsto x^{3 \cdot k}, \quad \psi_4 : x^k \mapsto x^{4 \cdot k}$$

for  $k = 0, 1, 2, 3, 4$ . Therefore,  $\text{Aut}(G) = \{\psi_1, \psi_2, \psi_3, \psi_4\}$ .

- $n = 6$ :  $(\mathbb{Z}/6\mathbb{Z})^\times = \{\bar{1}, \bar{5}\}$ . Then

$$\psi_1 : x^k \mapsto x^{1 \cdot k} = x^k, \quad \psi_5 : x^k \mapsto x^{5 \cdot k}$$

for  $k = 0, 1, 2, 3, 4, 5$ . Therefore,  $\text{Aut}(G) = \{\psi_1, \psi_5\}$ . ■

**(\*) Exercise 4.4.18**

This exercise shows that for  $n \neq 6$  every automorphism of  $S_n$  is inner. Fix an integer  $n \geq 2$  with  $n \neq 6$ .

- Prove that the automorphism group of a group  $G$  permutes the conjugacy classes of  $G$ , i.e. for each  $\sigma \in \text{Aut}(G)$  and each conjugacy class  $\mathcal{K}$  of  $G$ , the set  $\sigma(\mathcal{K})$  is also a conjugacy class of  $G$ .
- Let  $\mathcal{K}$  be the conjugacy class of transpositions in  $S_n$  and let  $\mathcal{K}'$  be the conjugacy class of any element of order 2 in  $S_n$  that is not a transposition. Prove that  $|\mathcal{K}| \neq |\mathcal{K}'|$ . Deduce that any automorphism of  $S_n$  sends transpositions to transpositions. (See [Exercise 4.3.33](#).)
- Prove that for each  $\sigma \in \text{Aut}(S_n)$ ,

$$\sigma : (12) \mapsto (a b_2), \quad \sigma : (13) \mapsto (a b_3), \quad \dots, \quad \sigma : (1n) \mapsto (a b_n),$$

for some distinct integers  $a, b_2, b_3, \dots, b_n \in \{1, 2, \dots, n\}$ .

- Show that  $(12), (13), \dots, (1n)$  generate  $S_n$  and deduce that any automorphism of  $S_n$  is uniquely determined by its action on these elements. Use (c) to show that  $S_n$  has at most  $n!$  automorphisms and conclude that  $\text{Aut}(S_n) = \text{Inn}(S_n)$  for  $n \neq 6$ .

**Solution.**

- Let  $\mathcal{K}$  be a conjugacy class of  $G$  and let  $\sigma \in \text{Aut}(G)$ . For any  $x \in \mathcal{K}$ , we have  $\sigma(x) \in G$ . Consider the conjugacy class  $\mathcal{K}'$  of  $\sigma(x)$  in  $G$ . For any  $y \in \mathcal{K}$ , there exists some  $g \in G$  such that  $y = gxg^{-1}$ . Then

$$\sigma(y) = \sigma(gxg^{-1}) = \sigma(g)\sigma(x)\sigma(g)^{-1} \in \mathcal{K}'.$$

Therefore,  $\sigma(\mathcal{K}) \subseteq \mathcal{K}'$ .

Now let  $z \in \mathcal{K}'$ . Then there exists some  $h \in G$  such that  $z = h\sigma(x)h^{-1}$ . Since  $\sigma$  is an automorphism, there exists some  $g \in G$  such that  $\sigma(g) = h$ . Then

$$z = h\sigma(x)h^{-1} = \sigma(g)\sigma(x)\sigma(g)^{-1} = \sigma(gxg^{-1}) \in \sigma(\mathcal{K}).$$

Therefore,  $\mathcal{K}' \subseteq \sigma(\mathcal{K})$ . Hence,  $\sigma(\mathcal{K}) = \mathcal{K}'$ , so that  $\text{Aut}(G)$  permutes the conjugacy classes of  $G$ .

(b) By [Exercise 4.3.33](#), the formulas for each conjugacy class is given by

$$|\mathcal{K}| = \binom{n}{2} = \frac{n(n-1)}{2}, \quad \text{and} \quad |\mathcal{K}'| = \frac{n!}{2^k k! (n-2k)!}$$

for some  $k \geq 2$ . Assume, by way of contradiction, that  $|\mathcal{K}| = |\mathcal{K}'|$ . Equating the two expressions and cancellation, we obtain

$$(n-2)(n-3) \cdots (n-2k+1) = 2^k k!.$$

Observe that the left-hand side contains  $2k-2$  consecutive integers, which has at most  $k-1$  powers of 2. The right-hand side has at least  $k$  powers of 2, so that the two sides cannot be equal. Therefore,  $|\mathcal{K}| \neq |\mathcal{K}'|$ . Hence, by (a), any automorphism of  $S_n$  sends transpositions to transpositions.

(c) Let  $\sigma \in \text{Aut}(S_n)$ . By (b), we know that  $\sigma$  sends transpositions to transpositions. In particular,

$$\sigma((1\ 2)) = (a\ b_2), \quad \sigma((1\ 3)) = (c\ d_3)$$

for distinct integers. Note that if  $\{a, b_2\} \cap \{c, d_3\} = \emptyset$ , then

$$\sigma((1\ 2)(1\ 3)) = \sigma((1\ 2))\sigma((1\ 3)) = (a\ b_2)(c\ d_3)$$

is a product of two disjoint transpositions. However,  $(1\ 2)(1\ 3) = (1\ 3\ 2)$  is a 3-cycle, so this is impossible. Therefore,  $\{a, b_2\} \cap \{c, d_3\} \neq \emptyset$ . Without loss of generality, assume that  $a = c$  since we may interchange the order within the transposition. Repeating this argument for all transpositions  $(1\ k)$  for  $k = 4, \dots, n$ , we obtain the desired result. Moreover, these integers must be distinct since  $\sigma$  is injective.

(d) Note that  $(1\ 2), (1\ 3), \dots, (1\ n)$  generate  $S_n$  since any transposition  $(i\ j)$  can be written as

$$(i\ j) = (1\ i)(1\ j)(1\ i).$$

Therefore, any automorphism of  $S_n$  is uniquely determined by its action on these elements. By (c), there are at most  $n!$  ways to assign images to these transpositions since there are  $n$  choices for the image of  $(1\ 2)$ ,  $n-1$  choices for the image of  $(1\ 3)$ , and so on. Therefore,  $|\text{Aut}(S_n)| \leq n!$ . However,  $|\text{Inn}(S_n)| = |S_n/Z(S_n)| = n!$  since  $Z(S_n)$  is trivial for  $n \geq 3$ . Since  $\text{Inn}(S_n) \leq \text{Aut}(S_n)$ , we must have  $\text{Aut}(S_n) = \text{Inn}(S_n)$  for  $n \neq 6$ . ■

(\*) **Exercise 4.4.19**

This exercise shows that  $|\text{Aut}(S_6) : \text{Inn}(S_6)| \leq 2$  (Exercise 10 in Section 6.3 shows that equality holds by exhibiting an automorphism of  $S_6$  that is not inner).

(a) Let  $\mathcal{K}$  be the conjugacy class of transpositions in  $S_6$  and let  $\mathcal{K}'$  be the conjugacy class of any element of order 2 in  $S_6$  that is not a transposition. Prove that  $|\mathcal{K}| \neq |\mathcal{K}'|$  unless  $\mathcal{K}'$  is the conjugacy class of products of three disjoint transpositions. Deduce that  $\text{Aut}(S_6)$  has a subgroup of index at most 2 which sends transpositions to transpositions.

(b) Prove that  $|\text{Aut}(S_6) : \text{Inn}(S_6)| \leq 2$ . (Follow the same steps as in (c) and (d) of the preceding exercise to show that any automorphism that sends transpositions to transpositions is inner.)

**Solution.**

(a) This is similar to (b) in the previous exercise. By [Exercise 4.3.33](#), the formulas for each conjugacy class is given by

$$|\mathcal{K}| = \binom{6}{2} = 15, \quad \text{and} \quad |\mathcal{K}'| = \frac{6!}{2^k k! (6-2k)!}$$

for some  $k \geq 2$ . Since we are working in  $S_6$ , we only need to check  $k = 2$  and  $k = 3$ . For  $k = 2$ , we have

$$|\mathcal{K}'| = \frac{6!}{2^2 \cdot 2! \cdot 2!} = 45 \neq 15 = |\mathcal{K}|.$$

For  $k = 3$ , we have

$$|\mathcal{K}'| = \frac{6!}{2^3 \cdot 3! \cdot 0!} = 15 = |\mathcal{K}|.$$



Therefore,  $|\mathcal{K}| \neq |\mathcal{K}'|$  unless  $\mathcal{K}'$  is the conjugacy class of products of three disjoint transpositions. Hence, by (a) in the previous exercise,  $\text{Aut}(S_6)$  has a subgroup of index at most 2 which sends transpositions to transpositions.

- (b) Following the same steps as in (c) and (d) of the previous exercise, we can show that any automorphism that sends transpositions to transpositions is inner. Therefore, by (a), we have  $|\text{Aut}(S_6) : \text{Inn}(S_6)| \leq 2$ . ■

(\*) **Exercise 4.4.20**

For any finite group  $P$  let  $d(P)$  be the minimum number of generators of  $P$  (so, for example,  $d(P) = 1$  if and only if  $P$  is a nontrivial cyclic group and  $d(Q_8) = 2$ ). Let  $m(P)$  be the maximum of the integers  $d(A)$  as  $A$  runs over all abelian subgroups of  $P$  (so, for example,  $m(Q_8) = 1$  and  $m(D_8) = 2$ ). Define

$$J(P) = \langle A \mid A \text{ is an abelian subgroup of } P \text{ with } d(A) = m(P) \rangle.$$

( $J(P)$  is called the *Thompson subgroup* of  $P$ .)

- (a) Prove that  $J(P)$  is a characteristic subgroup of  $P$ .  
 (b) For each of the following groups  $P$ , list all abelian subgroups  $A$  of  $P$  that satisfy  $d(A) = m(P)$ :  $Q_8$ ,  $D_8$ ,  $D_{16}$ , and  $QD_{16}$  (where  $QD_{16}$  is the quasidihedral group of order 16 defined in [Exercise 2.5.11](#)). (Use the lattices of subgroups for these groups in Section 2.5.)  
 (c) Show that  $J(Q_8) = Q_8$ ,  $J(D_8) = D_8$ ,  $J(D_{16}) = D_{16}$ , and  $J(QD_{16})$  is a dihedral subgroup of order 8 in  $QD_{16}$ .  
 (d) Prove that if  $Q \leq P$  and  $J(P)$  is a subgroup of  $Q$ , then  $J(P) = J(Q)$ . Deduce that if  $P$  is a subgroup (not necessarily normal) of the finite group  $G$  and  $J(P)$  is contained in some subgroup  $Q$  of  $P$  such that  $Q \trianglelefteq G$ , then  $J(P) \trianglelefteq G$ .

**Solution.**

- (a) Let  $\sigma \in \text{Aut}(P)$ . Let  $A$  be an abelian subgroup of  $P$  such that  $d(A) = m(P)$ . Then  $\sigma(A)$  is also an abelian subgroup of  $P$ . Moreover, it is clear that  $d(\sigma(A)) = d(A) = m(P)$ . Therefore,  $\sigma(J(P)) = J(P)$ , so that  $J(P)$  is characteristic in  $P$ .  
 (b) •  $Q_8$ : From the lattice, we see that every abelian subgroup is cyclic so that  $m(Q_8) = 1$ . The abelian subgroups  $A$  with  $d(A) = 1$  are  $\langle i \rangle$ ,  $\langle j \rangle$ , and  $\langle k \rangle$ .  
 •  $D_8$ : From the lattice, we see that the abelian subgroups with maximum number of generators are those isomorphic to  $V_4$ , hence  $m(D_8) = 2$ . The abelian subgroups  $A$  with  $d(A) = 2$  are  $\langle r^2, s \rangle$  and  $\langle r^2, rs \rangle$ .  
 •  $D_{16}$ : From the lattice, we see that the abelian subgroups with maximum number of generators are those isomorphic to  $V_4$ , hence  $m(D_{16}) = 2$ . The abelian subgroups  $A$  with  $d(A) = 2$  are  $\langle r^4, s \rangle$ ,  $\langle r^4, r^5s \rangle$ ,  $\langle r^4, r^2s \rangle$ , and  $\langle r^4, r^3s \rangle$ .  
 •  $QD_{16}$ : From the lattice, we see that the abelian subgroups with maximum number of generators are those isomorphic to  $V_4$ , hence  $m(QD_{16}) = 2$ . The abelian subgroups  $A$  with  $d(A) = 2$  are  $\langle \tau\sigma^2, \sigma^4 \rangle$  and  $\langle \tau, \sigma^4 \rangle$ .  
 (c) • Any pair of the abelian subgroups in (b) generate  $Q_8$ , so that  $J(Q_8) = Q_8$ .  
 • Note that  $r, s \in \langle r^2, s \rangle \cup \langle r^2, rs \rangle$ , so that  $J(D_8) = D_8$ .  
 • Note that  $r^3 = r^3s \cdot s$  and  $r = r^4 \cdot r^{-3}$  so that the join of all abelian subgroups in (b) generates  $D_{16}$ . Therefore,  $J(D_{16}) = D_{16}$ .  
 • Observe that

$$\langle \tau\sigma^2, \sigma^4 \rangle = \{1, \sigma^4, \tau\sigma^2, \tau\sigma^6\} \quad \text{and} \quad \langle \tau, \sigma^4 \rangle = \{1, \sigma^4, \tau, \tau\sigma^4\}.$$

The join of these two subgroups is the subgroup  $\langle \sigma^2, \tau \rangle$ , which has order 8. Since  $\tau^2 = (\sigma^2)^4 = 1$ , and  $\sigma^2\tau = \tau\sigma^6$ , then the mapping  $\varphi : D_8 \rightarrow \langle \sigma^2, \tau \rangle$  defined by  $\varphi(r) = \sigma^2$  and  $\varphi(s) = \tau$  is an isomorphism. Therefore,  $J(QD_{16})$  is a dihedral subgroup of order 8 in  $QD_{16}$ .

- (d) Note that any abelian subgroup of  $Q$  is also an abelian subgroup of  $P$ . Therefore,  $m(Q) \leq m(P)$ . Since  $J(P) \leq Q$ , then every abelian subgroup  $A$  of  $P$  with  $d(A) = m(P)$  is also an abelian subgroup of  $Q$ . Therefore,  $m(P) \leq m(Q)$ . Hence,  $m(P) = m(Q)$ . Moreover, any abelian subgroup  $B$  of  $Q$  with  $d(B) = m(Q) = m(P)$  is also an abelian subgroup of  $P$ . Therefore,  $J(P) = J(Q)$ .

Using that  $J(P) = J(Q)$ , then part (a) implies that  $J(P) \text{ char } Q$ . Since  $Q \trianglelefteq G$ , then  $J(P) \trianglelefteq G$ . ■



## 4.5 Sylow's Theorem

Let  $G$  be a finite group and let  $p$  be a prime.

### Exercise 4.5.1

Prove that if  $P \in \text{Syl}_p(G)$  and  $H$  is a subgroup of  $G$  containing  $P$  then  $P \in \text{Syl}_p(H)$ . Give an example to show that in general, a Sylow  $p$ -subgroup of a subgroup of  $G$  need not be a Sylow  $p$ -subgroup of  $G$ .

**Solution.** Let  $|G| = p^n m$  where  $p \nmid m$ . Since  $P \leq H \leq G$ , then  $|H| = p^k t$  for some  $k \leq n$  and  $p \nmid t$ . Since  $P \in \text{Syl}_p(G)$ , then  $|P| = p^n$ . However, since  $P \leq H$ , then  $|P| = p^n \leq |H| = p^k t$ . Therefore, we must have  $k = n$ , so that  $|H| = p^n t$ . Hence,  $P \in \text{Syl}_p(H)$ .

Consider  $G = S_4$  and its subgroup  $H = A_4$ . Then  $|G| = 24$  and  $|H| = 12$ . A Sylow 2-subgroup of  $H$  is isomorphic to  $V_4$ , which has order 4. However, a Sylow 2-subgroup of  $G$  has order 8, so that the Sylow 2-subgroup of  $H$  is not a Sylow 2-subgroup of  $G$ . ■

### Exercise 4.5.2

Prove that if  $H$  is a subgroup of  $G$  and  $Q \in \text{Syl}_p(H)$  then  $gQg^{-1} \in \text{Syl}_p(gHg^{-1})$  for all  $g \in G$ .

**Solution.** Let  $|H| = p^n m$  where  $p \nmid m$ . Then  $|Q| = p^n$ . Note that  $|gHg^{-1}| = |H| = p^n m$  since conjugation is an isomorphism. Moreover,  $|gQg^{-1}| = |Q| = p^n$ . Therefore,  $gQg^{-1} \in \text{Syl}_p(gHg^{-1})$ . ■

### Exercise 4.5.3

Use Sylow's Theorem to prove Cauchy's Theorem. (Note that we only used Cauchy's Theorem for abelian groups—Proposition 3.21—in the proof of Sylow's Theorem so this line of reasoning is not circular.)

**Solution.** Let  $|G| = p^n m$  where  $p \nmid m$ . By Sylow's Theorem, there exists some  $P \in \text{Syl}_p(G)$  such that  $|P| = p^n$ . Since  $n \geq 1$ , then  $P$  is nontrivial. By Theorem 4.8, we know that the center  $Z(P)$  is nontrivial. Since  $Z(P)$  is abelian, then by Proposition 3.21, there exists some element  $x \in Z(P)$  such that  $|x| = p$ . Therefore,  $x \in G$  has order  $p$ . ■

### Exercise 4.5.4

Exhibit all Sylow 2-subgroups and Sylow 3-subgroups of  $D_{12}$  and  $S_3 \times S_3$ .

**Solution.** For  $D_{12}$ , we have  $|D_{12}| = 12 = 2^2 \cdot 3$ . Therefore, the Sylow 2-subgroups have order 4 and the Sylow 3-subgroups have order 3. The Sylow 2-subgroups are

$$\langle r^3, s \rangle, \quad \langle r^3, rs \rangle, \quad \langle r^3, r^2s \rangle.$$

The Sylow 3-subgroup is

$$\langle r^2 \rangle.$$

where this is the only Sylow 3-subgroup, since if  $n_3 = 4$ , then there would be 8 elements of order 3, which is impossible.

We first show that the Sylow  $p$ -subgroups of a direct product are the direct products of the Sylow  $p$ -subgroups of each factor. Let  $G$  and  $H$  be finite groups, and let  $P \in \text{Syl}_p(G)$  and  $Q \in \text{Syl}_p(H)$ . Then  $|P| = p^n$  and  $|Q| = p^m$  for some  $n, m \geq 0$ . Therefore,  $|P \times Q| = p^{n+m}$ . Moreover, since  $|G \times H| = |G||H|$ , then the highest power of  $p$  dividing  $|G \times H|$  is also  $p^{n+m}$ . Hence,  $P \times Q \in \text{Syl}_p(G \times H)$ . For the converse, let  $R \in \text{Syl}_p(G \times H)$ . Then  $|R| = p^k$  for some  $k \geq 0$ . Consider the projections  $\pi_G : G \times H \rightarrow G$  and  $\pi_H : G \times H \rightarrow H$ . Then  $\pi_G(R)$  is a  $p$ -subgroup of  $G$  and  $\pi_H(R)$  is a  $p$ -subgroup of  $H$ . Therefore, there exist Sylow  $p$ -subgroups  $P$  of  $G$  and  $Q$  of  $H$  such that  $\pi_G(R) \leq P$  and  $\pi_H(R) \leq Q$ . Hence,  $R \leq P \times Q$ . Since  $|R| = p^k$  is the highest power of  $p$  dividing  $|G \times H|$ , then we must have  $|P \times Q| = p^k$ . Therefore,  $R = P \times Q$ . This shows that the Sylow  $p$ -subgroups of a direct product are the direct products of the Sylow  $p$ -subgroups of each factor.

We now proceed with  $S_3 \times S_3$ . Note that  $|S_3 \times S_3| = 36 = 2^2 \cdot 3^2$ . Therefore, the Sylow 2-subgroups have order 4 and the Sylow 3-subgroups have order 9. Moreover, note that  $\text{Syl}_2(S_3)$  is the set of subgroups generated by transpositions, and  $\text{Syl}_3(S_3)$  is the unique subgroup generated by a 3-cycle. Therefore, we have

$$\text{Syl}_2(S_3 \times S_3) = \{\langle \alpha \rangle \times \langle \beta \rangle \mid \alpha, \beta \in \{(1\ 2), (1\ 3), (2\ 3)\}\}$$

and  $\text{Syl}_3(S_3 \times S_3)$  consists of  $A_3 \times A_3$ . ■

#### Exercise 4.5.5

Show that a Sylow  $p$ -subgroup of  $D_{2n}$  is cyclic and normal for every odd prime  $p$ .

**Solution.** Let  $|D_{2n}| = 2n = 2p^k m$  where  $p \nmid m$  and  $k \geq 1$ . Then the Sylow  $p$ -subgroups have order  $p^k$ . Note that  $r^m$  has order  $p^k$ , so that  $\langle r^m \rangle$  is a Sylow  $p$ -subgroup and is cyclic. Moreover, conjugacy of Sylow  $p$ -subgroups imply that any Sylow  $p$ -subgroup is also cyclic.

To see that the Sylow  $p$ -subgroup is normal, note that  $n_p \equiv 1 \pmod p$  and  $n_p \mid 2m$ . Since  $p$  is odd and  $p \nmid m$ , then  $(2m, p) = 1$ . Therefore,  $n_p = 1$ , so that the Sylow  $p$ -subgroup is unique and hence normal. ■

#### Exercise 4.5.6

Exhibit all Sylow 3-subgroups of  $A_4$  and all Sylow 3-subgroups of  $S_4$ .

**Solution.** The Sylow 3-subgroups of both  $A_4$  and  $S_4$  have order 3. The Sylow 3-subgroups of  $S_4$  are

$$\langle (1\ 2\ 3) \rangle, \quad \langle (1\ 2\ 4) \rangle, \quad \langle (1\ 3\ 4) \rangle, \quad \langle (2\ 3\ 4) \rangle.$$

Note that these are also the Sylow 3-subgroups of  $A_4$  since they are all contained in  $A_4$ . ■

#### Exercise 4.5.7

Exhibit all Sylow 2-subgroups of  $S_4$  and find elements of  $S_4$  which conjugate one of these into each of the others.

**Solution.** The Sylow 2-subgroups of  $S_4$  have order 8. Note that we have calculated 2 of these subgroups in [Exercise 4.2.5](#), namely parts (a) and (b). Since  $n_2 \equiv 1 \pmod 2$  and  $n_2 \mid 3$ , then  $n_2 = 1$  or 3. However, there are more than one Sylow 2-subgroup, so that  $n_2 = 3$ . One of the Sylow 2-subgroups is

$$\langle (1\ 2\ 3\ 4), (2\ 4) \rangle = \{1, (1\ 2\ 3\ 4), (1\ 3)(2\ 4), (1\ 4\ 3\ 2), (2\ 4), (1\ 2)(3\ 4), (1\ 3), (1\ 4)(2\ 3)\}.$$

The other two can be reached by conjugation. For example, conjugating by  $(2\ 3)$  gives

$$\langle (1\ 3\ 2\ 4), (3\ 4) \rangle = \{1, (1\ 3\ 2\ 4), (1\ 2)(3\ 4), (1\ 4\ 2\ 3), (3\ 4), (1\ 3)(2\ 4), (1\ 2), (1\ 4)(2\ 3)\},$$

and conjugating by  $(3\ 4)$  gives

$$\langle (1\ 2\ 4\ 3), (2\ 3) \rangle = \{1, (1\ 2\ 4\ 3), (1\ 4)(2\ 3), (1\ 3\ 4\ 2), (2\ 3), (1\ 2)(3\ 4), (1\ 4), (1\ 3)(2\ 4)\}.$$

Note that these are all distinct but have nontrivial intersections. ■

#### Exercise 4.5.8

Exhibit two distinct Sylow 2-subgroups of  $S_5$  and an element of  $S_5$  that conjugates one into the other.

**Solution.** Use any of the Sylow 2-subgroups of  $S_4$  from [Exercise 4.5.7](#) and consider them as subgroups of  $S_5$ . ■

**Exercise 4.5.9**

Exhibit all Sylow 3-subgroups of  $\text{SL}_2(\mathbb{F}_3)$  (cf. [Exercise 2.1.9](#)).

**Solution.** Recall that  $|\text{SL}_2(\mathbb{F}_3)| = 24 = 3 \cdot 2^3$ . Therefore, the Sylow 3-subgroups have order 3. Moreover, recall that  $\text{SL}_2(\mathbb{F}_3)$  is generated by the two matrices as listed in [Exercise 2.4.9](#). Since every group of order 3 is cyclic, it is clear that two distinct Sylow 3-subgroups are

$$\langle A \rangle = \left\{ I_2, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \right\}, \quad \langle B \rangle = \left\{ I_2, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \right\}$$

Since  $n_3 \equiv 1 \pmod{3}$  and  $n_3 \mid 8$ , then  $n_3 = 4$  since we were able to exhibit more than one Sylow 3-subgroup. Conjugating  $A$  by  $B$ , we obtain another matrix  $C$  of order 3. Conjugating  $B$  by  $C$  gives another matrix  $D$  of order 3. Therefore, the other two Sylow 3-subgroups are

$$\langle C \rangle = \left\{ I_2, \begin{bmatrix} 0 & 1 \\ 2 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \right\}, \quad \langle D \rangle = \left\{ I_2, \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix} \right\}. \quad \blacksquare$$

**Exercise 4.5.10**

Prove that the subgroup of  $\text{SL}_2(\mathbb{F}_3)$  generated by

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

is the unique Sylow 2-subgroup of  $\text{SL}_2(\mathbb{F}_3)$  (cf. [Exercise 2.4.10](#)).

**Solution.** Let  $P$  be the subgroup generated by the two matrices. Note that  $P \cong Q_8$  by [Exercise 2.4.10](#) so that  $|P| = 8$ , and  $P \in \text{Syl}_2(\text{SL}_2(\mathbb{F}_3))$ . To show uniqueness, let  $Q \in \text{Syl}_2(\text{SL}_2(\mathbb{F}_3))$ . Since  $Q$  is conjugate to  $P$ , then  $Q \cong Q_8$  as well. We now consider  $|P \cap Q|$ , and let  $Z = \langle I \rangle$  be the subgroup of  $P$  associated with  $Z(Q_8)$ . For any  $g \in \text{SL}_2(\mathbb{F}_3)$ , then  $gPg^{-1} \cong Q_8$ . However,  $gZg^{-1} \leq gPg^{-1}$  is a subgroup of order 2, and  $Z \leq gPg^{-1}$  as well, since  $Z \leq Z(\text{SL}_2(\mathbb{F}_3))$ . It follows that  $gZg^{-1} = Z$ , hence  $Z \trianglelefteq \text{SL}_2(\mathbb{F}_3)$  and every Sylow 2-subgroup must contain  $Z$ .

Clearly, the intersection cannot be trivial since both  $P$  and  $Q$  are isomorphic to  $Q_8$ , hence they must both contain  $Z$ . If  $|P \cap Q| = 2$ , then  $P \cap Q = Z$ . Note that in  $P$ , there are 6 elements of order 4. However, taking the case of  $n_3 = 3$ , then there would be  $1 + 1 + 6 \cdot 3 = 20$  elements accounted for, where the first 1 is from the identity, the second 1 is from the unique element of order 2, and  $6 \cdot 3$  is from the elements of order 4 in each Sylow 2-subgroup. However, from [Exercise 4.5.9](#), there are at least 8 elements of order 3, giving a total of at least 28 elements, which is impossible. Therefore,  $|P \cap Q| \neq 2$ .

If  $|P \cap Q| = 4$ , first let  $P = \{I, -I, A, A^3, B, B^3, AB, A^3B^3\}$  where  $A$  and  $B$  generate  $P$ . Since  $P \cap Q \leq P \cong Q_8$  and has order 4, it must be one of the cyclic subgroups. Without loss of generality, let  $P \cap Q = \langle A \rangle$ . Since we are assuming distinct Sylow 2-subgroups, then we may let  $Q = \{I, -I, A, A^3, C, C^3, AC, A^3C^3\}$  for some  $C \notin P$ . Let  $R$  be the third Sylow 2-subgroup. Then  $|P \cap R|$  must also be 4 by the preceding reasons of ruling out 1 and 2, while 8 would imply  $P = R$ . Therefore,  $P \cap R$  is also a cyclic subgroup of order 4. Again, without loss of generality, let  $P \cap R = \langle B \rangle$ . Then we may let  $R = \{I, -I, B, B^3, D, D^3, BD, B^3D^3\}$  for some  $D \notin P$ . Now consider  $Q \cap R$ , which must also have order 4. However, by construction,  $Q$  and  $R$  only share the elements  $I, -I$ , so that  $|Q \cap R| = 2$ , a contradiction. Therefore,  $|P \cap Q| \neq 4$ .

Hence, we must have  $|P \cap Q| = 8$ , so that  $P = Q$ . Therefore,  $P$  is the unique Sylow 2-subgroup of  $\text{SL}_2(\mathbb{F}_3)$ .  $\blacksquare$

**Exercise 4.5.11**

Show that the center of  $\mathrm{SL}_2(\mathbb{F}_3)$  is the group of order 2 consisting of  $\pm I$ , where  $I$  is the identity matrix. Prove that  $\mathrm{SL}_2(\mathbb{F}_3)/Z(\mathrm{SL}_2(\mathbb{F}_3)) \cong A_4$ . [Use facts about groups of order 12.]

**Solution.** Consider the matrices

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad Z = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

in  $\mathrm{SL}_2(\mathbb{F}_3)$ , where  $Z \in Z(\mathrm{SL}_2(\mathbb{F}_3))$ . Then

$$AZ = \begin{bmatrix} a+c & b+d \\ c & d \end{bmatrix} = \begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix} = ZA$$

from which we can determine  $c = 0$  and  $a = d$ . Then

$$BZ = \begin{bmatrix} a & b \\ a & a+b \end{bmatrix} = \begin{bmatrix} a+b & b \\ a & a \end{bmatrix} = ZB$$

from which we can determine that  $a \in \mathbb{F}_3$  and  $b = 0$ . Then  $Z = aI_2$ . To prove the second part, we show that  $\mathrm{SL}_2(\mathbb{F}_3)/Z(\mathrm{SL}_2(\mathbb{F}_3))$  does not have a normal Sylow 3-subgroup. To that end, let  $P \in \mathrm{Syl}_3(\mathrm{SL}_2(\mathbb{F}_3))$ , and consider  $P \cap Z$ . Under the projection homomorphism  $\pi$  restricted to  $P$ , it is clear that  $\ker(\pi|_P) = P \cap Z$ , since  $\ker(\pi|_P)$  are elements  $p \in P$  such that  $\pi(p) = Z$ . By the First Isomorphism Theorem, then  $P/\ker(\pi|_P) \cong \pi(P)$ , or  $P/P \cap Z \cong \pi(P)$ . Since  $(|P|, |Z|) = 1$  and  $P \cap Z$  is a subgroup of both, then the intersection is trivial so that  $P \cong \pi(P)$ . Then  $|\pi(P)| = 3$  so that  $\pi(P) \in \mathrm{Syl}_3(\mathrm{SL}_2(\mathbb{F}_3)/Z(\mathrm{SL}_2(\mathbb{F}_3)))$ . This shows that any Sylow 3-subgroup of  $\mathrm{SL}_2(\mathbb{F}_3)$  is also a Sylow 3-subgroup of the quotient.

Since Exercise 4.5.9 showed that  $\mathrm{SL}_2(\mathbb{F}_3)$  has 4 Sylow 3-subgroups, then there are also 4 Sylow 3-subgroups in the quotient. Hence,  $n_3(\mathrm{SL}_2(\mathbb{F}_3)/Z(\mathrm{SL}_2(\mathbb{F}_3))) = 4 \neq 1$ . By the text's classification of groups of order 12, it must be that the quotient group is isomorphic to  $A_4$ . ■

**Exercise 4.5.12**

Let  $2n = 2^a k$  where  $k$  is odd. Prove that the number of Sylow 2-subgroups of  $D_{2n}$  is  $k$ . [Prove that if  $P \in \mathrm{Syl}_2(D_{2n})$  then  $N_{D_{2n}}(P) = P$ .]

**Solution.** We construct a Sylow 2-subgroup. Note that  $n = 2^{a-1}k = |r|$  so that  $|r^k| = 2^{a-1}$ . Since  $s$  has order 2, then  $P = \langle r^k, s \rangle$  has order  $2^a$  and is a Sylow 2-subgroup of  $D_{2n}$ . To show that  $N_{D_{2n}}(P) = P$ , let  $g \in N_{D_{2n}}(P)$ . We then have two cases:

- If  $g = r^i$  for some  $i$ , then it normalizes  $s$ . In particular,  $r^i s r^{-i} = r^{2i} s \in P$ . Then  $r^{2i} s = r^{km} s$  for some  $m$ . It follows that  $k \mid 2i$ . However,  $k$  is odd so that  $k \mid i$ , hence  $r^i \in \langle r^k \rangle \leq P$ .
- If  $g = sr^i$ , it also normalizes  $s$ . Then  $sr^i s (sr^i)^{-1} = sr^i s r^{-i} = r^{-2i} s \in P$ . Then  $r^{-2i} s = r^{km} s$  for some  $m$ . It follows that  $k \mid -2i$ . However,  $k$  is odd so that  $k \mid i$ , hence  $sr^i \in \langle r^k, s \rangle = P$ .

In both cases, we have  $g \in P$ , so that  $N_{D_{2n}}(P) = P$ .

Now we show that the number of Sylow 2-subgroups of  $D_{2n}$  is  $k$ . Note that the number of Sylow 2-subgroups is given by the index of the normalizer of any Sylow 2-subgroup. Since we have shown that the normalizer is just the subgroup itself, then the number of Sylow 2-subgroups is given by the index of any Sylow 2-subgroup, which is  $|D_{2n}|/|P| = 2^a k / 2^a = k$ . ■

**Exercise 4.5.13**

Prove that a group of order 56 has a normal Sylow  $p$ -subgroup for some prime  $p$  dividing its order.

**Solution.** Note that  $56 = 2^3 \cdot 7$ . Recall that  $n_7 \equiv 1 \pmod{7}$  and  $n_7 \mid 8$ , so that  $n_7 = 1$  or  $8$ . Moreover,  $n_2 \equiv 1 \pmod{2}$  and  $n_2 \mid 7$ , so that  $n_2 = 1$  or  $7$ . Assume, by way of contradiction, that both  $n_7 = 8$  and  $n_2 = 7$ . Then there are at least  $8 \cdot 6 = 48$  elements of order 7. For the Sylow 2-subgroups, their pairwise intersection can have at most 4 elements, so that there are at least  $7 \cdot 4 = 28$  elements of order 2-power order. This gives a total of at least 77 elements, which is impossible. Therefore, either  $n_7 = 1$  or  $n_2 = 1$ , so that there is a normal Sylow  $p$ -subgroup for some prime  $p$  dividing the order of the group. ■

**Exercise 4.5.14**

Prove that a group of order 312 has a normal Sylow  $p$ -subgroup for some prime  $p$  dividing its order.

**Solution.** Note that  $312 = 2^3 \cdot 3 \cdot 13$ . Recall that  $n_{13} \equiv 1 \pmod{13}$  and  $n_{13} \mid 24$ , so that  $n_{13} = 1$  or  $24$ . Moreover,  $n_3 \equiv 1 \pmod{3}$  and  $n_3 \mid 104$ , so that  $n_3 = 1, 13, 52$ , or  $104$ . Lastly,  $n_2 \equiv 1 \pmod{2}$  and  $n_2 \mid 39$ , so that  $n_2 = 1, 3, 13$ , or  $39$ . Assume, by way of contradiction, that  $n_{13} = 24$ ,  $n_3 > 1$ , and  $n_2 > 1$ . In particular, then  $n_3 \geq 13$  and  $n_2 \geq 3$ . The Sylow-3 and Sylow-13 subgroups intersect trivially, while the Sylow 2-subgroups can have at most 4 elements in common. Then there are at least  $24 \cdot 12 = 288$  elements of order 13, at least  $13 \cdot 2 = 26$  elements of order 3, and at least  $3 \cdot 4 = 12$  elements of order 2-power order, giving a total of at least 326 elements, which is impossible. Therefore, either  $n_{13} = 1$ , or  $n_3 = 1$ , or  $n_2 = 1$ , so that there is a normal Sylow  $p$ -subgroup for some prime  $p$  dividing the order of the group. ■

**Exercise 4.5.15**

Prove that a group of order 351 has a normal Sylow  $p$ -subgroup for some prime  $p$  dividing its order.

**Solution.** Note that  $351 = 3^3 \cdot 13$ . Recall that  $n_{13} \equiv 1 \pmod{13}$  and  $n_{13} \mid 27$ , so that  $n_{13} = 1$  or  $27$ . Moreover,  $n_3 \equiv 1 \pmod{3}$  and  $n_3 \mid 13$ , so that  $n_3 = 1$  or  $13$ . Assume, by way of contradiction, that both  $n_{13} = 27$  and  $n_3 = 13$ . The Sylow-13 subgroups intersect trivially, while the Sylow-3 subgroups can have at most 9 elements in common. Then there are at least  $27 \cdot 12 = 324$  elements of order 13 and at least  $13 \cdot 8 = 104$  elements of order 3, giving a total of at least 428 elements, which is impossible. Therefore, either  $n_{13} = 1$  or  $n_3 = 1$ , so that there is a normal Sylow  $p$ -subgroup for some prime  $p$  dividing the order of the group. ■

**Exercise 4.5.16**

Let  $|G| = pqr$ , where  $p, q$  and  $r$  are primes with  $p < q < r$ . Prove that  $G$  has a normal Sylow subgroup for either  $p, q$  or  $r$ .

**Solution.** Assume, by way of contradiction, that  $G$  does not contain a normal Sylow subgroup. In particular,  $n_r \neq 1$  so that  $n_r = pq$ . Then  $G$  contains  $pq(r-1)$  elements of order  $r$ . Moreover,  $n_q = 1$  or  $pr$ . If  $n_q = pr$ , then there are  $pr(q-1)$  elements of order  $q$ . Then there are

$$pq(r-1) + pr(q-1) = pqr - pq + prq - pr = 2pqr - p(q+r)$$

elements of order  $q$  or  $r$ . Observe that  $(q-1)(r-1) - 1 > 0$  since  $q \geq 3$  and  $r \geq 5$ . Then

$$(q-1)(r-1) - 1 = qr - q - r = qr - (q+r) > 0$$

so that  $qr > q+r$ . Then  $pqr > p(q+r)$ , so that  $2pqr - p(q+r) > pqr$ . This gives a contradiction since there are at most  $pqr$  elements in  $G$ . Therefore,  $n_q = 1$ , so that there is a normal Sylow subgroup in  $G$ . ■

**Exercise 4.5.17**

Prove that if  $|G| = 105$  then  $G$  has a normal Sylow 5-subgroup and a normal Sylow 7-subgroup.

**Solution.** Note that  $105 = 3 \cdot 5 \cdot 7$ . Recall that  $n_7 \equiv 1 \pmod{7}$  and  $n_7 \mid 15$ , so that  $n_7 = 1$  or  $15$ . Moreover,  $n_5 \equiv 1 \pmod{5}$  and  $n_5 \mid 21$ , so that  $n_5 = 1$  or  $21$ . Assume, by way of contradiction, that both  $n_7 = 15$  and  $n_5 = 21$ . Then there are at least  $15 \cdot 6 = 90$  elements of order 7 and at least  $21 \cdot 4 = 84$  elements of order 5, giving a total of at least 174 elements, which is impossible. Therefore, either  $n_7 = 1$  or  $n_5 = 1$ , so that there is a normal Sylow subgroup for either prime.

Without loss of generality, let  $P \in \text{Syl}_5(G)$  be normal. Then  $G/P$  has order 21, and  $G/P$  contains a normal Sylow 7-subgroup  $Q/P$ . Taking the preimage of  $Q/P$  under the projection homomorphism  $\pi$ , then  $\pi^{-1}(Q/P)$  contains a normal Sylow 7-subgroup  $Q$  of  $G$ . From Corollary 4.20, we know that  $Q$  is characteristic in  $\pi^{-1}(Q/P)$ , and  $\pi^{-1}(Q/P)$  is normal in  $G$ , so that  $Q$  is normal in  $G$  as well. Therefore, there is a normal Sylow 5-subgroup and a normal Sylow 7-subgroup in  $G$ . ■